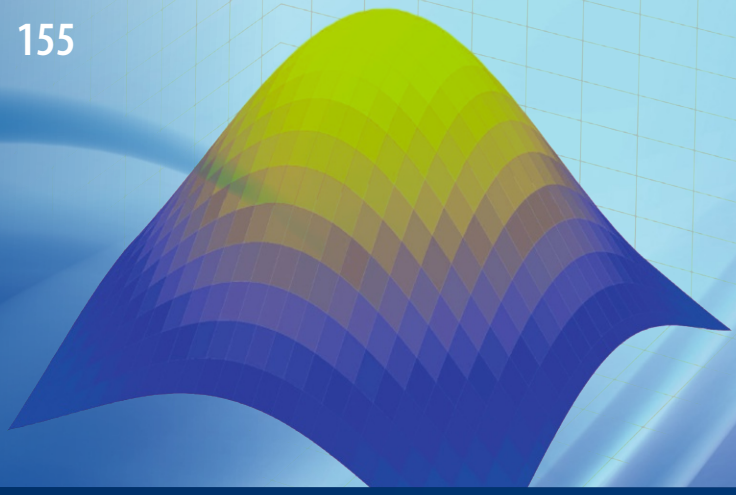
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Natalya Vaysfeld · Zinaida Zhuravlova

Mixed Boundary Problems in Solid Mechanics

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Cover illustration: It demonstrates the movement of points inside the semi-strip under an applied load.

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THIS BOOK IS DEDICATED TO THE MEMORY OF PROFESSOR GENNADY POPOV

Preface

Target Audience

The book is aimed at researchers, primarily but not exclusively graduate students, postdoctoral researchers, specialists from Aerospace and Civil Engineering, Materials Science, and Engineering Mechanics and should naturally also be of interest to specialists of Physics and Applied Mathematics. The book covers a wide range of subjects and techniques related to mixed boundary problems of elasticity from basic concepts to special techniques that are unlikely to appear in traditional university graduate courses. This book may also be of interest to industrial researchers who encounter defects such as cracks and inclusions of different materials in mechanisms under different localization and type of loading. So the topics present the application of mathematical mechanics of solid bodies notably in elasticity, showing the interconnection of elasticity and temperature that would normally be treated independently. Theoretical and experimental results are expected to be useful for researchers investigating a wide range of materials including metals, composites, ceramics, polymers, biomaterials, and nanomaterials under different mechanical and temperature loading. The aim of the book is to introduce an interdisciplinary audience to a variety of stress state phenomena occurring in elasticity near defects and edges of the bodies.

Annotation

The book is dedicated to new methods to solve mixed elasticity problems, which are among the most difficult problems in terms of their analytical solution. Mixed problems of mathematical physics are widely used in modeling problems in engineering, biology, geology, medicine, etc. and give a more realistic display of the problem described in the real engineering problem. An analytical approach to

solving modeling problems allows the finding of the most important qualitative characteristics of objects, to predict the development of destruction processes in them during the propagation of waves, temperature effects, mechanical pressure, and other mechanical phenomena that are essential for determining the strength and reliability of objects. From this point of view, the development of mathematical solving methods is an essential component of the development of modern applied programs for the study of such physical processes in bodies.

The book considers elastic objects of the canonical form under the influence of various loads. These problems are formulated as boundary value problems of mathematical physics with mixed boundary conditions. The authors propose a new approach for constructing analytical solutions of these problems in two-dimensional and three-dimensional formulations. The application of this method has been demonstrated in solving specific mixed problems of elasticity for the objects of different shapes, with various types of impacts on them. These analytical solutions are accompanied by the presented calculation results, allowing the drawing of significant conclusions about the nature of mechanical phenomena that occur in bodies. The book is useful for scientists in the field of applied mathematics, engineering, mechanics of deformed solids, geosciences, biology, and medicine.

Structure of the Book

Relevance and novelty of the proposed book Mixed problems of elasticity occupy an important place in the mechanics of deformable solids, which is due to their role in the modeling of various engineering problems. The main approaches for the analytical solving of such problems are based on the representation of solutions of equilibrium equations through auxiliary functions (harmonic, biharmonic, etc.). The main inconvenience of these approaches is that deriving expressions of real mechanical characteristics requires additional operations, which are often quite non-trivial.

The proposal in the book's approach uses direct integral transforms of the equilibrium equations. It allows construction in the transform domain of an analytical solution of the corresponding vector boundary problem relating to the searched displacement transforms. To simplify the calculations, Green's matrix-function was constructed in the form of bilinear expansion. This approach is demonstrated in the mixed elasticity problem for a semi-strip, which is the model object to identify patterns of stress-strain state of elastic bodies.

There are unsolved problems in the study of plane mixed problems of elasticity, which require the development of analytical methods to solve, which would simplify the construction of the solution and reveal the overall qualitative picture of the semi-strip's stress state.

This substantiates the relevance of developing a new method of analytical solution of plane mixed problems of elasticity for the semi-strip.

Here, for the first time, readers will learn:

- the methodology for solving mixed plane elasticity problems for a rectangular domain. It is based on the direct transformation of equilibrium equations. This technique allows to derive the analytical expressions for the searched mechanical characteristics as displacements and stress;
- how to construct Green's matrix-function in bilinear form and how to apply the matrix differential calculation method for the analytical solutions of mixed plane problems. In the case of the absence of a crack, the solution is reduced to one singular integral equation regarding the unknown displacement's jump. The main features of the dependence of the displacement and stress fields on the parameters of the load's interval on the short end of the semi-strip are established;
- how with the help of the proposed method to derive the effective approximate solution of a problem for an elastic semi-strip weakened by defects such as a transverse crack, longitudinal crack, etc.

The *first chapter* outlines the theoretical foundations on which the solving of the mixed problems of elasticity is based. This material contains both basic information from the theory of boundary value problems and material that is rarely included in traditional university courses. Along with basic concepts of the theory of boundary value problems of mathematical physics, non-traditional methods of constructing a scalar Green's function, as well as the theory of constructing Green's matrices, are outlined. Among the less known methods for solving boundary value problems with discontinuities within a domain is the generalized method of integral transforms, which is outlined in this chapter. The material of the first chapter is necessary and useful for understanding the essence of solving problems outlined in later chapters.

In the *second chapter*, a review of scientific works related to the solution of plane problems of the theory of elasticity and thermoelasticity for a semi-strip is carried out; stripes and semi-spaces with cracks; solution of integral equations with fixed singularities. Such studies are briefly analyzed.

The *third chapter* outlines the theoretical foundations on which the solving of the elasticity problems for a rectangular domain is based.

In *Sect. 3.1*, the general statement of mixed plane elasticity problems for a semi-infinite strip is formulated. The semi-strip's short edge is loaded by a normal load, and different boundary conditions for displacements and stress can be given on the semi-infinite boundaries. The stress state of the semi-strip should be found. The initial problem is reduced to a one-dimensional problem with the help of semi-infinite sin-cos Fourier transform. In *Sect. 3.2*, this problem is reformulated as a one-dimensional vector boundary problem. Its solution is searched as the superposition of the general solution of a homogeneous vector equation and the partial solution of an inhomogeneous vector equation. The general solution of the homogeneous vector equation is constructed in *Sect. 3.3* with the help of matrix differential calculation apparatus. The partial solution of the inhomogeneous vector equation is found in *Sect. 3.4* with the help of Green's matrix-function, which is constructed by the method of matrix integral transforms in the form of bilinear expansion. In *Sect. 3.5*, the inversion of derived transforms for the displacements and the summing

up of weakly-convergent integrals, which are in their expressions are done. Regarding the conditions on the semi-strip's short edge, the solving of the initial problem is reduced to the solving of a singular integro-differential equation (SIDE) or to the solving of an integral equation (SIE) with one or two fixing singularities. Their solution is presented in *Sect. 3.6*. The SIDE is solved by the orthogonal polynomials method. According to it, the unknown function is constructed as the expansion by Chebyshev polynomials of the second kind. Thus, the problem was reduced to the solving of an infinite system of linear algebraic equations. To solve the SIE, a transcendent equation is constructed. The unknown function is expanded in the sum of unknown constants multiplied by some functions dependent on the transcendent equation's roots. These constants are found from the system of linear algebraic equations.

In the *fourth chapter*, the procedure outlined in previous chapter is detailed for the semi-infinite strip in different cases of short-end load configuration: case A: mechanical load is applied at the center of the semi-strip's edge; case B: mechanical load acts from one side of the edge; case C: mechanical load is given on the whole edge.

The solving of the boundary problem in case A is reduced to the solving of the SIDE. In the case B, the problem is reduced to the solving the SIE with one fixed singularity. Unlike the previous cases, in case C the problem is reduced to the solving the SIE with two fixed singularities.

The values of normal stress on the semi-strip's boundaries regarding the size and location of the zone of application of mechanical loading by the short edge are analyzed. For cases A and B, the values of parameters under which the calculations become unstable, and the influence of additional fixed singularities should be taken into consideration, are found. For case C the stable results were derived along whole semi-strip's short edge.

The *fifth chapter* is dedicated to problems of uncoupled thermoelasticity for a rectangular domain. In *Sect. 5.1*, the exact solution for the problem of stationary thermoconductivity for a semi-strip is constructed. This solution is used in *Sect. 5.2* to solve the uncoupled thermoelasticity problem for a semi-infinite strip. The solution is constructed by the scheme shown in Chap. 1. The dependence of normal and tangential stress inside the semi-strip on the values of temperature and mechanical load for different load configurations (cases A, B, and C) is established. It is found that the temperature influence is essential on the semi-strip's stress state.

In the *sixth chapter*, the plane mixed elasticity problem for a semi-infinite strip with a transverse crack is considered. The semi-infinite boundaries are in fixed conditions, and the mechanical load is applied by configurations A, B, and C by the short edge. The initial problem is reduced to a one-dimensional problem with the help of semi-infinite sin-cos Fourier transform applied by the generalized scheme.

The solving of the boundary problem in the case of the load by scheme A is reduced to the system of three integro-differential equations, which is solved by the orthogonal polynomials method. In the case of load applied by the scheme B or C, the solving of the initial problem is reduced to the system of three integral equations, which is solved taking into account one or two fixed singularities in the first equation. The stress intensity factors (SIF) are calculated for all three cases. The

numerical analysis of the SIF changing during the change of the crack's length is done. The smallest distances between the crack tips and semi-strip's boundaries are found for all three cases of loading.

In the case of symmetric location of the crack relative to the lateral sides and symmetrical mechanical load, the simplified case when one unknown jump of displacements on the crack equals zero can be considered. Then the solving of the problem is reduced to the system of two singular integral equations. The SIF in this case is equal to the SIF in the general case under symmetric conditions.

In *Appendix* chapter, the material necessary for a better understanding of the book's content is presented. The authors considered it useful to include this material to explain some of the details of these problems' solving.

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List of Abbreviations

μ	Poisson's ratio
κ	Muskhelishvili coefficient
SIDE	Singular integro-differential equation
SIE	Singular integral equation
SIF	Stress intensity factor
SSIDE	System of singular integro-differential equations
SSIE	System of singular integral equations

Chapter 1
Mixed Boundary Value Problems of Solid Mechanics



This chapter presents the theory of mixed boundary value problems of solid mechanics on the basis of the elasticity theory models. The different approaches for these models' construction and solving are proposed. The apparatus for solving mixed problems using the Green's function construction technique for both the scalar and vector cases is considered in the chapter. The properties and main characteristics of the Green's function are given and formulated in the form of theorems.

Non-traditional methods of integral transforms' application in the case of discontinuities inside a medium are discussed here. The generalized method of integral transforms is presented for the case when inside a domain an unknown function and its derivative have discontinuities of the first kind with given jumps. Derived discontinuity problems lead to singular integral equations, which are solved by the orthogonal polynomial method. The properties of this method and conditions of its application are considered in the chapter. This effective approximate method usually is not covered by university courses.

Not widely known method of the Green's function's construction using the apparatus of the so-called basis functions is presented in the chapter, as well as the generalization of this method to a vector case, where it is necessary to construct the Green's function matrix. It allows to solve systems of equations in partial derivatives which are often met in solid mechanics modeling.

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1.1 The Formulation of Mixed Boundary Value Problems of Elasticity: Elements of the Theory

The apparatus of the classical boundary value problems theory was created in well-known works of Lame [80], N. I. Muskhelishvili [93], V. Novatsky [104], S. Timoshenko [142] and developed by Babeshko [164], F.D. Gakhov [43], Popov [112].

Here and further, the authors propose a rectangular area as the simplest model object to demonstrate mathematical methods for solving mixed boundary value problems. The formulation of problems in this case can be considered both in displacements and in stress, which depends on the boundary conditions.

As it is known [80], in terms of stress, the equilibrium equations have the following form:

$$\begin{aligned}
&\frac{\partial \sigma_x}{\partial x}+\frac{\partial \tau_{yx}}{\partial y}+\frac{\partial \tau_{zx}}{\partial z}+X=0,\\
&\frac{\partial \tau_{xy}}{\partial x}+\frac{\partial \sigma_y}{\partial y}+\frac{\partial \tau_{zy}}{\partial z}+Y=0,\\
&\frac{\partial \tau_{xz}}{\partial x}+\frac{\partial \tau_{yz}}{\partial y}+\frac{\partial \sigma_z}{\partial z}+Z=0
\end{aligned} \tag{1.1}$$

Here, σ_x, σ_y, σ_z are normal stress along the axes x, y, z correspondingly. τ_{ij} are tangential stress, $i, j \in \{x, y, z\}$, X, Y, Z are volume forces.

In terms of displacements the motion equations have the following form [80]:

$$\begin{aligned}
&\Delta u+\mu_0\frac{\partial \theta}{\partial x}+\frac{X}{G}=\frac{\rho}{G}\frac{\partial^2 u}{\partial t^2},\\
&\Delta v+\mu_0\frac{\partial \theta}{\partial y}+\frac{Y}{G}=\frac{\rho}{G}\frac{\partial^2 v}{\partial t^2},\ ,\\
&\Delta w+\mu_0\frac{\partial \theta}{\partial z}+\frac{Z}{G}=\frac{\rho}{G}\frac{\partial^2 w}{\partial t^2}
\end{aligned} \tag{1.2}$$

where $\Delta=\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$ is the Laplace operator, $\theta=\frac{\partial u}{\partial x}+\frac{\partial y}{\partial y}+\frac{\partial w}{\partial z}$ is a volume expansion, G is a shear modulus, $\mu_0=\frac{1}{1-2\mu}$, μ is the Poisson ratio, ρ is a density.

Depending on the type of boundary conditions, boundary value problems are divided into basic and mixed boundary value problems.

Basic problems are understood as problems where the boundary conditions are uniformly specified along the entire boundary [141]:

- In the case when the function $u(x)|_s = f(x),\ x \in s$ is given on the boundary of the domain, the problem is called the first basic problem of mathematical physics.

- In the case when the normal derivative of the searched function $\left.\frac{\partial u(x)}{\partial \nu}\right|_s = f(x),\ x \in s$ (ν is a normal vector to the surface s) is given on the boundary, the problem is called the second basic problem of mathematical physics.
- In the case when a combination of the function and its normal derivative $\left[\alpha(x)u(x) + \beta(x)\frac{\partial u(x)}{\partial \nu}\right]\Big|_s = f(x),\ x \in s$ is given on the boundary, the problem is called the third basic problem of mathematical physics.

Mixed problems of mathematical physics are problems where there is a change in the type of boundary conditions at the boundary. For example, on one part of the boundary the displacements are specified, and on the other part the derivatives of the displacements are given [141].

$$u|_{s_1} = f_1(x),\ x \in s_1,\ \left.\frac{\partial u}{\partial \nu}\right|_{s_2} = q_2(x),\ x \in s_2,$$

where s is the boundary, $s = s_1 \cup s_2$.

It is worth noting that all problems of mathematical physics are divided into correctly and incorrectly stated problems. A problem is called correctly stated according to Hadamard definition if three conditions are met, namely:

1. a solution of a problem exists;
2. the solution of a problem is the unique one;
3. the solution of a problem is the stable solution.

If at least one of the conditions is violated, then the problem is incorrectly stated.

The initial conditions are formulated to describe the value of functions and their derivatives relatively the time variable at the initial moment of time.

The existence and uniqueness of a boundary value problem solution are ensured by the uniqueness theorem for the solution of problems of mathematical physics [141].

1.2 Methods for Solving of Boundary Value Problems

Here and below, second-order partial differential equations will be considered, which is due to the fact that equations of this type are the most frequently used in modeling of many applied processes. The equations of the following form are considered:

$$
\begin{aligned}
&r_1(x)[p_1(x)u'(x,y)]' - q_1(x)u(x,y) + r_2(y)[p_2(y)u'(x,y)]' - \\
&- q_2(y)u(x,y) = f(x,y), \quad a_0 < x < a_1, \quad b_0 < y < b_1, \\
&V_j[u] \equiv \alpha_{j0}u(a_j,y) + \alpha_{j1}\frac{\partial u}{\partial x}(a_j,y) = 0, j = 0,1; b_0 < y < b_1, \\
&V_k[u] \equiv \beta_{j0}u(x,b_j) + \beta_{j1}\frac{\partial u}{\partial y}(x,b_j) = 0, \quad j = 0,1; \quad a_0 < x < a_1.
\end{aligned} \tag{1.3}
$$

Here, $r_j(x),\ p_j(x),\ q_j(x) \in C^1(a_0;a_1),\ j = 1, 2,\ r_j(y),\ p_j(y),\ q_j(y) \in C^1(b_0;b_1)$, $j = 1, 2$ are given functions, which must be continuous with their first derivatives, $V_j[u]$, $V_k[u]$ are boundary functionals, where $\alpha_{j0} + \alpha_{j1} \neq 0, \beta_{j0} + \beta_{j1} \neq 0$, $u = u(x, y)$ is a searched function that is continuous along with its derivatives.

Here, two main methods for solving problems in partial derivatives are considered: the Fourier method and the method of integral transforms. The idea of both methods is the reducing of the original boundary value problem to a one-dimensional one. Each of the methods significantly uses the Sturm-Liouville problem. Let's consider this problem and its properties.

1.2.1 The Sturm-Liouville Problem and Its Properties

The Sturm-Liouville boundary value problem has the following form [131]:

$$
\begin{cases}
-\left[p(x)y'_\lambda(x)\right]' + q(x)y_\lambda(x) = \dfrac{\lambda y_\lambda(x)}{r(x)}, \quad a_0 < x < a_1 \\
V_i^m[y_\lambda(x)] = 0, \quad i = 0,1
\end{cases} \tag{1.4}
$$

Here, $p(x) \in C^1(a_0;a_1),\ q_j(x),\ r(x) \in C(a_0;a_1)$, $V_i^m[y_\lambda] \equiv \alpha_{i0}y_\lambda(a_i) + \alpha_{i1}y_\lambda{}'(a_i) = 0, i = 0,1; a_0 < x < a_1$.

The form (1.4) is the standard form of the Sturm-Liouville boundary value problem. To solve this problem means to find all non-trivial solutions of the problem $X_k(x) = y_{\lambda_k}(x)$, which are called eigenfunctions, and the corresponding values of the parameter $\lambda = \lambda_k$ that are called eigenvalues of the Sturm-Liouville problem.

Let's introduce the Sturm-Liouville operator

$$
\begin{cases}
Sy_\lambda(x) = \dfrac{\lambda y_\lambda(x)}{r(x)}, \quad a_0 < x < a_1 \\
V_i^m[y_\lambda(x)] = 0, \quad i = 0,1
\end{cases} \tag{1.5}
$$

where $Sy_\lambda(x) = -\left[p(x)y'_\lambda(x)\right]' + q_\lambda(x)y_\lambda(x)$.

The two main cases of the Sturm-Liouville problem (1.5) are usually set off [143]. Conditions of first regular case demand: an interval where a variable change must be finite, and the ends of the interval are not singular points (the points where

the solution turning to infinity). If at least one of the conditions is violated, then this problem is called as irregular Sturm-Liouville problem.

The properties of eigenvalues and eigenfunctions of the Sturm-Liouville problem [141] are presented without proving.

1. The Sturm-Liouville boundary value problem has a countable set of eigenvalues, and all of them are real: $\lambda_n,\ n = 0, 1, 2, \ldots$.
2. Eigenfunctions corresponding to different eigenvalues are orthogonal:

$$\int_0^l X_n(x)X_m(x)dx = \begin{cases} 0, & n \neq m \\ \int_0^l X_n^2(x)dx = N_n^2, & n = m \end{cases}$$

3. If some function $\Phi(x)$ has a continuous derivative and a piecewise continuous second derivative for $x \in [0; l]$ and also satisfies the boundary conditions in (1.5), then it can be expanded into a uniform and absolutely convergent Fourier series—according to the eigenfunctions of the Sturm-Liouville boundary value problem $u_n(x)$:

$$\Phi(x) = \sum_{k=1}^{\infty} \Phi_k u_k(x),$$

$$u_k = \frac{1}{\|u_k\|^2} \int_0^l \xi(x)\Phi(x)u_k(x)dx, \quad \|u_k\|^2 = \int_0^l \xi(x)u_k^2(x)dx,$$

where $\xi(x)$ is the weight with which the eigenfunctions on the interval $[0; l]$ are orthogonal.

This property of the Sturm-Liouville boundary value problem was formulated and proved by V.G. Steklov and is called as Steklov's theorem [131].

The Sturm-Liouville boundary value problem is the self-conjugate boundary value problem. Before to prove this fact let's consider main definitions of a self-conjugated boundary value problem.

Two differential operators of n-th order are input:

$$L_n y(x) = \sum_{j=0}^{n} p_j(x) y^{(n-j)}(x) = f(x).$$

$$L_n^* y(x) = \sum_{j=0}^{n} (-1)^j \left[p_j(x)y(x)\right]^{(n-j)}.$$

If the equality $L_n y(x) = L_n^* y(x)$ has place, the operator is called self-conjugate.

A boundary value problem

$$\begin{cases} L_n y(x) = f(x), a < x < b \\ V_i[y(x)] = 0, i = \overline{0, n-1} \end{cases}$$

is called self-conjugate if two conditions are met: (1) the operator of the problem is self-conjugate; (2) if two functions $u(x)$ and $v(x)$ satisfy the original boundary value problem, then the integral condition is fulfilled

$$\int_a^b [v(x)L_n u(x) - u(x)L_n v(x)]dx = 0.$$

Let's prove that the Sturm-Liouville operator is self-conjugated. With this aim let's consider the auxiliary boundary value problem

$$\begin{cases} Sy(x) = f(x), \quad a < x < b \\ V_i^m[y(x)] = 0, \quad i = 0, 1 \end{cases} \tag{1.6}$$

and prove that this problem is self-conjugated. To do this, one must prove that the differential operator

$$Sy(x) = -[p(x)y'(x)]' - q(x)y(x) = -p(x)y''(x) - p'(x)y'(x) + q(x)y(x)$$

is self-conjugate. The general form of the differential operator of the second order is

$$L_2 y(x) = p_0(x)y''(x) + p_1(x)y'(x) + p_2(x)y(x).$$

The conjugate form can be written as follows:

$$\begin{aligned} &L_2^* y(x) = [p_0(x)y(x)]'' - [p_1(x)y(x)]' + p_2(x)y(x) = \\ &= p_0(x)y'' + y'\left[2p_0'(x) - p_1(x)\right] + y(x)\left[p_0''(x) - p_1'(x) + p_2(x)\right]. \end{aligned}$$

After comparing of the differential operator $L_2 y(x)$ with the Sturm-Liouville operator $Sy(x)$ can be seen that $p_0(x) = -p(x)$, $p_1(x) = -p'(x)$, $p_2(x) = q(x)$.

Now let's compare the coefficients of the Sturm-Liouville operator with the coefficients of the conjugate operator. The coefficients of the higher derivatives coincide:

$$-p'(x) = 2(-p'(x)) + p'(x) = -p'(x),$$

$$q(x) = (-p(x))'' + (p'(x))' + q(x) = q(x).$$

The conditions when the Sturm-Liouville operator is self-conjugate are: (1) $p_1(x) = p_0'(x)$; (2) $p_0''(x) - p_1'(x) = 0$.

Now it remains to check the integral condition. Let there exist two functions $u(x)$ and $v(x)$ satisfying the following two semihomogeneous boundary value problems

$$\begin{cases} S\begin{bmatrix} u(x) \\ v(x) \end{bmatrix} = f(x), \quad x \in (a;b) \\ V_i^m \begin{bmatrix} u(x) \\ v(x) \end{bmatrix} = 0, \quad i = 0,1 \end{cases}$$

Let us prove that the equality $\int_a^b [v(x)S(u(x)) - u(x)Sv(x)]dx = 0$ holds. Indeed, instead of the abbreviated notation of the Sturm-Liouville operator, let us substitute its full form:

$$\int_a^b \left[v(x)[(-pu') + q(x)u(x)] - u(x)\left[(-pv')' + q(x)v(x)\right]\right]dx =$$

$$= \int_a^b \left[-v(x)(pu')' + u(x)(pv')'\right]dx.$$

After integration by parts of the obtained integral one derives

$$-v(x)\beta u'(x)\Big|_a^b + u(x)pv'(x)_a^b\Big| + \int_a^b v'(x)pu'(x)dx - \int_a^b u'(x)pv'(x)dx \qquad (1.7)$$

In order to calculate substitutions in expression (1.7), boundary functionals are written:

$$V_i^m \begin{bmatrix} u(x) \\ v(x) \end{bmatrix} \equiv \begin{bmatrix} \alpha_{i0}u(a_i) - \alpha_{i1}u'(a_i) \\ \alpha_{i0}v(a_i) - \alpha_{i1}v'(a_i) \end{bmatrix} \equiv 0, \quad i = 0,1; \quad \alpha_{i1} \neq 0; \quad a_0 = a, \quad a_1 = b.$$

where $\begin{bmatrix} u'(a_i) \\ v'(a_i) \end{bmatrix} = -\frac{\alpha_{i0}}{\alpha_{i1}} \begin{bmatrix} u(a_i) \\ v(a_i) \end{bmatrix}$. The derived relations are substituted into the expression (1.7):

$$
\begin{aligned}
&-v(b)p(b)u'(b)+v(a_0)p(a_0)u'(a_0)+\\
&+u(b)p(b)v'(b)-u(a)p(a)v'(a)=-v(b)p(b)\left(-\frac{\alpha_{10}}{\alpha_{11}}u(b)\right)+\\
&+v(a)p(a)\left(-\frac{\alpha_{00}}{\alpha_{01}}u(a)\right)+u(b)p(b)\left(-\frac{\alpha_{10}}{\alpha_{11}}v(b)\right)-u(a)p(a)\left(-\frac{\alpha_{00}}{\alpha_{01}}v(a)\right)=0
\end{aligned}
$$

As it is seen, the integral condition is fulfilled.

This proves that the Sturm-Liouville boundary value problem is self-conjugate.

The solving of the Sturm-Liouville problem is the most important component of the following methods proposed for solving of partial derivative equations.

1.2.2 The Fourier Method

Consider the general scheme of the Fourier method using the example of the initial-boundary value problem of thermal conductivity.

The following thermal conductivity problem is formulated in the form:

$$
\begin{aligned}
&\frac{\partial u(x,t)}{\partial t}=a^2\frac{\partial^2 u(x,t)}{\partial x^2},\quad x\in(0;l),\quad t>0\\
&u(0,t)=0,\quad u(l,t)=0,\quad t>0\\
&u(x,0)=f(x),\quad x\in(0;l)
\end{aligned}
\tag{1.8}
$$

It is necessary to construct the solution of the initial-boundary value problem (1.8). It is supposed that the unknown function $u(x,t)$ allows the representation in the form:

$$u(x,t)=X(x)T(t) \tag{1.9}$$

Substitution of the representation (1.9) in the equation from (1.8) leads to the correspondence:

$$X(x)T'(t)-a^2X''(x)T(t)=0$$

After division of the both parts by $X(x)T(t)$ the equation takes the form:

$$\frac{T'(t)}{T(t)}-a^2\frac{X''(x)}{X(x)}=0 \tag{1.10}$$

As it is seen, $\frac{X'(x)}{X(x)}$ has to coincide with a constant, for example, $\frac{X''(x)}{X(x)}=-\lambda^2$. After making such a substitution in (1.10), two ordinary differential equations are derived:

$$\begin{aligned} &T'(t) - a^2\lambda^2 T(t) = 0, \quad t > 0 \\ &X''(x) + \lambda^2 X(x) = 0, \quad x \in (0; l) \end{aligned} \tag{1.11}$$

After taking into account the representation (1.9), the boundary conditions from (1.8) are written in the form

$$\begin{aligned} &u(0,t) = X(0)T(t) = 0 \\ &u(l,t) = X(l)T(t) = 0 \end{aligned} \tag{1.12}$$

As it is seen, the second equation in (1.11) and boundary conditions (1.12) lead to the Sturm-Liouville problem (1.4).

Taking into account the properties of the eigenvalues and eigenfunctions of the Sturm-Liouville problem (1.4), the solution can be taken as

$$u(x,t) = \sum_{n=0}^{\infty} A_n e^{-\lambda_n^2 a^2 t} X_n(x) \tag{1.13}$$

Each term of the series (1.13) satisfies the differential equation and the boundary conditions in (1.8). The initial conditions remained unfulfilled. The transition to $t = 0$ is done

$$u(x,0) = \sum_{n=0}^{\infty} A_n X_n(x) = f(x), \quad x \in (0; l) \tag{1.14}$$

Using the orthogonality property of eigenvalues and eigenfunctions of the Sturm-Liouville problem, both parts of equality (1.14) are multiplied by the eigenfunction $X_m(x)$ and are integrated over the interval [0;1]

$$\begin{aligned} &\sum_{n=0}^{\infty} A_n \int_0^l X_n(x) X_m(x) dx = \int_0^l f(x) X_m(x) dx \\ &A_m N_m^2 = f_m \end{aligned} \tag{1.15}$$

where $A_m = \frac{f_m}{N_m^2}$.

Finally, the solution of stated problem (1.8) is derived:

$$u(x,t) = \sum_{n=0}^{\infty} \frac{f_n}{N_n^2} e^{-\lambda_n^2 a^2 t} X_n(x) \tag{1.16}$$

where N_n^2 is the square of the eigenfunctions' norm:

$$N_m^2 = \int_0^l X_m^2(x)dx = \int_0^l \left(\sin^2 \lambda_n x\right)dx\Big|_0^l = \int_0^l \frac{\cos 2\lambda_n x - 1}{2}dx =$$
$$= \frac{\sin 2\lambda_n x}{2 \cdot 2\lambda_n}\Big|_0^l + \frac{1}{2}l = \frac{1}{2}l$$

The solution of the original initial-boundary value problem (1.8) is finally derived.

This is typical example of Fourier method application. The second method of partial differential equations' solving is the method of integral transforms, which can be applied either for problems with a continuous domain (the classical integral transform method) or for problems with discontinuities inside the domain (the generalized integral transform method).

1.2.3 The Method of Integral Transforms

The boundary value problem is considered in the form

$$\begin{aligned}
&r_1(x)[p_1(x)u'(x,y)]' - q_1(x)u(x,y) + r_2(y)[p_2(y)u'(x,y)]' - \\
&- q_2(y)u(x,y) = f(x,y), \\
&u = u(x,y), \quad a_0 < x < a_1, \quad b_0 < y < b_1, \\
&V_j[u] \equiv \alpha_{j0}u(a_j,y) + \alpha_{j1}\frac{\partial u}{\partial x}(a_j,y) = 0, j = 0,1; b_0 < y < b_1, \\
&V_k[u] \equiv \beta_{j0}u(x,b_j) + \beta_{j1}\frac{\partial u}{\partial y}(x,b_j) = 0, \quad j = 0,1; \quad a_0 < x < a_1.
\end{aligned} \tag{1.17}$$

Here, the case of integral transform method application is proposed for the continuous statement of the problem (1.17).

1.2.4 The Classical Integral Transform Method

The main idea of the method is as follows: one needs to find an integral transform that allows to represent the original unknown function in the form $u_\alpha(y) = \int_{a_0}^{a_1} \times \frac{K_1(x,\alpha)}{r_1(x)}u(x,y)dx$—the transform of function. The multiplication of both parts of the equation from (1.17) by a factor $\frac{K_1(x,\alpha)}{r_1(x)}$ and consequently integration of all terms of the equation over the interval $x \in (a_0, a_1)$ should be done for it.

Let's consider the first summand after integration by parts relatively the variable x:

$$\int_{a_0}^{a_1} K_1(x,\alpha)[p_1(x)u'(x,y)]'dx = N_\alpha(y) + \int_{a_0}^{a_1} \frac{r_1(x)u(x,y)}{r_1(x)}\left(p_1(x)K_1'(x,\alpha)\right)'d\alpha$$

where $N_\alpha(y) = [K_1(x,\alpha)p_1(x)u'(x,y)]\big|_{a_0}^{a_1} - \left[p_1(x)u(x,y)K_1'(x,\alpha)\right]\big|_{a_0}^{a_1}$.

Let's consider the second summand:

$$-\int_{a_0}^{a_1} \frac{K_1(x,\alpha)}{r_1(x)} q_1(x)u(x,y)dx = -\int_{a_0}^{a_1} \frac{u(x,y)}{r_1(x)}[q_1(x)K_1(x,\alpha)]dx.$$

The integration of the third term led to equality:

$$\int_{a_0}^{a_1} r_2(y)\frac{K_1(x,\alpha)}{r_1(x)}[p_2(y)u'(x,y)]'dx = r_2(y)\left(p_2(y)u_\alpha'(y)\right)'.$$

Thus, after integration, the equation in (1.17) will take the form:

$$\begin{aligned} &N_\alpha(y) + \int_{a_0}^{a_1} \frac{u(x,y)}{r_1(x)}\left[r_1(x)\left(p_1(x)K_1'(x,\alpha)\right)' - q_1(x)K_1(x,\alpha)\right]dx + \\ &+r_2(y)\left(p_2(y)u_\alpha'(y)\right)' - q_2(y)u_\alpha(y) = f_\alpha(y) \end{aligned} \tag{1.18}$$

Let's demand that the so far unknown function $K_1(x,\alpha)$ satisfies the differential equation

$$r_1(x)\left[p_1(x)K_1'(x,\alpha)\right]' - q_1(x)K_1(x,\alpha) = -\alpha^2 K_1(x,\alpha), \quad a_0 < x < a_1 \tag{1.19}$$

If such a solution is found, equation (1.18) will be formulated with respect to one unknown function of the transform $K_\alpha(y)$. The following Sturm-Liouville problem for the function $K(x,\alpha)$ follows:

$$\begin{cases} r_1(x)\left[K_1'(x,\alpha)p_1(x)\right]' - q_1(x)K_1(x,\alpha) = -\alpha^2 K_1(x,\alpha), \quad a_0 < x < \alpha_1 \\ V_i[K_1(x,\alpha)] \equiv \alpha_{i0}K_1(a_i,\alpha) + \alpha_{i1}K_1'(a_i,\alpha) = 0, \quad i = 0,1 \end{cases} \tag{1.20}$$

The boundary conditions in (1.20) after simplification can be derived in the form

$$N_\alpha(y) = -\frac{p_1(a_1)u(a_1,y)}{\alpha_{11}}\left[\alpha_{10}K_1(a_1,\alpha)+\alpha_{11}K_1'(a_1,\alpha)\right]+ \\ +\frac{p_1(a_0)u(a_0,y)}{\alpha_{01}}\left[\alpha_{00}K_1(a_0,\alpha)+\alpha_{01}K_1'(a_0,\alpha)\right] \tag{1.21}$$

Let's suppose that Sturm-Liouville problem (1.20) is solved. Then the equation in (1.18) takes the form:

$$r_2(y)\left[p_2(y)u_\alpha'(y)\right]' - q_2(y)u_\alpha(y) - \alpha^2 u_\alpha(y) = f_\alpha(y), \quad y \in [b_0, b_1] \tag{1.22}$$

Boundary conditions are transformed into expressions: $\beta_{i0}u_\alpha(b_i) + \beta_{i1}u_\alpha'(b_i) = 0, \ \ i - 0, 1,$ or in a short form

$$V_i[u_\alpha(y)] = 0, \quad i = 0, 1 \tag{1.23}$$

Thus, relations (1.22) and (1.23) define a one-dimensional boundary value problem in transform domain.

1.2.5 The Generalized Integral Transform Method

In the previous subchapter, it was assumed that the region where the boundary value problem is given does not contain defects and the searched function is continuous.

In the case when in the region $a_0 < x < a_1$ there is a defect, which is understood as the line $x = c$, $a_0 < c < a_1$, then the previous approach is not applicable. It is assumed that when passing through defect the unknown function and its first derivative have discontinuities of the first kind with the following jumps:

$$\langle u(c,y)\rangle = u(c-0,y) - u(c+0,y) = X_0(y); b_0 < y < b_1$$
$$\left\langle \frac{\partial u}{\partial x}(c,y)\right\rangle = \frac{\partial u}{\partial x}(c-0,y) - \frac{\partial u}{\partial x}(c+0,y) = X_1(y); b_0 < y < b_1$$

In such case the generalization of classical integral transform method is proposed. The general method of integral transformations implies the continuity of the function $u(x, y)$, while the generalized method of integral transforms proposes to carry out transforms for a function that has discontinuities of the first kind with given jumps.

Now the boundary value problem is posed in a discontinuous statement:

$$r_1(x)[p_1(x)u'(x,y)]' - q_1(x)u(x,y) + r_2(y)[p_2(y)u^{\cdot}(x,y)]^{\cdot} - \\ - q_2(y)u(x,y) = f(x,y)$$

$$u = u(x,y), \quad a_0 < x < a_1, \quad b_0 < y < b_1,$$
$$V_j[u] \equiv \alpha_{j0} u(a_j, y) + \alpha_{j1} \frac{\partial u}{\partial x}(a_j, y) = 0, j = 0, 1; b_0 < y < b_1,$$
$$V_k[u] \equiv \beta_{j0} u(x, b_j) + \beta_{j1} \frac{\partial u}{\partial y}(x, b_j) = 0, \quad j = 0, 1; \quad a_0 < x < a_1.$$

$$\langle u(c,y) \rangle = u(c-0,y) - u(c+0,y) = X_0(y); b_0 < y < b_1$$
$$\left\langle \frac{\partial u}{\partial x}(c,y) \right\rangle = \frac{\partial u}{\partial x}(c-0,y) - \frac{\partial u}{\partial x}(c+0,y) = X_1(y); b_0 < y < b_1$$

Let's multiply both parts of the equation from (1.17) by a factor $\frac{K_1(x,\alpha)}{r_1(x)}$ and consequently integrate all terms of the equation, dividing this interval into the sum of two subintervals $x \in (a_0, c-0)$, $x \in (c+0, a_1)$.

Consider the first summand, after the consecutive double integration by parts it has the form

$$\int_{a_0}^{a_1} K_1(x,\alpha)[p_1(x)u'(x,y)]'dx =$$
$$= \int_{a_0}^{c-0} K_1(x,\alpha)[p_1(x)u'(x,y)]'dx + \int_{c+0}^{a_1} K_1(x,\alpha)[p_1(x)u'(x,y)]'dx =$$
$$= M_\alpha(y) + N_\alpha(y) + \int_{a_0}^{a_1} \frac{r_1(x)u(x,y)}{r_1(x)} \left(p_1(x)K_1'(x,\alpha)\right)' d\alpha$$

where $N_\alpha(y) = [K_1(x,\alpha)p_1(x)u'(x,y)]\big|_{a_0}^{a_1} - \left[p_1(x)u(x,y)K_1'(x,\alpha)\right]\big|_{a_0}^{a_1}$, $M_\alpha(y) = K_1(c,\alpha)p_1(c)X_1(y) - K_1'(c,\alpha)p_1(c)X_0(y)$.

The second, third, and fourth summands are integrated analogically to the case A. Thus, after integration, the equation in (1.17) has the form

$$M_\alpha(y) + N_\alpha(y) + \int_{a_0}^{a_1} \frac{u(x,y)}{r_1(x)} \left[r_1(x)\left(p_1(x)K_1'(x,\alpha)\right)' - q_1(x)K_1(x,\alpha)\right]dx + \tag{1.24}$$
$$+ r_2(y)\left(p_2(y)u_\alpha'(y)\right)' - q_2(y)u_\alpha(y) = f_\alpha(y)$$

If the Sturm-Liouville problem (1.20) is solved, the equation in (1.24) will take the form:

$$r_2(y)\left[p_2(y)u_\alpha'(y)\right]' - q_2(y)u_\alpha(y) - \alpha^2 u_\alpha(y) = f_\alpha(y) - M_\alpha(y), \quad y \in [b_0, b_1] \tag{1.25}$$

In this situation, the right side of equation (1.25) contains unknown jumps of the searched function and its first derivative, which are either given according to the formulation of the boundary value problem or need to be further defined.

Boundary conditions are transformed into expressions (1.23).

1.3 The Green's Function: The Definition and Properties

The material of the next two subchapters essentially relies on the results obtained by Popov G.Ya. in [110].

1.3.1 The Main Properties of the Green's Function

Let's define three main types of boundary value problems:

$$\begin{cases} L_n(y(x)) = f(x), & a < x < b, \\ V_i[y(x)] = \gamma_i, & i = \overline{0, n-1} \end{cases} \tag{1.26}$$

The first type is a heterogeneous problem (1a) if $f(x) \neq 0, \gamma_i \neq 0, 0 \leq i \leq n-1$; the second type is a semihomogeneous problem (1b) if $f(x) \neq 0, \gamma_i = 0, i = \overline{0, n-1}$; the third type is a homogeneous problem (1c) if $f(x) \equiv 0, \gamma_i = 0, i = \overline{0, n-1}$.

The Green's function is a function of two variables $G(x, \xi)$ defined in a rectangular domain $a \leq x, \xi \leq b$ and through which the solution of the boundary value problem (1b) for any function $f(x)$ is determined by the formula:

$$y(x) = \int_a^b G(x, \xi) f(\xi) d\xi \tag{1.27}$$

This definition of the Green's function allows it to be interpreted as a solution to the following boundary value problem:

$$\begin{cases} L_n[G(x, \xi)] = \delta(x - \xi), a < x < b \\ V_i[G] = 0, \quad i = \overline{0, n-1.} \end{cases} \tag{1.28}$$

Here, $L_n(y(x))$ is the differential operator of the n-th order $L_n(y(x)) = \sum_{j=0}^{n} p_i(x) y^{(n-j)}(x)$, $V_i[y(x)] = V_i^a[y(x)] + V_i^b[y(x)] = \sum_{j=0}^{n-1} \alpha_{ij} y^{(j)}(a) + \sum_{j=0}^{n-1} \beta_{ij} y^{(j)}(b)$, $\delta(x)$ is the Dirac function having the property

$$\int_a^b f(x)\delta(x-\xi)dx = \begin{cases} 0, \xi \notin [a;b], \\ f(\xi), \xi \in [a;b] \end{cases} \tag{1.29}$$

Here, coefficients $P_j(x)$, α_{ij}, β_{ij} are real, in particular $p_0(x) \neq 0$, $p_i(x), f(x) \in C(a;b)$.

Here and further, the boundary conditions of a problem are supposedly nonsingular. It means that Wronskian of the fundamental system of solutions (in the following FSS) is not equal to zero

$$W\left[y_j(x)\right] = \begin{vmatrix} y_0(x) & y_1(x) \ldots\ldots y_{n-1}(x) \\ y_0'(x) & y_1'(x) \ldots\ldots y_{n-1}'(x) \\ \vdots & \vdots \\ y_0^{(n-1)}(x) & y_1^{(n-1)}(x) \ldots\ldots y_{n-1}^{(n-1)}(x) \end{vmatrix} \neq 0, \tag{1.30}$$

where $y_j(x), j = \overline{0, n-1}$ is FSS.

In the case of a nonhomogeneous (1a) one-dimensional boundary value problem, in order to construct a solution, it is necessary, in addition to the FSS, to define a basic system of solutions (BSS) $\{\psi_k(x)\}$, $k = \overline{0, n-1}$, each term of which must satisfy the boundary value problem:

$$\begin{cases} L_n(\psi_k(x)) = 0, \\ V_i[\psi_k(x)] = \delta_{ik}, \quad k = \overline{0, n-1}, \end{cases} \tag{1.31}$$

where δ_{ik} is the Kronecker symbol.

Having obtained the FSS, Green's function, and the BSS, the solution of the inhomogeneous problem (1a) is presented in the form:

$$y(x) = \int_a^b G(x,\xi) f(\xi) d\xi + \sum_{k=0}^{n-1} \gamma_k \psi_k(x) \tag{1.32}$$

Let us prove this.

Theorem 1 *The expression (1.32) is a solution of a nonhomogeneous one-dimensional boundary value problem (1a).*

Proof The proof will consist of two parts. First, it will be demonstrated that the expression (1.32) is a solution of the corresponding differential equation:

$$L_n(y(x)) = L_n\left(\int_a^b G(x,\xi)f(\xi)d\xi + \sum_{k=0}^{n-1}\gamma_k\psi_k(x)\right) = \\ = L_n\left(\int_a^b G(x,\xi)f(x)\right)d\xi + \sum_{k=0}^{n-1}\gamma_k L_n(\psi_k(x)). \tag{1.33}$$

The last summand in formula (1.33) is equal to zero due to the BSS properties, and the first due to formula (1.27) is equal to $f(x)$.

Thus, it is proved that the function (1.32) satisfies the differential equation. Let us now check that it satisfies the boundary conditions. For this purpose, the boundary functional is applied to it

$$V_i[y(x)] = V_i\left[\int_a^b G(x,\xi)f(\xi)d\xi\right] + V_i\left[\sum_{k=0}^{n-1}\gamma_k\psi_k(x)\right].$$

Due to the fact that the Green's function satisfies the semi-uniform boundary value problem (1b), the first term is equal to zero. The second term can be rewritten in the form

$$V_i\left[\sum_{k=0}^{n-1}\gamma_k\psi_k(x)\right] = \sum_{k=0}^{n-1}\gamma_k[V_i(\psi_k(x))] = \sum_{k=0}^{n-1}\psi_k\delta_{ik} = \gamma_i.$$

So, it is derived that $V_i[y(x)] = \gamma_i$.

The theorem is proved.

Theorem 2 *If the boundary conditions of a homogeneous one-dimensional boundary value problem (1c) are nonsingular, then the homogeneous boundary value problem has only a trivial solution.*

Proof The general solution of the differential equation can be represented as a superposition of its FSS: $y(x) = \sum_{j=0}^{n-1} C_j y_j(x)$. The boundary conditions are satisfied:

$$V_i[y(x)] = V_i\left[\sum_{j=0}^{n-1} C_j y_j(x)\right] = \sum_{j=0}^{n-1} C_j V_i\left[y_j(x)\right] = 0.$$

Thus, a system of linear algebraic equations with respect to unknown coefficients $C_j, j = \overline{0, n-1}$ was derived. The determinant of this system is different from zero due to the property of the FSS (1.30). That is, all coefficients $C_j = 0, j = \overline{0, n-1}$. So, $y(x) \equiv 0$.

The theorem is proved.

As it is seen, the existence of the Green's function is fundamentally important in the construction of the boundary value problem's solution. Let's consider main properties of the Green's function.

1.3.2 Theorem of the Unity of Existence of the Green's Function

Theorem 3 *If a one-dimensional boundary value problem has nonsingular boundary conditions, then the Green's function of such a problem is unique.*

Proof It will be conducted from the opposite. Suppose that there exist two different Green's functions $G_1(x,\xi)$, $G_2(x,\xi)$ such that $y_1(x) = \int\limits_a^b G_1(x,\xi)f(\xi)d\xi$, $y_2(x) = \int\limits_a^b G_2(x,\xi)f(\xi)d\xi$ are solutions of the boundary value problem (1.26). Let's construct a function $G(x,\xi) = G_1(x,\xi) - G_2(x,\xi)$. Due to the fact that $y_1(x)$, $y_2(x)$ are solutions of the equations, it can be written $L_n(y_1(x)) = f(x)$, $L_n(y_2(x)) = f(x)$. Let's construct the function $y(x) = y_1(x) - y_2(x)$ and apply the differential operator to it

$$L_n(y(x)) = L_n(y_1(x)) - L_n(y_2(x)) = f(x) - f(x) = 0$$

On the other hand,

$$L_n(y(x)) = \int\limits_a^b G_1(x,\xi)f(\xi)d\xi - \int\limits_a^b G_2(x,\xi)f(\xi)d\xi = \int\limits_a^b G(x,\xi)f(\xi)d\xi.$$

It is obtained that $\int\limits_a^b G(x,\xi)f(\xi)d\xi = 0$ for any right-hand side of the equation $f(x)$. Due to the fact that the function is arbitrary, it was chosen it in the form $f(x) = \begin{cases} 0, & x \in (a;\xi-\Delta\xi)\cup(\xi+\Delta\xi;b) \\ \neq 0, & x \in (\xi-\Delta\xi,\xi+\Delta\xi) \end{cases}$. The solution of the problem in this case, applying the average theorem, will be written as a correspondence

$$y(x) = \int\limits_{\Delta\xi} G(x,\xi)f(\xi)d\xi = G(x,x_{mid}) \int\limits_{\Delta\xi} f(\xi)d\xi$$

Recall that the function $y(x)$ is a solution of a homogeneous boundary value problem (1c), $y(x)$ must equal zero. Since the integral on the right-hand side is nonzero, it follows that $G(x,x_{mid}) = 0$. The function $f(x)$ is chosen arbitrarily, that is,

it can be concluded that $G(x,\xi) \equiv 0$. From which it follows that $G_1(x,\xi) = G_2(x,\xi)$. The uniqueness of the Green's function has been proved.

In order to find out the conditions for the Green's function existence, the theorem of four defining properties of the Green's function is considered.

1.3.3 Theorem on the Four Defining Properties of the Green's Function

Theorem 4 *If the function $G(x,\xi)$ given in the rectangle $a < x < b$, $a < \xi < b$ has the following four properties, then this function is the Green's function of a semihomogeneous boundary value problem (1b):*

1. *Green's function and its derivatives up to n−2 order $\frac{\partial^m G(x,y)}{\partial x^m} = G^{(m)}(x), m = \overline{0, n-2}$ are continuous throughout the region;*
2. *The derivative of the Green's function of the (n−1)-th order is discontinuous on the line $x = \xi$, and its jump is equal to the inverse value of the coefficient at the higher derivative $G^{(n-1)}(x, x-0) - G^{(n-1)}(x, x+0) = \frac{1}{p_0(x)}$;*
3. *Any derivative with respect to the variable x of the Green's function is continuous in the domains $a < x < \xi$ and $\xi < x < b$. In each of these triangles, the equation will be fulfilled $L_n G(x,\xi) = 0$.*
4. *The Green's function must satisfy the homogeneous boundary conditions in the domains $a < x < \xi$ and $\xi < x < b$.*

$$V_i[G(x,\xi)] = 0, \quad i = \overline{0, n-1}, \quad a < x < \xi, \quad \xi < x < b$$

Proof Let's input the notation for areas $a < x < \xi$ and $\xi < x < b$—Δ_1 and Δ_2 correspondingly. Let us represent the solution of the semi-inhomogeneous problem (1.26) (1b) in the form:

$$y(x) = \int_a^x G(x,\xi) f(\xi) d\xi + \int_x^b G(x,\xi) f(\xi) d\xi \tag{1.34}$$

In each of the regions Δ_1 and Δ_2, both the Green's function and its derivative are continuous, therefore, when differentiating terms in the right-hand side of the equality (1.34), the derivative can be entered under the integral sign, taking into account the well-known formulas: $\frac{\partial}{\partial x} \int_a^x F(x,\xi) d\xi = \int_a^x \frac{\partial}{\partial x} F(x,\xi) d\xi + F(x,x)$ and

$\frac{\partial}{\partial x}\int\limits_x^b F(x,\xi)d\xi = \int\limits_x^b \frac{\partial}{\partial x}F(x,\xi)d\xi - F(x,x)$ [131]. Keeping this in mind, let's calculate the first derivative:

$$y'(x) = \int\limits_a^x G^{(l)}(x,\xi)f(\xi)d\xi + G(x,x-0)f(x) +$$
$$+ \int\limits_x^b G^{(l)}(x,\xi)f(\xi)d\xi - G(x,x+0)f(x)$$

Let's simplify the resulting expression:

$$y'(x) = \int\limits_a^x G^{(l)}(x,\xi)f(\xi)d\xi + \int\limits_x^b G^{(l)}(x,\xi)f(\xi)d\xi + f(x)$$
$$\times (G(x,x-0) - G(x,x+0)). \tag{1.35}$$

From the condition of continuity of the Green's function, the last term in equality (1.35) is equal to zero. Using a similar scheme, the derivatives of Green's functions up to $n-1$ order are calculated

$$y^{(k)}(x) = \int\limits_a^x G^{(k)}(x,\xi)f(\xi)d\xi + \int\limits_x^b G^{(k)}(x,\xi)f(\xi)d\xi +$$
$$+f(x)\left[G^{(k-1)}(x,x-0) - G^{(k-1)}(x,x+0)\right], \quad k=\overline{0,n-1}.$$

The last terms in all expressions were equal to zero due to the continuity of the Green's function and its derivatives according to the conditions of the theorem. But, in the case of the n-th order derivative, the last term is not equal to zero:

$$y^{(n)}(x) = \int\limits_a^x G^{(n)}(x,\xi)f(\xi)d\xi + \int\limits_x^b G^{(n)}(x,\xi)f(\xi)d\xi +$$
$$+f(x)\left[G^{(n-1)}(x,x-0) - G^{(n-1)}(x,x+0)\right] =$$
$$= \int\limits_a^x G^{(n)}(x,\xi)f(\xi)d\xi + \int\limits_x^b G^{(n)}(x,\xi)f(\xi)d\xi + f(x)\frac{1}{p_0(x)}, \quad k=\overline{0,n-1}.$$

The last equality is obtained with the help of condition 2 of the theorem.

Let us now substitute all the calculated derivatives into the differential equation:

$$L_n(y(x)) = p_0(x)y^{(n)}(x) + p_1(x)y^{(n-1)}(x) + \ldots + p_n(x)y(x) =$$
$$= \int_a^x L_n(G(x,\xi))f(\xi)d\xi + \int_x^b L_n(G(x,\xi))f(\xi)d\xi + f(x).$$

Due to condition 3 of the theorem, the first two terms are equal to zero. Therefore, it is obtained $L_n(y(x)) = f(x)$, i.e. $y(x)$ is the solution of the differential equation of the problem.

Let us show that the function $y(x)$ satisfies the boundary conditions. For this purpose, the boundary functional is written in the form:

$$V_i[y(x)] = V_i^a[y(x)] + V_i^b[y(x)] = \sum_{j=0}^{n-1} \alpha_{ij} y^{(j)}(a) + \sum_{j=0}^{n-1} \beta_{ij} y^{(j)}(b) \tag{1.36}$$

The calculated derivatives are substituted in the boundary conditions

$$y^{(j)}(a) = \int_a^b G^{(j)}(a,\xi)f(\xi)d\xi = \int_a^b V_i^a[G(x,\xi)]f(\xi)d\xi,$$
$$\mathrm{K}(x,\varepsilon) : |x-y| \le \varepsilon$$

and these correspondences are substituted into conditions (1.36):

$$V_i[y(x)] = \int_a^b V_i[G(x,\xi)]f(\xi)d\xi = 0, \quad i = \overline{0, n-1} \tag{1.37}$$

Equality (1.37) is obtained due to the condition 4 of the theorem.

It is proved that the function satisfying the conditions 1–4 is the Green's function.

Corollary If out of the four conditions of the theorem, only the first three conditions are fulfilled, then the function $G(x,\xi)$ will be a fundamental function:

$$y(x) = \int_a^b \Phi(x,\xi)f(\xi)d\xi.$$

1.3.4 Methods of Green's Function's Construction

Theorem 5 *If the boundary conditions of the boundary problem (1b) are nonsingular, then the Green's function for such a problem always exists.*

Proof The theorem will be proved by a constructive way. Assume that the fundamental system of solutions (FSS) $y_j(x), j=\overline{0,n-1}$ of the differential equation is known. Let's construct a function of two variables in such a way that as many as possible properties of the Green's function would be fulfilled:

$$G(x,\xi)=\begin{cases}\sum\limits_{j=0}^{n-1} a_j(\xi)y_j(x), & x<\xi\\ \sum\limits_{j=0}^{n-1} b_j(\xi)y_j(x), & x>\xi\end{cases} \tag{1.38}$$

Here, $a_j(\xi), b_j(\xi), j=\overline{0,n-1}$ are continuous functions in the domains Δ_1 and Δ_2 correspondingly.

Note that the condition 3 of the theorem for the domains Δ_1 and Δ_2 is fulfilled due to the fact that the fundamental solutions have continuous derivatives up to the n-th order.

Let's fulfill the remaining properties of theorem about the four main properties of Green's function. For this purpose, let's calculate the m-th derivative of the function:

$$G^{(m)}(x,\xi)=\begin{cases}\sum\limits_{j=0}^{n-1} a_j(\xi)y_j^{(m)}(x), & x<\xi\\ \sum\limits_{j=0}^{n-1} b_j(\xi)y_j^{(m)}(x), & x>\xi\end{cases} \qquad m=\overline{0,n-1}$$

In order to guarantee the continuity of all derivatives up to the n-th order to hold, one must require

$$G^{(m)}(x,x-0)-G^{(m)}(x,x+0)=\sum_{j=0}^{n-1} b_j(x)y_j^{(m)}(x)-\sum_{j=0}^{n-1} a_j(x)y_j^{(m)}(x)= \\ =\sum_{j=0}^{n-1}\big(b_j(x)-a_j(x)\big)y_j^{(m)}(x) \tag{1.39}$$

when $m=\overline{0,n-2}$.

$$G^{(m)}(x,x-0)-G^{(m)}(r,x+0)=\sum_{j=0}^{n-1}\left(b_j(x)-a_j(x)\right)y_j^{(m)}(x)=\frac{1}{p_0(x)} \tag{1.40}$$

when $m=\overline{n-1}$.

Equalities (1.39), (1.40) form the nonhomogeneous system of n equations relatively to n unknown coefficients

$$c_j(x)=b_j(x)-a_j(x),\ \ j=\overline{0,n-1}. \tag{1.41}$$

The determinant of the obtained system is set aside from zero $\det\left\|y_j^{(m)}\right\|_{j,m=0}^{n-1}\neq 0$ due to the fact that $\{y_j(x)\}$ is the FSS. Therefore, the system has a unique solution, that is, the coefficients $c_j(x)$ can be found in a unique way. The first and second conditions of the theorem are satisfied.

In order to fulfill condition 4 the derivatives of the function $G(x,\xi)$ relatively the first variable must be calculated at the ends of interval $x\in(a;b)$:

$$G^{(m)}(a,\xi)=\sum_{j=0}^{n-1}a_j(\xi)y_j^{(m)}(a),\ \ G^{(m)}(b,\xi)=\sum_{j=0}^{n-1}b_j(\xi)y_j^{(m)}(b).$$

Let's form now boundary functionals $V_i^a[G(x,\xi)]$ and $V_i^b[G(x,\xi)]$:

$$V_i^a[G(x,\xi)]=\sum_{m=0}^{n-1}\alpha_{im}G^{(m)}(a,\xi),$$
$$V_i{}^b[G(x,\xi)]=\sum_{m=0}^{n-1}\beta_{im}G^{(m)}(b,\xi).$$

With the help of the correspondences (1.36) the boundary functional will be rewritten in the form:

$$\begin{aligned}V_i[G(x,\xi)]&=V_i^a[G(x,\xi)]+V_i^b[G(x,\xi)]=\sum_{m=0}^{n-1}\alpha_{im}\sum_{j=0}^{n-1}a_j(\xi)y_j^{(m)}(a)+\\&+\sum_{m=0}^{n-1}\beta_{im}\sum_{j=0}^{n-1}b_j(\xi)y_j^{(m)}(b)=\sum_{j=0}^{n-1}a_j(\xi)V_i^a\left[y_j(x)\right]+\sum_{j=0}^{n-1}b_j(\xi)V_i^b\left[y_j(x)\right]\end{aligned} \tag{1.42}$$

To fulfill the fourth condition, it is required:

$$V_i[G(x,\xi)]=0.$$

Let's add and subtract the expression $\sum_{j=0}^{n-1} a_j(\xi) V_i^b\left[y_j(x)\right]$ at the right-hand side of the correspondence (1.42):

$$V_i[G(x,\xi)] = \sum_{j=0}^{n-1} a_j(\xi) V_i^a\left[y_j(x)\right] + \sum_{j=0}^{n-1} c_j(\xi) V_i^b\left[y_j(x)\right] - \\ - \sum_{j=0}^{n-1} a_j(\xi) V_i^b\left[y_j(x)\right] = \sum_{j=0}^{n-1} a_j(\xi) V_i\left[y_j(x)\right] + \sum_{j=0}^{n-1} c_j(\xi) V_i^b\left[y_j(x)\right] = 0. \tag{1.43}$$

Thus, the following equality is obtained:

$$\sum_{j=0}^{n-1} a_j(\xi) V_i\left[y_j(x)\right] = -\sum_{j=0}^{n-1} c_j(\xi) V_i^b\left[y_j(x)\right] \tag{1.44}$$

where coefficients $c_j(\xi) = b_j(\xi) - a_j(\xi)$ were determined earlier.

System (1.44) is a system of linear algebraic equations with respect to unknown coefficients $a_j(\xi)$. The determinant of this system is different from zero $det\left\|V_i\left[y_j(x)\right]\right\|_0^{n-1} \neq 0$. This is ensured by the fact that the boundary conditions are nonsingular. Therefore, the coefficients $a_j(\xi)$ are determined as well. To find coefficients $b_j(\xi)$ the correspondence (1.41) is used.

Thus, all four defining properties of the Green's function are satisfied, and the function (1.38) is indeed the Green's function of a semihomogeneous boundary value problem (1b).

1.3.5 Self-Conjugate Boundary Value Problems

Let a linear differential operator of the n-th order be given

$$L_n y(x) = \sum_{j=0}^{n} p_j(x) y^{(n-j)}(x) = f(x). \tag{1.45}$$

Then the operator conjugate to it has the form:

$$L_n^* y(x) = \sum_{j=0}^{n} (-1)^j \left[p_j(x) y(x)\right]^{(n-j)}. \tag{1.46}$$

In the case if $L_n y(x) = L_n^* y(x)$, the operator is called self-conjugate. Let's introduce the definition of the self-conjugate boundary value problem.

A boundary value problem $\begin{cases} L_n y(x) = f(x), a < x < b, \\ V_i[y(x)] = 0, i = \overline{0, n-1} \end{cases}$ is called self-conjugate if two conditions are fulfilled: (1) the operator of the problem is self-conjugate; (2) if there are two functions $u(x)$ and $v(x)$, which satisfy the original boundary value problem, then the integral condition is fulfilled for them

$$\int_a^b [v(x)L_n u(x) - u(x)L_n v(x)]dx = 0.$$

Theorem 6 *The Green's function of the self-conjugate boundary value problem is symmetric function:* $G(x, \xi) = G(\xi, x)$.

Proof Let us have a self-conjugate boundary value problem

$$\begin{cases} L_n y(x) = f(x), & x \in (a, b), \\ V_i[y(x)] = 0, & i = \overline{0, n-1}. \end{cases} \tag{1.47}$$

By the definition of the self-conjugate boundary value problem, it follows that: $L_n(y(x)) = L_n^*(y(x))$, that is, the differential operator of the problem is self-conjugate; the integral condition is fulfilled for two functions satisfying the boundary value problem (1.47):

$$\begin{cases} L_n \begin{bmatrix} u(x) \\ v(x) \end{bmatrix} = f(x), \quad x \in (a, b), \quad i = \overline{0, n-1}, \\ V_i \begin{bmatrix} u(x) \\ v(x) \end{bmatrix} = 0, \end{cases}$$

namely

$$\int_a^b [v(x)L_n u(x) - u(x)L_n v(x)] = 0. \tag{1.48}$$

The solutions of both problems can be written with the help of the Green's function in the form

$$u(x) = \int_a^b G(x, \xi) f(\xi) d\xi, \quad v(x) = \int_a^b G(x, \xi) f(\xi) d\xi.$$

Then, the substitution of these expressions into formula (1.48) leads to the equality:

$$\begin{aligned}&\int_a^b [v(x)L_n u(x) - u(x)L_n v(x)]dx = \int_a^b f(x)(v(x) - u(x))dx = \\ &= \int_a^b f(x) \int_a^b G(x,\xi)f(\xi)d\xi dx - \int_a^b f(x) \int_a^b G(x,\xi)f(\xi)d\xi dx = 0.\end{aligned} \tag{1.49}$$

The repeated integrals can be rewritten as double integrals:

$$\int_a^b \int_a^b f(x)f(\xi)G(x,\xi)dxd\xi - \int_a^b \int_a^b f(x)f(\xi)G(x,\xi)d\xi dx = 0 \tag{1.50}$$

In the first term of expression (1.50), the order of integration is changed:

$$\begin{aligned}&\int_a^b \int_a^b f(\xi)f(x)G(\xi,x)dxd\xi - \int_a^b \int_a^b f(x)f(\xi)G(x,\xi)dxd\xi = \\ &= \int_a^b \int_a^b f(\xi)f(x)[G(\xi,x) - G(x,\xi)]d\xi dx.\end{aligned}$$

The equality (1.50) has the form

$$\int_a^b \int_a^b f(x)f(\xi)F(x,\xi)dxd\xi = 0. \tag{1.51}$$

Here, $F(x,\xi) = G(x,\xi) - G(\xi,x)$. It does not yet follow from the equality of the integral (1.51) to zero that $F(x,\xi) \equiv 0$. Let's prove this fact by contradiction. It is supposed that $F(x,\xi) \neq 0$, then there exists some point $x_0 \in (a;b)$ such that $F(x_0,\xi_0) \neq 0$, and, for example, $F(x_0,\xi_0) > 0$. Since the solution of the boundary value problem represented by the Green's function can be applied for any continuous function $f(x)$, it can be chosen in such form: $\begin{cases} f(x_0) \neq 0, & x_0 \in (a;b) \\ f(x) \equiv 0, & x \in (a;x_0) \cup (x_0;b) \end{cases}$. This choice of functions is based on the fact that a solution exists for any continuous function at the right-hand side of the differential equation. In this case, without limiting the generality of reasoning, it can be assumed that $f(x_0) > 0$ and $f(\xi_0) > 0$. Then equality (1.51) takes the form

$$\int\limits_{\Delta(x_0)} \int\limits_{\Delta(\xi_0)} f(x)f(\xi)F(x,\xi)dxd\xi = 0$$

Under the integral here there is a function whose value at the point (x_0, ξ_0) is positive $F(x_0)f(\xi_0)F(x_0, \xi_0) > 0$. Therefore, the integral of it in the left-hand side of the equality must be positive $\int\limits_{\Delta(x_0)} \int\limits_{\Delta(\xi_0)} f(x)f(\xi)F(x,\xi)dxd\xi > 0$.

The contradiction is obtained, from which it follows that the function $F(x, \xi) = 0$, and from this, in turn, it follows that $G(x, \xi) \equiv G(\xi, x)$.

The apparatus of the Green's function is effective in solving a boundary value problem of the form (1.26). In the case when the differential operator has a matrix structure, that is, a system of differential equations is solved, the apparatus of the scalar Green's function is inapplicable, and it is required to involve the apparatus of the matrix Green's function, which is much less widely known. The main provisions can be found in [110]. Here, we briefly present the main properties and methods for constructing the matrix Green's function.

1.4 Main Properties of the Matrix Green's Function

1.4.1 Fundamental Matrix. Fundamental Solution. Nonsingular Boundary Conditions and the Basis Matrix

The role of the scalar Green's function in the case of systems of differential equations is played by Green's matrix-function [110]. Consider a system of differential equations, which can be written as:

$$P_0(x)z'(x) + P_1(x)z(x) = f(x), \quad a < x < b \tag{1.52}$$

where $z(x) = \left\|z_j(x)\right\|_0^{n-1}, f(x) = \left\|f_j(x)\right\|_0^{n-1}, P_m(x) = \left\|P_{j,k}^m\right\|_0^{n-1}$ $\left(m = 0, 1; P_{j,k}^0 = 0, j \neq k\right)$. It is assumed that the elements of all matrices are also continuous on the interval (a, b).

Matrix

$$\Phi(x, \xi) = \left\|\varphi_{j,k}(x, \xi)\right\|_0^{n-1} \tag{1.53}$$

is called fundamental if the column matrix (a vector)

$$z(x) = \int_a^b \Phi(x,\xi) f(\xi) d\xi \tag{1.54}$$

is a solution of equation (1.52) for the column matrix $f(x)$.

The fundamental solution of the matrix differential equation (1.52) is the matrix

$$Z(x) = \left\| z_{j,k}(x) \right\|_0^{n-1} \tag{1.55}$$

with nonzero determinant

$$\det Z(x) = \det \left\| z_{j,k}(x) \right\|_0^{n-1} \neq 0 \tag{1.56}$$

and satisfying the homogeneous matrix equation

$$P_0 Z' + P_1 Z = 0, \quad a < x < b \tag{1.57}$$

If Z is a fundamental matrix solution, then, obviously, ZC ($c = \left\| c_j \right\|_0^{n-1}$ is a constant vector) will also be a solution of equation (1.57). By direct substitution, one can verify that

$$z = Zc, \quad c = \left\| c_j \right\|_0^{n-1} \tag{1.58}$$

is a solution of a homogeneous version of equations (1.52), that is,

$$P_0 z' + P_1 z = 0 \tag{1.59}$$

Moreover, it can be shown that (1.58) is a general solution of equation (1.59), i.e., any of its solutions has the form (1.58).

Let's introduce numerical matrices

$$\gamma = \left\| \gamma_j \right\|_0^{n-1}, A = \left\| \alpha_{j,k} \right\|_0^{n-1}, B = \left\| \beta_{j,k} \right\|_0^{n-1} \tag{1.60}$$

and define the boundary functional

$$U = Az(a) + Bz(b) \tag{1.61}$$

where its action on matrices of (1.53) type is given by the formulae

$$\begin{aligned} U[\Phi] &= A\Phi(a,\xi) + B\Phi(b,\xi) \\ \widetilde{U}[\Phi] &= \Phi(x,a)A + \Phi(x,b)B \end{aligned} \tag{1.62}$$

As in the scalar case, an inhomogeneous, semihomogeneous, and homogeneous boundary value problems for equation (1.52) are posed:

$$P_0 z' + P_1 z = f, \quad a < x < b, \quad U[z] = \gamma \tag{1.63}$$

$$P_0 z' + P_1 z = f, \quad a < x < b, \quad U[z] = 0 \tag{1.64}$$

$$P_0 z' + P_1 z = 0, \quad a < x < b, \quad U[z] = 0 \tag{1.65}$$

and the notation of nonsingular boundary conditions is introduced. The latter are called nonsingular if

$$\det U[z] \neq 0, \quad U[z] = Az(a) + Bz(b) \tag{1.66}$$

The boundary conditions are understood as nonsingular boundary conditions if the homogeneous boundary value problem (1.65) has only a trivial solution and vice versa. If the homogeneous boundary value problem (1.65) has only a trivial solution, then (1.66) holds. To verify this, the general solution (1.58) should be substituted into the boundary condition $U[z] = 0$.

By analogical way to the scalar case solution $\Psi(x) = \left\|\psi_{j,k}\right\|_0^{n-1}$ of the boundary value problem

$$P_0 \psi' + P_1 \psi = 0, \quad a < x < b, \quad U[\psi] = I, \left(I = \left\|\delta_{j,k}\right\|_0^{n-1}\right) \tag{1.67}$$

will be called the basis matrix of boundary value problems (1.63)–(1.65). Such a matrix exists under nonsingular boundary conditions and, as can be easily seen, it is determined by the formulae

$$\Psi(x) = ZC, \quad C = (U[z])^{-1} \tag{1.68}$$

1.4.2 Existence and Uniqueness of the Green's Matrix

The Green's matrix of the boundary value problem (1.64) is the matrix $G(x,\xi) = \left\|g_{j,k}(x,\xi)\right\|_0^{n-1}$ through which its solution for an arbitrary $f(x)$ is expressed by the formula

$$z(x) = \int_a^b G(x,\xi) f(\xi) d\xi \tag{1.69}$$

If such a matrix is constructed and the basis matrix is known, then the solution of the inhomogeneous boundary value problem (1.63) can be written in the form

$$z(x) = \int_a^b G(x,\xi) f(\xi) d\xi + \Psi(x)\gamma \tag{1.70}$$

which is easy to verify by substituting (1.70) into (1.63) with (1.67) taken into account. As in the scalar case, it is proved that under nonsingular boundary conditions (1.66), there can be only one Green's matrix of the boundary value problem (1.64), that is, Theorem 3 also holds for the system of equations.

The analog of Theorem 4 is the following theorem.

Theorem 7 *The matrix* $G(x,\xi) = \left\| g_{j,k}(x,\xi) \right\|_0^{n-1}$ *with the following three properties is the fundamental matrix of equation (1.52).*

1. *Matrix* $G(x,\xi)$ *is continuous in both variables in each of the triangular regions* $a \le x < \xi$, $\xi < x \le b$.
2. *On the line* $x = \xi$ *the matrix* $G(x,\xi)$ *suffers a jump, determined by the formulas*

$$\begin{aligned} G(\xi+0,\xi) - G(\xi-0,\xi) &= P_0^{-1}(\xi) \\ G(x,x-0) - G(x,x+0) &= P_1^{-1}(x) \end{aligned} \tag{1.71}$$

3. *In each of the named triangular regions, the matrix* $G(x,\xi)$ *by* x *satisfies equation (1.57), i.e.*

$$P_0 G'(x,\xi) + P_1 G(x,\xi) = 0$$

If, in addition to the listed properties, the matrix satisfies the boundary condition

$$U[G] = 0 \tag{1.72}$$

then it is the Green's matrix of the boundary value problem (1.64).

The proof of this theorem is carried out in the same way as Theorem 4, taking into account the known rules for differentiation, integration, multiplication, and addition of matrices.

The analog of Theorem 5 is the following theorem.

Theorem 8 *If the fundamental matrix $\Phi(x,\xi)$ of equation (1.52) is continuous in each of the triangular regions ($a \le x < \xi$, $\xi < x \le b$) and satisfies the boundary condition (1.72), then it is the Green's matrix of the boundary value problem (1.64).*

Proof Under nonsingular boundary conditions, there always exists a Green's matrix. To do this, one should perform constructions similar to those done in the scalar case (1.54). To satisfy condition *(1.71)*, we define the Green's matrix in the form

$$G(x,\xi) = \begin{cases} Z(x)A^0(\xi), & x < \xi \\ Z(x)B^0(\xi), & x > \xi \end{cases} \tag{1.73}$$

where $Z(x)$ is the fundamental solution of equation (1.52), and $A^0(\xi)$ and $B^0(\xi)$ are arbitrary matrices depending on the parameter ξ. Then, we fulfill conditions 2 and 4, having disposed of the matrices $A^0(\xi)$ and $B^0(\xi)$. Substituting (1.73) into the left-hand side of the first equality from *(1.71)* and then into *(1.72)* gives

$$\begin{aligned} &Z(\xi)\left[B^0(\xi) - A^0(\xi)\right] = P_0^{-1}(\xi) \\ &AZ(a)A^0(\xi) + BZ(b)B^0(\xi) = 0 \end{aligned} \tag{1.74}$$

The resulting equations for the matrices $A^0(\xi)$ and $B^0(\xi)$ are uniquely solvable due to (1.56) and (1.66). Thus, we have proved the following proposition, which is an analog of Theorem 5.

Theorem 9 *If the homogeneous boundary value problem (1.65) has only a trivial solution (that is, the boundary conditions are nonsingular), then the boundary value problem (1.64) has one and only one Green's matrix, which has all four properties listed in Theorem 7. It can be determined by formula (1.73), in which the matrices $A^0(\xi)$ and $B^0(\xi)$ are found from equations (1.74).*

1.4.3 Techniques for Deriving the Green's Matrix-Function

The deriving of Green's matrix-function is implemented with the help of the matrix integral transform method [110]. Let's construct Green's matrix-function $G(x,\xi)$ for the vector boundary problem of the following structure:

$$\begin{cases} L_2\vec{y}\,(x) = \vec{f}\,(x) \\ V_i\left[\vec{y}\,(x)\right] = 0, i = 0, 1, \end{cases} \tag{1.75}$$

where L_2 is a differential operator of the second order $L_2\vec{y}\,(x) = I\vec{y}\,''(x) + 2\beta Q\vec{y}\,'(x) - \beta^2 P\vec{y}\,(x)$, I is an identity matrix,

$P = \begin{pmatrix} \frac{\kappa-1}{\kappa+1} & 0 \\ 0 & \frac{\kappa+1}{\kappa-1} \end{pmatrix}$, $Q = \begin{pmatrix} 0 & \frac{1}{\kappa+1} \\ -\frac{1}{\kappa-1} & 0 \end{pmatrix}$, $\vec{f}(x)$ is some known vector-function, $\vec{y}(x)$ is unknown vector-function, V_0, V_1 are boundary functionals of the following form:

$$V_0\left[\vec{y}(x)\right] = \alpha_n \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \vec{y}(0) - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \vec{y}'(0),$$
$$V_1\left[\vec{y}(x)\right] = \alpha_n \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \vec{y}(a) - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \vec{y}'(a).$$

The kernel of the integral transform is chosen as follows:

$$H(x, \alpha_n) = (\sin \alpha_n x 00 \cos \alpha_n x), \alpha_n = \frac{n\pi}{a}, n = 0, 1, 2 \ldots \tag{1.76}$$

Both parts of equation in (1.75) are multiplied by the kernel (1.76) and they are integrated by parts on the segment $[0; a]$. Considering that

$$H'(x, \alpha_n) = \alpha_n \begin{pmatrix} \cos \alpha_n x & 0 \\ 0 & -\sin \alpha_n x \end{pmatrix},$$
$$H'(0, \alpha_n) = \alpha_n \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, H'(a, \alpha_n) = \alpha_n \begin{pmatrix} \cos n\pi & 0 \\ 0 & 0 \end{pmatrix},$$
$$H(0, \alpha_n) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, H(a, \alpha_n) = \begin{pmatrix} 0 & 0 \\ 0 & \cos n\pi \end{pmatrix} = \cos n\pi \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$
$$H''(x, \alpha_n) = -\alpha_n^2 H(x, \alpha_n),$$

one can obtain that

$$\int_0^a H(x, \alpha_n) I \vec{y}''(x) dx = \cos n\pi \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \vec{y}'(a) - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \vec{y}'(0)$$
$$- \alpha_n \cos n\pi \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \vec{y}'(a) + \alpha_n \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \vec{y}'(0) - \alpha_n^2 \vec{y}_n.$$

So

$$\int_0^a H(x,\alpha_n) I \vec{y}''(x)dx = V_0\left[\vec{y}(x)\right] - \cos n\pi V_1\left[\vec{y}(x)\right] - \alpha_n^2 \vec{y}_n.$$

Due to the fulfillment of the boundary conditions in (1.75), the following expressions are derived:

$$\int_0^a H(x,\alpha_n) I \vec{y}''(x)dx = -\alpha_n^2 \vec{y}_n, \quad \int_0^a H(x,\alpha_n) Q \vec{y}'(x)dx = \alpha_n \widetilde{Q} \vec{y}_n,$$

$$\int_0^a H(x,\alpha_n) P \vec{y}(x)dx = \int_0^a H(x,\alpha_n) \vec{f}(x)dx = \vec{f}_n.$$

Equation in (1.75) takes the following form:

$$\Omega_\beta(\alpha_n)\vec{y}_n = \vec{f}_n, \tag{1.77}$$

here, $\vec{y}_n = \int_0^a \vec{y}(x) H(x,\alpha_n)dx$, $\widetilde{Q} = \begin{pmatrix} 0 & \frac{1}{\kappa+1} \\ \frac{1}{\kappa-1} & 0 \end{pmatrix}$, $\Omega_\beta(\alpha_n) = -I\alpha_n^2 -$

$2\beta\alpha_n\widetilde{Q} - \beta^2 P = \; = \begin{pmatrix} -\alpha_n^2 - \beta^2\frac{\kappa-1}{\kappa+1} & -\frac{2\beta\alpha_n}{\kappa+1} \\ -\frac{2\beta\alpha_n}{\kappa-1} & -\alpha_n^2 - \beta^2\frac{\kappa+1}{\kappa-1} \end{pmatrix}$.

The solution of equation (1.77) has the form

$$\vec{y}_n = \Omega_\beta^{-1}(\alpha_n)\vec{f}_n$$

where

$$\Omega_\beta^{-1}(\alpha_n) = \frac{1}{\left(\alpha_n^2+\beta^2\right)^2}\begin{pmatrix} -\frac{\alpha_n^2(\kappa-1)+\beta^2(\kappa+1)}{\kappa-1} & \frac{2\beta\alpha_n}{\kappa+1} \\ \frac{2\beta\alpha_n}{\kappa-1} & -\frac{\alpha_n^2(\kappa+1)+\beta^2(\kappa-1)}{\kappa+1} \end{pmatrix}.$$

The inverse integral transform is applied to the solution $\vec{y}_n = \begin{pmatrix} y_{n1} \\ y_{n2} \end{pmatrix}$. In component form it can be written so

$$y_1(x) = \frac{2}{a}\sum_{n=1}^{\infty} y_{n1}\sin(\alpha_n x) = \frac{2}{a}\sum_{n=0}^{\infty}{}' y_{n1}\sin(\alpha_n x),$$

$$y_2(x) = \frac{y_{02}}{a} + \frac{2}{a}\sum_{n=1}^{\infty} y_{n2}\cos(\alpha_n x) = \frac{2}{a}\sum_{n=0}^{\infty} \prime y_{n2}\cos(\alpha_n x).$$

It is rewritten in vector form as $\vec{y}(x) = \frac{2}{a}\sum_{n=0}^{\infty} \prime H(x,\alpha_n)\vec{y}_n$, here stroke means that the zeroth term is multiplied by ½. From another point of view, the expression $\vec{y}_n = \int_0^a H(x,\alpha_n)\vec{y}(x)dx$ takes place. After uniting of these two results, the formula for the desirable vector was constructed $\vec{y}(x) = \frac{2}{a}\sum_{n=0}^{\infty} \prime H(x,\alpha_n)\Omega_\beta^{-1}(\alpha_n)\vec{f}_n$. Using the formula $\vec{f}_n = \int_0^a H(\xi,\alpha_n)\vec{f}(\xi)d\xi$, the following correspondence is derived:

$$\vec{y}(x) = \frac{2}{a}\sum_{n=0}^{\infty} \prime H(x,\alpha_n)\Omega_\beta^{-1}(\alpha_n) \int_0^a H(\xi,\alpha_n)\vec{f}(\xi)d\xi =$$

$$= \int_0^a \left[\frac{2}{a}\sum_{n=0}^{\infty} \prime H(x,\alpha_n)\Omega_\beta^{-1}(\alpha_n)H(\xi,\alpha_n)\right]\vec{f}(\xi)d\xi.$$

From the last formula it follows that Green's matrix-function of the boundary problem (1.75) has the form

$$G(x,\xi) = \frac{2}{a}\sum_{n=0}^{\infty} \prime H(x,\alpha_n)\Omega_\beta^{-1}(\alpha_n)H(\xi,\alpha_n). \tag{1.78}$$

The expression (1.78) is the bilinear expansion of Green's matrix-function. For this Green's matrix-function all properties of a Green's function are fulfilled, in particular the boundary conditions in (1.75) are fulfilled

$$V_0[G(x,\xi)] = 0, V_1[G(x,\xi)] = 0$$

The Green's matrix has the following form:

$$G(x,\xi) = \frac{2}{a}\sum_{n=0}^{\infty} \prime \begin{pmatrix} -\frac{\sin(\alpha_n\xi)\sin(\alpha_n x)\left[(\kappa-1)\alpha_n^2+(\kappa+1)\beta^2\right]}{(\kappa-1)\left(\alpha_n^2+\beta^2\right)^2} & \frac{2\alpha_n\beta\cos(\alpha_n\xi)\sin(\alpha_n x)}{(\kappa+1)\left(\alpha_n^2+\beta^2\right)^2} \\ \frac{2\alpha_n\beta\sin(\alpha_n\xi)\cos(\alpha_n x)}{(\kappa-1)\left(\alpha_n^2+\beta^2\right)^2} & -\frac{\cos(\alpha_n\xi)\cos(\alpha_n x)\left[(\kappa+1)\alpha_n^2+(\kappa-1)\beta^2\right]}{(\kappa+1)\left(\alpha_n^2+\beta^2\right)^2} \end{pmatrix}.$$

The simplification of the matrix and the final component formula for Green's function is shown in Appendix B.

1.5 Orthogonal Polynomials Method

Boundary value problems are often reduced to solving of singular integral equations. In some cases, it is possible to solve these equations exactly and find these solutions in the explicit form. Unfortunately, it is not very often met. So why one of the effective approximate solution methods, which is proposed here, is the orthogonal polynomials method [123, 133, 147]. The material of the next subchapter essentially relies on the results obtained by Popov G.Ya. in [112].

1.5.1 The Scheme of the Method. Polynomial Kernels. Spectral Correspondences

The orthogonal polynomials method is based on a remarkable property of classical polynomials: they are eigenfunctions of many integral operators.

Let's consider the algorithm of this method on the following example of the integral equation

$$\frac{1}{\pi}\int_{-1}^{1}\ln\frac{1}{|t-\tau|}\varphi(\tau)d\tau = f(t), \quad |t|\le 1 \tag{1.79}$$

Here, $\varphi(\tau)$ is unknown functions, $f(t)\in C_{[-1;1]}$ is known right-hand side of (1.79), which is continuous.

The presence of the spectral correspondence makes it easy to construct an explicit solution of the integral equation (1.79). This correspondence, which is given at [112],

$$\begin{gathered}\frac{1}{\pi}\int_{-1}^{1}\ln\frac{1}{|t-\tau|}\frac{T_n(\tau)}{\sqrt{1-\tau^2}}d\tau = \mu_n^0 T_n(t), \quad n=\overline{0,\infty},\\ \mu_0^0 = \ln 2, \mu_n^0 = 1/n, n=\overline{1,\infty}, \quad t\le 1\end{gathered} \tag{1.80}$$

shows that the Chebyshev polynomials $T_n(t)$ are eigenfunctions of an integral operator with a logarithmic kernel, and μ_n^0 are its eigenvalues (therefore, it seems natural to call such correspondences as spectral ones).

Such spectral correspondence allows to search the desirable function in the form of the following decomposition

$$\varphi(t) = \sum_{m=0}^{\infty} \varphi_m \frac{T_n(t)}{\sqrt{1-t^2}} \tag{1.81}$$

After substituting of (1.81) at (1.79), and using (1.80), it was derived

$$\sum_{m=0}^{\infty} \mu_m^0 \varphi_m T_m(t) = f(t), \quad |t| \le 1 \tag{1.82}$$

The both sides of equality (1.82) are multiplied by $(1-t^2)^{-1/2} T_n(t)$ and integrated over the interval $t \in (-1;1)$. The subsequent use of the Chebyshev polynomials orthogonality leads to the formula for the expansion coefficients at (1.81):

$$\varphi_n = \frac{f_n}{\mu^0 \|T_n\|^2}, \quad f_n = \int_{-1}^{1} \frac{f(x) T_n(x)}{\sqrt{1-x^2}} dx \tag{1.83}$$

Formulas (1.81), (1.83) determine the required explicit solution of equation (1.79).

The spectral correspondence can also be applied to the solving of the singular integral equation of the second kind:

$$\varphi(t) - \lambda \int_{-1}^{1} \ln\left(\frac{1}{|t-\tau|}\right) \frac{\varphi(\tau) d\tau}{\rho(\tau)} = f(t), \quad |t| \le 1 \tag{1.84}$$

where $\rho(\tau)$ is a given function. For this, relation (1.80) should be represented in the form

$$\begin{gathered} \int_{-1}^{1} K(t,\tau) \varphi_n(\tau) d\tau = \mu_n^0 \varphi_n(t), \\ K(t,\tau) = \frac{1}{\pi} \ln\left(\frac{1}{|t-\tau|}\right) \frac{(1-t^2)^{-1/4}}{(1-\tau^2)^{1/4}}, \quad \varphi_n(t) = \frac{T_n(t)}{\|T_n\|(1-t^2)^{1/4}} \end{gathered} \tag{1.85}$$

whence, based on the Hilbert-Schmidt theorem [166], a bilinear expansion follows:

$$K(t,\tau) = \sum_{m=0}^{\infty} \mu_m^0 \varphi_m(t) \varphi_m(\tau), \quad 1 \le t, \tau \le 1$$

which, based on presentation (1.85), can also be written as:

$$\frac{1}{\pi}\ln\frac{1}{|t-\tau|}=\sum_{m=0}^{\infty}\frac{\mu_m^0}{\pi}T_m(t)T(\tau) \tag{1.86}$$

Substitution of this expansion into equation (1.84) leads to the expression

$$\varphi(t)=f(t)+\frac{\lambda}{\pi}\sum_{m=0}^{\infty}\mu_m^0\varphi_m T_m(t),\quad \left(\varphi_m=\int_{-1}^{1}\frac{\varphi(\tau)T_m(\tau)d\tau}{\rho(\tau)}\right) \tag{1.87}$$

To determine the coefficients φ_n, both parts of equality (1.87) should be multiplied by $T_n(t)\rho^{-1}(t)$ and integrated over the interval $[-1, 1]$.

As a result, the infinite system of algebraic equations is obtained

$$\begin{gathered}\varphi_n-\frac{\lambda}{\pi}\sum_{m=0}^{\infty}\mu_m^0 a_{nm}\varphi_m=f_n,\quad n=\overline{0,\infty}\\ \left(a_{nm}=\int_{-1}^{1}\frac{T_n(t)T_m(t)dt}{\rho(t)},\quad f_n=\int_{-1}^{1}\frac{f(t)T_n(t)dt}{\rho(t)}\right)\end{gathered} \tag{1.88}$$

An explicit solution of the infinite system (1.88) can be derived in exceptional cases. Therefore, they are solved approximately using the reduction method, i.e. replacing system (1.88) with a finite one

$$\varphi_n-\frac{\lambda}{\pi}\sum_{m=0}^{N}\mu_m^0 a_{nm}\varphi_m=f_n,\quad n=\overline{0,N} \tag{1.89}$$

and achieving the necessary accuracy by increasing the number N.

1.5.2 *Scheme for the Solving of the First Kind Integral Equation in the General Case*

Let us first state the scheme for solving of the first kind integral equation for one special, but rather widespread case

$$\int_l k_i(x-\xi)p(\xi)d\xi=g(x),\quad x\in l \tag{1.90}$$

where a kernel has the form

$$k_i(y) = \frac{1}{\pi} \int_0^\infty \frac{h_i(s)}{s} \cos(ys) ds \tag{1.91}$$

Here, l is the segment [-a, a] and it is supposed that

$$h_i(0) = 0, h_i(t) = 1 + O\left(t^{-\delta}\right), \quad t \to \infty, \delta > 0 \tag{1.92}$$

Let us transform equation (1.90) to the canonical segment $[-1, 1]$ by making the substitutions of variables in (1.90) and (1.91)

$$x = at, \xi = a\tau, s = \frac{\sigma}{a}; ap(a\tau) = \varphi(\tau), g(at) = f(t) \tag{1.93}$$

As a result of this substitution, instead of (1.90) it is derived

$$\int_{-1}^{1} k_a^{(i)}(t-\tau)\varphi(\tau)d\tau = f(t), \quad |t| \le 1, \tag{1.94}$$

$$k_a^{(i)}(t-\tau) = \frac{1}{\pi} \int_0^\infty h_i\left(\frac{\sigma}{a}\right) \frac{\cos[\sigma(t-\tau)]}{\sigma} d\sigma \tag{1.95}$$

The kernel of the resulting integral equation is representated in the form of two terms. With this aim the representation is used [112]

$$\frac{1}{\pi} \ln \frac{1}{|t-\tau|} = \frac{1}{\pi} \int_0^\infty \frac{\cos[\sigma(t-\tau)] - e^{-\sigma}}{\sigma} d\sigma \tag{1.96}$$

The function (1.96) is added and subtracted from the kernel (1.95). As a result, the required representation is obtained

$$\pi k_a^{(i)}(t-\tau) = -\ln|t-\tau| + d_a^{(i)}(t-\tau)\pi \tag{1.97}$$

$$d_a^{(i)}(y) = \int_0^\infty \frac{[h_i(\sigma/a) - 1]\cos(\sigma y) + e^{-\sigma}}{\pi\sigma} d\sigma \tag{1.98}$$

Taking into account (1.97) and the existence of the spectral correspondence (1.80), the solution of the integral equation is searched in the form (1.81). Analogically to the operations used for deriving of formula (1.83), an infinite system of algebraic equations must be solved

$$\|T_n\|^2 \mu_n^0 \varphi_n + \sum_{m=0}^{\infty} d_{nm}^a \varphi_m = f_n, \quad n = \overline{0, \infty}, \tag{1.99}$$

$$d_{nm}^a = \int_{-1}^{1} \int_{-1}^{1} d_a^{(i)}(t-\tau) \frac{T_n(t)}{\sqrt{1-t^2}} \frac{T_m(\tau)}{\sqrt{1-\tau^2}} dt d\tau \tag{1.100}$$

Moreover, the coefficients f_n are determined by the formula from (1.83). The double integral (1.100) can be transformed into a single integral after substituting (1.98) into (1.100)

$$\frac{d_{nm}^a}{\pi} = \int_0^{\infty} \frac{[h_i(\sigma/a) - 1] J_n(\sigma) J_m(\sigma) + e^{-\sigma} \delta_{n0} \delta_{m0}}{(-1)^{1/2(m+n)}} d\sigma \tag{1.101}$$

The above scheme for solving the integral equation (1.94) can be easily transferred to the general case [112]. The justification of the orthogonal polynomials method can be found in [112].

Chapter 2
Review



The study of the stress-deformed state of elastic bodies began to develop actively in the nineteenth century and has not lost its relevance to the present day. This is due to the wide range of applications in various engineering applications. The classical linear theory of elasticity is the most important basis for most strength calculations in engineering. During operation, building and other structures are exposed to mechanical, temperature, and other types of influence. Therefore, when designing, it is necessary to calculate the strength of such structures. Product strength characteristics can be obtained based on the analysis of the stress-strain state of their elastic models. One such model is a semi-infinite strip. Therefore, the development of analytical and numerical methods for studying its stress-strain state is an urgent problem.

2.1 Plane Problems of Elasticity for a Semi-strip

With regard that the present work is dedicated to the analytical-numerical solving of the problems for a semi-strip, the review is focused on analytical methods. These methods are the reducing to the integral Fredholm's equation, the usage of the expression in the form of Fourier series and integrals, the reducing to the infinite system of algebraic equations, variatious, energetic methods, methods which are based on the construction of the bioorthogonal equations' system, on the diagonalization of the equilibrium equation's system, and the usage of expansions by system of Fadle-Papkovich functions. As it can be seen, the majority of methods use some auxiliary functions, which complicates the construction of the solution. The usage of expansions by Fadle-Papkovich functions is impossible for the mixed problems.

The detailed review of indicated approaches for the solving of the plane elasticity problems is presented below.

I.I. Vorovich and V.V. Kopasenko [165] reduced the solving of the problem for a symmetrically loaded semi-strip fixed by its short edge to the Fredholm integral equation of the first kind regarding normal load in the fixing zone. As the authors

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N. Vaysfeld, Z. Zhuravlova, *Mixed Boundary Problems in Solid Mechanics*, La Matematica per il 3+2 155, https://doi.org/10.1007/978-3-031-37826-3_2

mentioned, this singularity complicates the investigation of the equation and its numerical solving. H. Zorski [181] reduced the solving of the problem for a semi-strip to the singular integral equation. D. B. Bogy reduced solving of the problem for bonded semi-strips [16] to the system of two singular integral equations. They were solved numerically. The solution of the problem for a semi-strip [15] was constructed as a superposition of solutions of the problem for a semi-plane and the problem for an infinite strip. A new method which allow to obtain an exact solution for the first and second moments of displacements for statically determinated beams with spatially random stiffness was developed in [36].

P. P. Teodorescu [136] solved the problem for a semi-strip when the stresses are given at the boundary. The stress function in this problem is presented as a combination of Fourier series and integrals. The analogical approach was applied in the papers [61, 86, 107] for the solving of the problems for a semi-strip. D. G. Crowdy and A. S. Fokas [27] constructed integral solutions for an elastostatic semi-strip.

W. T. Koiter and J. Alblas [67] applied integral transforms to the differential equation for the load functions. This allowed to reduce the problem for the stretching of the semi-strip with the free boundaries and the fixed short edge to the solving of the infinite system of linear algebraic equations. L. A. Aglovyan and R. S. Gevorkyan [4] reduced the solving of the problem for a semi-strip when its boundaries are free and the displacements are given at the short edge to the infinite system of linear algebraic equations. J. P. Benthem [13] considered the problem for a semi-strip when its lateral sides are free and the stresses are given at the short edge and in the case when the short edge is fixed and the stresses are given at infinity. The application of Laplace's transform to the equations for stress functions reduced the problem to the solving of infinity algebraic system, which regularity wasn't proved.

The variation method was used by L.P. Trapeznikov in [145] for the solving of the problem for a semi-infinite strip compressed by two concentrated forces with free short edge. V.G. Suchevan [134] solved the mixed problems of elasticity for a semi-strip with the help of variation Castigliano's method. The varying parameters were found from the infinity system of linear inhomogeneous algebraic equations.

P. Thecaris [137] used the energetic method during the solving of the problem for a semi-strip with free lateral sides and the concentrated force acting by the semi-strip's short edge.

The special system of biorthogonal functions is constructed in [62]. It was used for the solving of the problem for a semi-strip, which is loaded by its short edge and any conditions on the longitudinal sides could be given. M. I. Gussein-Zade [55] studied the conditions in which the stresses or displacements should satisfy to derive the solution which would fade away when moving away off this end.

V. M. Zamyatin, A. V. Makhov, and A. A. Svetashkov [170] proposed the new method for solving the plane problems of elasticity. It is based on the diagonalization of the equilibrium system's equations.

G. Horvay and J. Born [57] considered the problem of the semi-strip's stress-deformable state when the lateral sides are free from the load and the displacements are given at semi-strip's short edge in the form of the polynomials. The orthogonal polynomial method is used during the problem's solving.

The main aspects of the P.F. Papkovich algorithm in the method of homogeneous solutions for the two-dimensional biharmonic problem in rectangular area are outlined in [92].

O. S. Gogoleva [48] constructed the solution of the classical boundary problem of elasticity for the stress distribution in the rectangular semi-strip with free lateral sides and normal and tangential stresses given at the short edge. The solution is constructed as the decomposition by Fadle-Papkovich functions. M. D. Kovalenko and his apprentices [71, 122] applied decompositions by Fadle-Papkovich functions for the solving the problem for a semi-strip. R. D. Gregory [53] solved the problem for a semi-infinite strip with the help of Fadle-Papkovich functions.

I. P. Gavrilkyuk and L. V. Makarov [45] investigated the class of problems in the generalized statement in the case of the semi-strip. The difference scheme on the finite lattice was constructed in this paper and the convergence rate was estimated.

A. M. Gomilko [50] considered the two-dimensional biharmonic problem for a semi-strip with Dirichlet-type boundary conditions. It was solved with the help of superposition method. The analysis of the series convergence and the proof of the separation singularities method for the method of homogeneous solutions on the example of the mixed problem for a semi-strip were conducted in [51].

R. D. Gregory and F. Y. M. Wan [54] investigated the plane semi-strip's stress state under different boundary conditions. Some exact solutions for the semi-infinite crack in the strip were derived with the help of conformal mapping technique in [139]. The solution for an elastic semi-strip with lateral sides free from stress and short edge in conditions of ideal contact was presented in [82].

In the paper [164] the methods for solving of the mixed problem of linear elasticity, which forms the basis of the calculation of the contacting details' fortress, were considered. The approach for the solving of the boundary problems for the spatial canonical areas based on the integral transforms for the harmonic functions was proposed and developed in [102].

2.2 Plane Thermoelasticity Problems

The following papers are dedicated to the solving of the plane thermoelasticity problems.

V. Nowacki [104] constructed the solutions for the majority of thermoelasticity problems as the sum of two solutions, one of which is in the nature of thermoelastic potential and the another one contains the components of Galerkin displacement vector.

The paper [68] is dedicated to the solving of the plane and spatial quasistatic problem of linear thermoelasticity for the rectangular areas by the variation methods with the use of special orthogonal polynomials.

The boundary problem of the stationary thermal conductivity for the semi-strip with the circular hole was reduced to the infinite system of linear algebraic equations, which allowed application on the reduction method, in [29]. The temperature field

and the state of the temperature stresses in the semi-strip were investigated in [108]. The boundary problem of the temperature state of rigid semi-strip made from the different isotropic materials was solved in [125].

The new method for the solving of the plane quasistatic thermoelasticity problem in the displacement terms was proposed in [163] for the rectangular area. This method allowed to construct the analytical solution in the form of trigonometric series.

I. A. Abbas [1] considered the thermoelastic semi-infinite environment. The system of equation in the Laplace's transform space is solved with the help of analytic method and finite-element method. The approximate solutions of the problem were derived there during the inversion of the Laplace's transformants. R. V. N. Melnik [94] investigated the properties of the discrete approximation for the mathematical models of coupled thermoelasticity in the stressed-temperature statement.

M. S. Abou-Dina and A. F. Ghaleb [2] solved the plane problem of thermoelasticity with the help of harmonic functions. M. Hvozdara and K. Rosa [58] deduced the formulae for thermoelastic stresses and displacements caused by the stationary point heat source in an elastic semi-space. A. D. Shamrovskiy and G. V. Merkotan [127] solved the generalized coupled dynamic problem of thermoelasticity for the semi-space with the help of the theory of invariant-group properties of differential equations. In G. Gabrusev's paper [42] the algorithm for investigation of the stress state, which appears in transversally-isotropic layer as a result of the heat exchange by Newton's law between the layer and the exterior environment in the presence of distribution lines of the third kind conditions for the temperature and boundary planes of the layer was developed.

A. V. Rychahivskyy and Yu. V. Tokovyy [118] solved the thermoelasticity problem for the semi-plane with conditions of fixing, displacement, and mixed boundary conditions with the help of the direct integration method. V. Seremet [124] derived two new solutions for the boundary problems of thermoelasticity for a semi-plane. In the paper [88] the problem of a thermal contact between the semi-strip and strip was solved. This problem was reduced to the solving of the singular integral equation with additional condition by the use of Fourier's transform.

A. I. Veremeychik, V. V. Garbachevsky, and V. M. Khvisevich [162] constructed integral equations for the plane boundary problem of inhomogeneous thermoelasticity and developed the algorithm of numerical solving for the problem. The problem of coupled thermoelasticity under the cyclical load was solved in [171].

Yu. M. Kolyano and L. M. Zatvarska [70] constructed the solution for the thin semi-strip-plate, which is free from the external load and it is heated by heat source, by the method of functions' continuation. In the paper [168] the semi-infinite plate under the laser impulse local heat was investigated with the help of L-S generalized thermoelastic theory, where control finite-element equations were solved explicitly in the time area. The temperature fields and thermal stresses on the plates from the open array of heating sources were investigated in [169].

The boundary problem of coupled thermoelasticity for a semi-layer with the hole and mixed boundary conditions was solved by L. A. Fil'shtinskii and A. V. Bondar

in [39]. This problem was reduced to the system of four singular integral equations, which were solved numerically by the mechanical quadrature method. Yu. Tokovyy and Chien-Ching Ma presented an analytical approach to solve the plane elasticity and thermoelasticity problems for inhomogeneous, orthotropic planes, half-planes, and strips in [144]. Solution of these problems was reduced to the governing Volterra integral equation, which was solved with the use of the resolvent-kernel technique. S. Davidson [28] considered the problem for the determination of thermal stresses in the thermoelastic strip with the collinear array of the cracks parallel to the boundaries. The solution of the Duhamel-Neumann equation was constructed with the help of harmonic functions there. H. Qing, W. Yang, J. Lu, and D.-F. Li [115] solved the thermoelastic problem for an infinite strip, which contains an infinite set of periodically distributed boundary cracks. The integral equation to which this problem's solving for reduced was solved numerically. The plane thermoelasticity problem for the bodies with cracks was investigated in [66]. The basis of the mathematical theory for the thermoelastic equilibrium deformable solid bodies with thin inclusions was presented in [133]. The investigation of the thermoelastic fields in the bodies with initial deformations was investigated in the works of Kushnir R. M. and his apprentices [79].

2.3 Plane Problems of Elasticity for the Strip and Semi-plane with Cracks

The solving of the plane problems with cracks has the big value.

The method of singular integral equations as the method for solving of the two-dimensional elasticity problems for the bodies with cracks was developed by M. P. Savruk, P. N. Osiv, and I. V. Prokopchuk [121]. Two-dimensional static problems of the crack theory were considered in the paper of N. F. Morozov [99]. L. I. Slepyan [129] studied static, dynamic crack and slowly growing crack in solid bodied, also he discussed the criteria of the crack's growing. M. P. Savruk [120] considered the solving methods for the two-dimensional boundary problems of mathematic crack theory for the isotropic bodies and studied examples of the solving the problem for a semi-plane and strip with cracks.

The problem for a semi-infinite crack parallel to the semi-plane's border was reduced to the matrix Riemann problem with the help of the two-sided Laplace's transform by K. B. Ustinov in [148]. X.-F. Wu, E. Lilla and W.-S. Zou [167] considered the problem for a semi-infinite crack between two bonded dissimilar elastic strips with equal thickness. This boundary value problem was reduced by means of the conformal mapping technique to the standard Riemann-Hilbert problem. Non-standard boundary problem for the Laplace's equation in the infinity strip, which contained the finite crack was reduced by the Fourier transformation to the integral equation, and then to the vector Riemann-Hilbert problem in [10]. C. R. Chiang [25] developed the numerical procedure based on the Williams

eigen functions for the solving of the elastic problems destruction and demonstrated its application for the solving of the problem with the infinite strip containing central transverse crack.

R. Isida [59] reduced the solving of the problem for an infinite elastic strip with one boundary crack to the solving of the integral Fredholm equation of the second kind.

E. V. Borisova [17] solved the problem of the stress concentration near the finite crack's tip in the semi-plane with free boundaries by the modification of the discontinuous solutions' method. This method reduced the problem to the singular integral equation. M. B. Civelek and F. Erdogan [26] considered the problem for the infinite strip with the set of crack orthogonal to the boundaries. The problem's solution was reduced to the system of singular integral equations. M. R. Gecit [46] reduced the solution of the problem for a strip, one side of which is fixed and another is under conditions of the first main problem of elasticity, with the crack to the solving of the singular integral equation. It was solved by the collocation method. SIF values were given in the paper [74]. They were derived by the singular integral equation method.

H. Itou and A. Tani [60] reduced the solving of the problem for an infinite strip with a semi-infinite crack to the solving of the singular integral equation by the use of potentials of simple and double layer. E. N. Theotokoglou and G. J. Tsamasphyros [138] presented the method for the elastic strip weakened by the cracks and holes. The solution of this problem was reduced to the singular integral equation with the help of complex potentials. F. Tian-You [140] proposed the procedure for the stress intensity factor's calculation for the infinite crack in the strip. This procedure was based on the Muskhelishvili's complex potential formulation and combining conformal mapping technique.

H. Tada, P. C. Paris, and G. R. Irwin [135] found the approximate formula for SIF calculation by the method of asymptotic interpolation. The approximate formulae for SIF under the normal load acting by the crack's shores were given in the papers [22, 23].

S. D. Akbarov, N. Yahnioglu, and A. Turan [3] considered the problem for an infinite strip with a crack. For the problem's solving the methods of three-dimensional linearized elasticity were used, and for the calculation of stress intensity factors the energetic method was used.

The problem for a plane stressed elasticity semi-plane with the edge crack located along the line of connection of two absolutely-jointed heterogeneous quarter-planes was solved in [9]. The mixed boundary problem was further reduced to the system of two functional equations of Wiener-Hopf type. R. Ballarini and H. A. Lou [12] considered the application of Green's function to the solving of the plane problems of elasticity with crack on the examples of solving the problems for bimorphous strips with cracks. Stress intensity factors were calculated with regard to the load, geometry, and materials' properties. The solution for the finite strip, one boundary of which was bounded to the rigid support, and another was under the uniform stretching load, was derived by the consideration of the infinite strip which contained transverse rigid inclusion in the center and two-symmetrically loaded transverse

cracks in [47]. The Fourier transform and asymptotical analysis were applied for the reduction of the problem for the crack of the third kind, which was perpendicular to the conjunction of composite bimorphal strip, to the singular integral equation, which was solved numerically with the help of quadrature Gauss-Chebyshev formula in [91].

In the paper of X. Li and S. H. Guo [85] the effects of nonhomogeneity of dynamic stress intensity factors for the antiplane interfacial crack in a functionally-graduated material, bounded to the elastic semi-strip, were considered. The use of Fourier and Laplace transforms allowed to reduce the initial mixed boundary problem to three equations in series, and then to the singular integral equation of the first kind. The antiplane analysis of the destruction impact was presented for the weakly-continuous conjugation in the symmetric functionally-graduated composite strip in [84]. With the help of integral Laplace and Fourier transforms the problem was reduced to the pair integral equation, and the asymptotically analytic solution for the stress field near the crack tips was derived. In the paper of N.N. Antonenko [11] the solving approach for the problem of the central longitudinal crack with a filler in a strip was proposed. The integral Fourier transform was used for the problem's solving. The problem was reduced to the system of integro-differential equations regarding the derivatives of the displacement jumps on the crack's shores.

V. M. Alexandrov and D. A. Pozharskii [6] investigated the plane problem for the interfacial crack between an elastic strip and an elastic semi-plane made of different material. For the solving of the system of integral equation for the displacements' jumps in the crack zone the regular asymptotic method was used. This method is effective when the crack is essentially narrow. In the paper A. V. Dyskin, L. N. Germanovich, and K. B. Ustinov [35] two configurations of interfacial crack modeling were considered: the crack in a semi-plane and the crack in the center of an infinite strip. The interfacial crack on the border of division of the circular inclusion and the matrix was investigated in the paper [147].

L.V. Stepanova [132] analyzed the stress state near the stationary crack for transverse displacements in a plastic body. O. L. Bowie and C. E. Freese [18] solved the problem of bending of the strip with a crack. H. Chai [21] investigated the distributions of the crack in an elastic linked strip. L. I. Kal'muk, M. G. Stashchuk, and V. I. Pokhmurs'kii [64] solved the problem for a rigidly linked strip and the strip with free boundaries with a crack and rigid inclusion, and they calculated stress intensity factors near the crack tips and the ends of the rigid inclusion. The papers [100, 101] are dedicated to the investigation of cracks and rigid inclusions in dynamic statement.

Chen Y.-zh. [24] studied analysis of the stress in an infinite strip, which is weakened by periodical cracks. The boundary problem was reduced to the mixed complex problem. For its solving the extension of the variational method of eigenfunctions was used. R. S. Dhaliwal and B. M. Singh [31, 128] solved the problem for an infinite rigidly linked strip with two and three coplanar Griffith cracks. It is supposed that the cracks were opened from the inner pressure. With the help of Fourier transform, the problem's solving was reduced to the system of integral equations, which were solved exactly. D.-S. Lee [83] considered the

elastically-static problem for an infinite strip with a cylindrical hole and two symmetrically-located inner cracks perpendicular to the boundary. The problem's solving was reduced to the system of three integral equations.

N. A. Dorosh and G. S. Kit [32] considered the equilibrium problem for an infinite strip with a rectilinear transverse crack under the temperature influence. T. Menouillard and T. Belytschko [95] analyzed the propagation of cracks in the heated strip. M. V. Mirsalimov [98] reduced the thermoelasticity problem for a strip with a crack to the singular integral equation.

F. Erdogan and K. Arin [37] proposed the technology for the solving of elasticity problems with arbitrary located inner crack. As an example, the solution for an infinite crack was given there. F. Erdogan and H. Terada [38] considered the problem for a semi-strip with a side crack. The semi-strip was loaded by a rigid edge without friction, the edge was applied to the crack. The contact problem was reduced to the system of singular integral equations. O. V. Poberezhnyi [109] investigated the influence of instant heating sources on the stress state and stress intensity factors for a semi-strip with a boundary crack.

R. V. Goldstein, I. N. Ryskov, and R. L. Salganik [49] considered the problem for an infinite strip with central transverse crack and investigated stress intensity factors. In the paper [65] some dynamic problems for thin elastic bodies were considered.

V. D. Lamzyuk, V. I. Mossakovshii, and S. D. Sotnikova [81] solved the problem for a stressed crack in an infinite strip by the superposition method with the solving of three problems: about stressed crack in an infinite place, about the first main problem for an infinite homogeneous strip and about an infinite strip free from the load.

P. A. Martynyuk and E. B. Polyak [90] gave the general solution for the problem for a strip with a crack parallel to the boundaries at the arbitrary distance to the boundaries. In the paper [56] the interfacial cracks in elastic bodies under the harmonic load were investigated.

R. R. Parmerter and B. Mukherji [106] solved the problem for strip with a boundary crack by the finite-element method.

The method based on the formulae of mutual reexpansion of the solutions of Lame's system of equations in polar and Cartesian coordinates was proposed for the solving of the problem for an elastic fixed by its edge semi-strip with circular rigid inclusion in [30].

The approach of the method of crack lines stress field was proposed in [116] for the calculation of stress intensity factors of an infinite strip, which is stretching, with eccentric crack. The finite crack with constant length (Joffe-type crack), which developed in functionally-graduated strip under plane load was investigated with the help of Shmidt method in [89]. With the help of Fourier transform and defining of displacement components' jumps along the crack's surface as unknown functions, two pairs of dual integral equations were derived. For the solving of the pair integral equations the displacement components' jumps along the crack's surface were expanded in series by Jacobi polynomials. The problem for an infinite elastic strip which contained arbitrary oriented cracks and pair of concentrated forces was formulated in terms of the system of singular integral equations in [87]. The

symmetric problem about two transverse cracks in an elastic strip with reinforced surfaces was formulated in terms of singular integral equation in [75]. The special cases of one central crack or two boundary cracks were discussed. In the paper [5] the methods and results of investigations of contact interaction between thin-walled elements such as overlays (stringers) and inclusions with massive bodies, and also the action of stamps on the body with coatings and layers was proposed.

The static problems of elasticity for a semi-plane and strip weakened by a rectilinear transverse crack were investigated in [72]. The method of generalized integral transforms allowed to reduce the solving of the problem for equilibrium equations to the solving of singular integral equation of the first kind with Cauchy kernel regarding the derivative of the crack's opening function. The stress intensity factors' values at the crack tips were derived.

2.4 Consideration of Unknown Function's Fixed Singularities During the Solving of Integral Equations

The paper of R. V. Duduchava [33] is dedicated to the investigation of the singular integral equation of the form

$$c_0\varphi(x)+\frac{c_1}{\pi i}\int\limits_0^1\frac{\varphi(y)}{y-x}dy+\sum_{k=0}^{n}\frac{c_{k+2}x^{n_k}}{\pi i}\int\limits_0^1\frac{y^{k-n_k}\varphi(y)}{(y+x)^{k+1}}dy=f(x),\qquad(2.1)$$
$$x\in I=[0;1],0\le \operatorname{Re} n_k\le k,k=0,1,\ldots,n$$

and also form

$$A\varphi(x)=c_0\varphi(x)+\frac{c_1}{\pi i}\int\limits_{-1}^1\frac{\varphi(y)}{y-x}dy+\sum_{k=0}^{n}\frac{1}{\pi i}\int\limits_{-1}^1\frac{c_{k+2}(x,y)(1-x)^{m_k^+}(1+x)^{m_k^-}\varphi(y)}{(1-y)^{m_k^+-k}(1+y)^{m_k^--k}(1-xy)^{k+1}}dy$$
$$++\int\limits_{-1}^1K(x,y)\varphi(y)dy=f(x),$$
$$0\le\operatorname{Re}m_k^+\le k\qquad(2.2)$$

The equations with homogeneous kernels of the first power, which were complicated by shift operators, were studied by A. P. Soldatov [130] in the spaces of Hölder functions with complex infinite weight.

When $c_1 = 0$ equation (1.1), relying on work of M.G. Krein [73], was investigated by L.G. Mikhailov [96].

In the case when $n = 0$, $c_0 = 0$, $c_1 = 1$, Im $c_2 = 0$ equation (1.1) was fully solved by H. F. Bueekner [19] (the case when $n = 0$, $c_0 = 1$, $c_1 = 0$ was considered earlier in [73]).

Equation (1.1) was investigated by G. I. Bierman [14], but his investigation was unfinished; Equation (2.1) was solved in the function class $\bigcup_{0<\alpha<1} L_1(I, x^\alpha)$, which is non-invariant regarding operators that are in equation (2.1); by the substitution $x = e^{-t}$, $y = e^{-\tau}$ equation (2.1) was reduced to the equation of convolution on the half-axis, but it was solved formally by the Wiener-Hopf method [103], the integrability of derived solutions was not investigated; the Wiener-Hopf method in its classical form is suitable only for equations with continuous presymbols and when $c_1 = 0$ it is unsuitable for (1.1).

In the case when $n = 0$, $c_2 = -c_1$ equation (2.2) was fully solved in the papers of F. Tricomi [146] and S.G. Mikhlin [97].

Nowadays the investigations of the singular integral equations with fixed singularities continue.

In the paper of V.G. Popov [113] the solving of the dynamic problem for an elastic semi-strip, which is joined to the elastic semi-space, was reduced to the singular integral equation with two fixed singularities. It was solved with the help of special quadrature formulae for singular integrals. In [114] the solving of the problem for the determination of stress state in elastic isotropic semi-space with a crack, which intersected with its boundary under harmonic longitudinal shear vibrations, was reduced to the singular integro-differential equation regarding unknown displacement jump on the crack's shores. The solving of this equation was complicated by the existence of the fixed singularity in its kernel. The effective approximate method for the solving of such equations, which took into consideration real asymptotics of unknown function, was developed there. O.F. Kryvyi [77] solved the problem about mutual influence of interfacial tunnel crack and inclusion in partially-anisotropic space, where the method, which allowed to consider fixed singularities of unknown functions, was used. In the paper [105] the approach for the solving of singular integral equations with fixed singularities was proposed.

N. S. Gabbasov [40] considered linear integral equation of the third kind with fixed singularities in the kernel. For the approximate solving of that equation in the distribution space, the new generalized version of collocation method was introduced [41]. Symmetric characteristic singular equation with two fixed singularities at the ends in the class of functions limited at the ends was analyzed in [7, 8]. For Chebyshev polynomials of the first kind in the right-hand side, the solution of the integral equation was expressed in terms of two non-orthogonal polynomials with weights. The approximate solution of the full singular integral equation, which was based on that new generalized spectral correspondence for the singular operator with two fixed singularities, was derived by the reduction to the infinite system of linear algebraic equations of the second kind. The investigation of optimal quadrature formula for the singular integrals with fixed singularity was done in the paper [126]. The numeric solving for the hypersingular integral equation for toothed semi-plane problem was derived by the collocation methods proposed in

[20]. These methods searched the approximation of the solution's derivative of the original equation. The investigation of algebra, generated by singular integral Cauchy operators and integral operators with fixed singularities at the unit interval was done in the paper [34]. In [63] P. Junghanns and A. Rathsfeld considered the polynomial collocation method for the numerical solving of the singular integral Cauchy equations with fixed singularities on the interval, where the fixed singularities were of Mellin's convolution type. In the paper of A.V. Saakyan [119] the quadrature formulae of the highest algebraic accuracy were derived for singular integrals, and the efficiency of their application for the solving of singular integral equations with generalized Cauchy kernel, which are also called as equations with fixed singularity, was shown.

2.5 Conclusions to the Second Chapter

Modern authors use two main approaches: analytical and numerically-analytical, the choice between which is defined by the boundary conditions at the boundaries of the semi-strip. The common between all these approaches is the presentation of the solution of the equilibrium equations though harmonic, biharmonic, and other functions.

Here, the approach proposed by G.Ya. Popov [112] was used, where integral transforms are applied directly to the equilibrium equation and boundary conditions of the initial problem, which reduced the initial problem to the one-dimensional problem. It was solved with the help of vector boundary problems apparatus [110]. For the solving of the integral equation, to which the solving of the problem was reduced, the orthogonal polynomial method, proposed by G.Ya. Popov [112], was used.

The investigation of the papers, dedicated to the solving of plane elasticity problems for a semi-strip showed that the problem of investigation of the semi-strip's stress state in the case of existence of the transverse crack remains open. The fixed singularities of the displacement function in the corner points of the semi-strip were not considered in the papers.

From the analysis of scientific literature, it leads that, in spite of the major quantity of papers in research direction of mixed plane problems of elasticity, there are nonsolved problems of consideration of fixed singularities of the displacement function, namely for the elastic semi-infinite strip under external loads and existence of the crack.

Chapter 3
The Method of Construction of Solutions for the Mixed Elasticity Problems for a Semi-infinite Strip



This chapter contains the description of the analytical apparatus which is used for the solving of the mixed problems of elasticity for a semi-infinite strip. The method for the solving of the mixed plane problems which is based on the application of the integral transformations directly to the system of Lame's equilibrium equations and the boundary conditions, and which reduces the initial problem to the vector one-dimensional problem is described. The vector one-dimensional boundary problem is solved explicitly with the help of the apparatuses of the matrix differential calculation and Green's matrix-function.

3.1 Reducing of the Initial-Boundary Value Problem to the One-Dimensional Vector Boundary Value Problem

The elastic semi-strip (Fig. 3.1) (G is a share module, μ is a Poisson ratio), which occupies the area in Cartesian coordinates $0 < x < a$, $0 < y < \infty$ is considered.

The normal load is applied by the short edge of the semi-strip $y = 0, \quad 0 < x < a$

$$\sigma_y\big|_{y=0} = p(x), \quad \tau_{xy}\big|_{y=0} = 0, \quad 0 < x < a, \tag{3.1}$$

where $p(x)$ is the known function. The boundary conditions at the semi-infinite boundary sides $x = 0, \quad 0 < y < \infty$ and $x = a, \quad 0 < y < \infty$ are written in the following form:

$$U_0[f] = 0, \quad U_1[f] = 0, \quad 0 < y < \infty. \tag{3.2}$$

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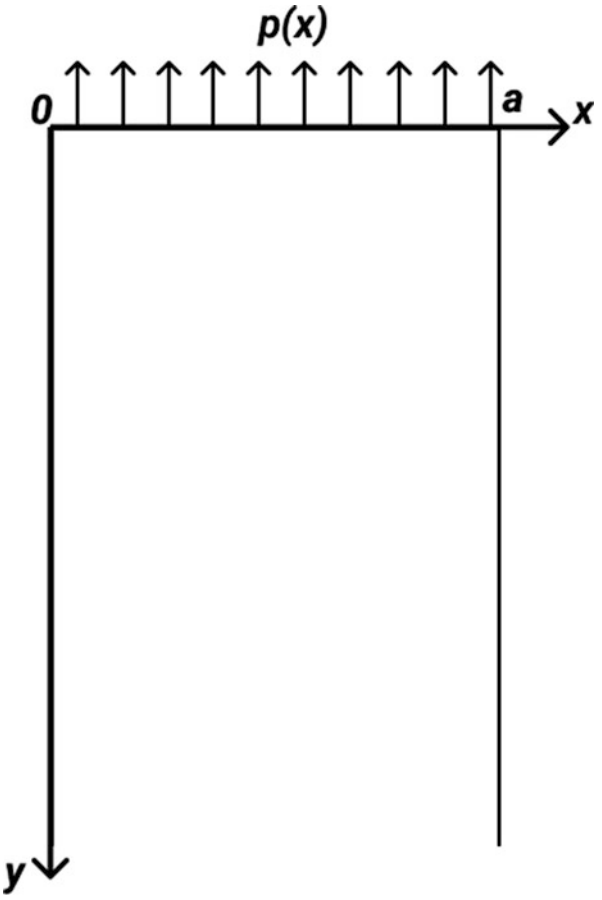


Fig. 3.1 Geometry and coordinate system of the semi-strip

Here, $U_0[f(x)] = \alpha_0 f(0, y) + \beta_0 \partial f/\partial x(0, y)$, $U_1[f(x)] = \alpha_1 f(a, y) + \beta_1 \partial f/\partial x(a, y)$ are boundary functionals of the general form (they will be detailed later for each particular problem), $f(x, y) = (u_x(x, y), u_y(x, y))^T$ is the displacements' vector.

Here and further, the notation for the displacements $u_x(x, y) = u(x, y)$, $u_y(x, y) = v(x, y)$ and the function derivatives $f(x, y)$: $f' = \partial f/\partial x$, $f^{\cdot} = \partial f/\partial y$ are inputted. The system of Lame's equilibrium equations can be written in the following form:

$$\begin{cases} \mu_* \dfrac{\partial^2 u(x,y)}{\partial x^2} + \dfrac{\partial^2 u(x,y)}{\partial y^2} + \mu_0 \dfrac{\partial^2 v(x,y)}{\partial x \partial y} + X_1(x,y) = 0 \\ \dfrac{\partial^2 v(x,y)}{\partial x^2} + \mu_* \dfrac{\partial^2 v(x,y)}{\partial y^2} + \mu_0 \dfrac{\partial^2 u(x,y)}{\partial x \partial y} + X_2(x,y) = 0, \end{cases} \tag{3.3}$$

where $\mu_0 = \frac{1}{1-2\mu}, \mu_* = \mu_0 + 1$, $X_1(x,y) = X_x(x,y)$, $X_2(x,y) = X_y(x,y)$ are the components of the volume force. Constants μ_0, μ_* are expressed by the Myschelishvili's constant $\kappa = 3 - 4\mu$ as the following: $\mu_0 = \frac{2}{\kappa - 1}, \mu_* = \frac{\kappa+1}{\kappa-1}, \frac{\mu_0}{\mu_*} = \frac{2}{\kappa+1}, \mu_0 - 1 = \frac{3-\kappa}{\kappa-1}$.

The system (3.3) is rewritten in the form:

$$\begin{cases} \dfrac{\partial^2 u(x,y)}{\partial x^2} + \dfrac{\kappa-1}{\kappa+1}\dfrac{\partial^2 u(x,y)}{\partial y^2} + \dfrac{2}{\kappa+1}\dfrac{\partial^2 v(x,y)}{\partial x \partial y} + \dfrac{\kappa-1}{\kappa+1}X_1(x,y) = 0 \\ \dfrac{\partial^2 v(x,y)}{\partial x^2} + \dfrac{\kappa+1}{\kappa-1}\dfrac{\partial^2 v(x,y)}{\partial y^2} + \dfrac{2}{\kappa-1}\dfrac{\partial^2 u(x,y)}{\partial x \partial y} + X_2(x,y) = 0. \end{cases} \tag{3.4}$$

Boundary conditions at the short edge (3.1) are reformulated in the terms of displacement functions

$$2G\mu_0\left(\mu\left.\frac{\partial u}{\partial x}\right|_{y=0} + (1-\mu)\left.\frac{\partial v}{\partial y}\right|_{y=0}\right) = p(x), 0 < x < a \tag{3.5}$$

$$\left.\frac{\partial u}{\partial y}\right|_{y=0} + \left.\frac{\partial v}{\partial x}\right|_{y=0} = 0, \quad 0 < x < a. \tag{3.6}$$

The unknown function $\chi(x) = v|_{y=0}, \chi'(x) = v'|_{y=0}$ is inputted. With regard to the boundary condition (3.6), the correspondence $\left.\partial u/\partial y\right|_{y=0} = -\chi'(x)$ is derived. So, the condition (3.6) is fulfilled automatically.

The problem (3.2), (3.4)–(3.6) is reduced to the one-dimensional boundary problem with the help of the integral sin-cos Fourier transform applied to equations (3.3) by the variable y

$$\begin{bmatrix} u_\beta(x), X_{1\beta}(x) \\ v_\beta(x), X_{2\beta}(x) \end{bmatrix} = \int_0^\infty \begin{bmatrix} u(x,y), X_1(x,y) \\ v(x,y), X_2(x,y) \end{bmatrix} [\cos \beta y \sin \beta y] dy \tag{3.7}$$

with the corresponding inverse formulae

$$\begin{bmatrix} u(x,y), X_1(x,y) \\ v(x,y), X_2(x,y) \end{bmatrix} = \frac{2}{\pi} \int_0^\infty \begin{bmatrix} u_\beta(x), X_{1\beta}(x) \\ v_\beta(x), X_{2\beta}(x) \end{bmatrix} [\cos \beta y \sin \beta y] d\beta \tag{3.8}$$

The integration of equations (3.4) step-by-step is shown in appendix A. The integral transform is also applied to the boundary conditions. As the result, the boundary problem in the transform domain is derived

$$\begin{cases} \dfrac{d^2u_\beta(x)}{dx^2} - \dfrac{\beta^2(\kappa-1)}{\kappa+1}u_\beta(x) + \dfrac{2\beta}{\kappa+1}\dfrac{dv_\beta(x)}{dx} = \dfrac{3-\kappa}{\kappa+1}\chi'(x) - \dfrac{\kappa-1}{\kappa+1}X_{1\beta}(x), \\ \dfrac{d^2v_\beta(x)}{dx^2} - \dfrac{\beta^2(\kappa+1)}{\kappa-1}v_\beta(x) - \dfrac{2\beta}{\kappa-1}\dfrac{du_\beta(x)}{dx} = -\beta\dfrac{\kappa+1}{\kappa-1}\chi(x) - X_{2\beta}(x), \\ U_0\left[\vec{y}_\beta\right] = 0, U_1\left[\vec{y}_\beta\right] = 0. \end{cases} \tag{3.9}$$

Here, $\vec{y}_\beta(x) = \begin{pmatrix} u_\beta(x) \\ v_\beta(x) \end{pmatrix}$.

3.2 Formulation and Solving of the Vector One-Dimensional Boundary Value Problem in the Transform Domain

The following vector $\vec{f}(x) = \begin{pmatrix} \dfrac{3-\kappa}{\kappa+1}\chi'(x) - \dfrac{\kappa-1}{\kappa+1}X_{1\beta}(x) \\ -\beta\dfrac{\kappa+1}{\kappa-1}\chi(x) - X_{2\beta}(x) \end{pmatrix}$ and matrices $P = \begin{pmatrix} \dfrac{\kappa-1}{\kappa+1} & 0 \\ 0 & \dfrac{\kappa+1}{\kappa-1} \end{pmatrix}$, $Q = \begin{pmatrix} 0 & \dfrac{1}{\kappa+1} \\ -\dfrac{1}{\kappa-1} & 0 \end{pmatrix}$ are inputted. Equations (3.9) in the vector form can be written as the vector equation $L_2\vec{y}_\beta(x) = \vec{f}(x)$, where L_2 is a differential operator of the second order $L_2\vec{y}_\beta(x) = I\vec{y}_\beta''(x) + 2\beta Q\vec{y}_\beta'(x) - \beta^2 P\vec{y}_\beta(x)$, I is an identity matrix. As the result, the vector one-dimension boundary problem is derived

$$\begin{aligned} &L_2\vec{y}_\beta(x) = \vec{f}(x) \\ &U_0\left[\vec{y}_\beta\right] = 0, U_1\left[\vec{y}_\beta\right] = 0. \end{aligned} \tag{3.10}$$

The solution of the vector one-dimensional boundary problem is searched as the superposition of the vector homogeneous equation's general solution $\vec{y}^{\,0}_\beta(x)$ and the vector inhomogeneous equation's partial solution $\vec{y}^{\,1}_\beta(x)$

$$\vec{y}_\beta(x) = \vec{y}^{\,0}_\beta(x) + \vec{y}^{\,1}_\beta(x) \tag{3.11}$$

These solutions are constructed with the help of the matrix differential calculation apparatus [44, 110], which required construction of the fundamental matrix system of the corresponding matrix homogeneous equation.

3.3 Construction of the Fundamental Matrix Solutions' System

As it was proved in [110], in order to obtain the solution of the homogeneous vector equation in (3.10), one should first construct the solution of the corresponding matrix equation

$$L_2 Y_\beta(x) = 0, 0 < x < a, \tag{3.12}$$

here, $Y_\beta(x)$ is the matrix of order 2×2. The matrix $Y_\beta(x)$ is chosen in the form $Y_\beta(x) = e^{\xi x} I$ and it is substituted into the matrix equation (3.12). As the result, the equality $L_2 e^{\xi x} I = M(\xi) e^{\xi x}$ is derived, where the matrix $M(\xi)$ has the following form:

$$M(\xi) = I\xi^2 + 2\beta Q\xi - \beta^2 P = \begin{pmatrix} \xi^2 - \beta^2 \dfrac{\kappa - 1}{\kappa + 1} & \dfrac{2\beta\xi}{\kappa + 1} \\ -\dfrac{2\beta\xi}{\kappa - 1} & \xi^2 - \beta^2 \dfrac{\kappa + 1}{\kappa - 1} \end{pmatrix}.$$

According to [111] the solution of the matrix homogeneous equation is constructed as the following:

$$Y(x) = \frac{1}{2\pi i} \oint_C e^{\xi x} M^{-1}(\xi) d\xi, \tag{3.13}$$

where $M^{-1}(\xi)$ is the matrix inverse to $M(\xi)$. The closed loop C covers all special points of the matrix $M^{-1}(\xi)$. The fact that the inverse matrix can be expressed as a transpose matrix of cofactor multiplied by the determinant of the initial matrix was used in the calculations. The determinant of the matrix $M(\xi)$ is found as a polynomial of the fourth degree

$$\det M(\xi) = \xi^4 + 2\beta^2\xi^2 + \beta^4 = (\xi - \beta)^2 (\xi + \beta)^2.$$

The inverse matrix has the following form:

$$M^{-1}(\xi)=\frac{1}{(\xi-\beta)^2(\xi+\beta)^2}\begin{pmatrix}\xi^2-\beta^2\dfrac{\kappa+1}{\kappa-1} & -\dfrac{2\beta\xi}{\kappa+1}\\ \dfrac{2\beta\xi}{\kappa-1} & \xi^2-\beta^2\dfrac{\kappa-1}{\kappa+1}\end{pmatrix}. \tag{3.14}$$

After the substitution of the constructed inverse matrix (3.14) into the expression (3.13) one can notice that the expression under the contour integral has two multiple poles $\xi=\beta$, $\xi=-\beta$ of the second order. The following matrix system of fundamental matrix solutions was derived after the calculation of the contour integral with the help of the residue theorem and the calculation of the residues in the multiple poles

$$Y_1(x)=\frac{e^{\beta x}}{2}\begin{pmatrix}\dfrac{\kappa-\beta x}{\beta(\kappa-1)} & -\dfrac{x}{\kappa+1}\\ \dfrac{x}{\kappa-1} & \dfrac{\kappa+\beta x}{\beta(\kappa+1)}\end{pmatrix}\quad Y_2(x)=\frac{e^{-\beta x}}{2}\times\begin{pmatrix}-\dfrac{\kappa+\beta x}{\beta(\kappa-1)} & \dfrac{x}{\kappa+1}\\ -\dfrac{x}{\kappa-1} & -\dfrac{\kappa-\beta x}{\beta(\kappa+1)}\end{pmatrix}.$$

After the substitution of the found matrices $Y_1(x)$, $Y_2(x)$ into formula (3.11) the solution of the vector equation can be written in the following form [111]:

$$\vec{y}_\beta(x)=Y_1(x)\begin{pmatrix}c_1\\ c_2\end{pmatrix}+Y_2(x)\begin{pmatrix}c_3\\ c_4\end{pmatrix}+\vec{y}^{\,1}_\beta(x), \tag{3.15}$$

where constants c_i, $i=\overline{1,4}$ are found from the boundary conditions on the semi-infinite sides of the semi-strip.

The construction of Green's matrix-function is done in Sect. 1.4. The solution of the boundary problem (3.10) takes the form

$$\vec{y}_\beta(x)=Y_1(x)\begin{pmatrix}c_1\\ c_2\end{pmatrix}+Y_2(x)\begin{pmatrix}c_3\\ c_4\end{pmatrix}+\int_0^a G(x,\xi)\vec{f}\,(\xi)d\xi. \tag{3.16}$$

3.4 The Construction of the Initial Problem's Solution

The expression for the displacements' transforms (3.16) is inverted by formula (3.8). As the result, the representation for the displacement functions which depend on the unknown function $\chi(\xi)$ or $\chi'(\xi)$ is derived. The unknown function can be found by

the fulfillment of the condition at the semi-strip's short edge $\sigma_y|_{y=0} = p(x)$, $0 < x < a$. But since the integrals which are included into the expressions for the displacement functions are weakly-convergent, their weakly-convergent parts should be separated first. For this the method, presented in appendix C, is used.

3.5 The Method of Solving of the Singular Integral or Integro-differential Equation

According to the conditions at the semi-strip's short edge the problem's solving is reduced to the solving of the singular integro-differential equation (SIDE) or the singular integral equation (SIE) with one or two singularities.

3.5.1 The Solving of SIDE

In the case when the load is applied to the center of the semi-strip's short edge $y = 0, \quad a_0 < x < a_1$

$$\sigma_y\big|_{y=0} = p(x), \quad \tau_{xy}\big|_{y=0} = 0, \quad a_0 < x < a_1. \tag{3.17}$$

On the segments $y = 0, \quad 0 < x < a_0, \ a_1 < x < a, y = 0, \quad a_1 < x < a$ the slide contact conditions are fulfilled

$$v\big|_{y=0} = 0, \quad \tau_{xy}\big|_{y=0} = 0, \quad 0 < x < a_0, a_1 < x < a \tag{3.18}$$

The solving of the problem is reduced to the solving of SIDE

$$\frac{d^2}{dx^2} \int_{a_0}^{a_1} \chi(\xi^*) \ln \frac{1}{|\xi^* - x^*|} d\xi^* + \int_{a_0}^{a_1} \chi(\xi^*) f(\xi^*, x^*) d\xi^* = r(x^*), \quad a_0 < x^* < a_1 \tag{3.19}$$

where $r(x), f(\xi, x)$ are known functions, which are regular when $a_0 \neq 0, \ a_1 \neq a$.

The substitution of the variables $\xi = \frac{2\xi^* - (a_0 + a_1)}{a_1 - a_0}, x = \frac{2x^* - (a_0 + a_1)}{a_1 - a_0}$ is done for the transfer to the integration interval $[-1; 1]$. As the result the integro-differential equation (3.19) takes the following form:

$$\frac{d^2}{dx^2}\int\limits_{-1}^{1}\widetilde{\chi}(\xi)\ln\frac{1}{|\xi-x|}d\xi+\int\limits_{-1}^{1}\widetilde{\chi}(\xi)\widetilde{f}(\xi,x)d\xi=\widetilde{r}(x),\qquad -1<x<1,\qquad (3.20)$$

here, $\widetilde{\chi}(\xi)=\chi\left(\frac{(a_1-a_0)\xi+(a_0+a_1)}{2}\right)$, $\widetilde{r}(x)=\frac{(a_1-a_0)}{2}r\left(\frac{(a_1-a_0)\xi+(a_0+a_1)}{2}\right)$, $\widetilde{f}(\xi,x)=\frac{(a_1-a_0)^2}{4}f\left(\frac{(a_1-a_0)\xi+(a_0+a_1)}{2},\frac{(a_1-a_0)x+(a_0+a_1)}{2}\right)$. The integral equation (3.20) is solved by the orthogonal polynomials method [112], which allows to consider the real order of the solution's singularity at the end of the integration interval. According to this method, the function $\widetilde{\chi}(\xi)$ is expanded into series by the Chebyshev polynomials of the second kind

$$\widetilde{\chi}(\xi)=\sum_{n=0}^{\infty}s_n\sqrt{1-\xi^2}U_n(\xi),\quad \xi\in[-1;1],\qquad (3.21)$$

where $U_n(x)$ are Chebyshev polynomials of the second kind. The expression (3.21) is substituted into SIDE (3.20), and the following spectral correspondence [112] is considered:

$$\frac{d^2}{dx^2}\int\limits_{-1}^{1}\ln\frac{1}{|x-y|}\sqrt{1-y^2}U_n(y)dy=-\pi(n+1)U_n(x),\ -1<x<1.\qquad (3.22)$$

The application of the standard scheme of the orthogonal polynomials method reduces the solving of SIDE to the solving of the linear algebraic equations' system in regarding to the unknown coefficients s_n, $n=0, 1, \ldots$

$$s_m+\sum_{n=0}^{\infty}s_nD_{m,n}=f_m, m=0,1,2\ldots,\qquad (3.23)$$

where $D_{m,n}=-\frac{1}{\pi(m+1)\|U_m\|^2}\int\limits_{-1}^{1}dx\int\limits_{-1}^{1}U_n(\xi)U_m(x)\sqrt{1-\xi^2}\sqrt{1-x^2}\widetilde{f}(\xi,x)d\xi$,

$f_m=-\frac{1}{\pi(m+1)\|U_m\|^2}\int\limits_{-1}^{1}\widetilde{r}(x)U_m(x)\sqrt{1-x^2}dx.$

The system (3.23) is solved by the reduction method, the applicability of which is proved by the scheme shown in [112].

3.5.2 The Solving of SIE in the Case of the Existence of One Fixed Singularity

In the case when the load is applied close to the left side of the semi-strip's short edge $y = 0, \quad 0 < x < a_1, a_1 \neq a$

$$\sigma_y\big|_{y=0} = p(x), \quad \tau_{xy}\big|_{y=0} = 0, \quad 0 < x < a_1. \tag{3.24}$$

On the line $y = 0, \quad a_1 < x < a$ the slide contact conditions are fulfilled

$$v\big|_{y=0} = 0, \quad \tau_{xy}\big|_{y=0} = 0, \quad a_1 < x < a. \tag{3.25}$$

The solving of the problem is reduced to the solving of SIE with one fixed singularity

$$\int_0^{a_1} \chi'(\xi^*)\left[\frac{1}{\xi^* - x^*} + f(\xi^*, x^*)\right] d\xi^* = r(x^*), \quad 0 < x^* < a_1, \tag{3.26}$$

where $r(x), f(\xi, x)$ are known functions, which are regular when $a_1 \neq a$. The function $f(\xi, x)$ has a fixed singularity at the left end of the integration interval.

The substitution of the variables $\xi = \frac{2\xi^* - a}{a}, x = \frac{2x^* - a}{a}$ is done for the transfer to the integration interval $[-1; 1]$. As the result SIE (3.26) can be rewritten in the following form:

$$\frac{1}{\pi}\int_{-1}^{1} \frac{\widetilde{\chi}(\xi)}{\xi - x} d\xi + \frac{1}{\pi}\int_{-1}^{1} \widetilde{\chi}(\xi)\left[\frac{h_1}{\xi + x + 2} + \frac{h_2(x+1)}{(\xi + x + 2)^2} + \frac{h_3(\xi+1)(x+1)}{(\xi + x + 2)^3}\right] d\xi +$$
$$+ \int_{-1}^{1} \widetilde{\chi}(\xi) K(x, \xi) d\xi = \widetilde{r}(x), \quad -1 < x < 1, \tag{3.27}$$

where $\widetilde{\chi}(\xi) = \chi'\left(\frac{a_1(\xi+1)}{2}\right)$, $K(x, \xi), \widetilde{r}(x)$ are known regular functions, $h_1 = -\frac{\kappa^2 - 3}{2\kappa}, h_2 = -\frac{2}{\kappa}, h_3 = \frac{4}{\kappa}$. Here, as it can be seen, the fixed singularity at the left end of the integration interval is present when $\xi = -1$.

For the existence of the unique solution of SIE (3.27) in the classes of the functions with integral singularities at the ends, the function $\widetilde{\chi}(\xi)$ has to fulfill the additional condition

Table 3.1 Roots of equation (3.30)

$\mathrm{Re}\lambda_k$	$\mathrm{Im}\lambda_k$
−0.711876621391612	0.578740005664639
0.307935318489597	0
1.692064681510401	0
2.711876621391613	0.578740005664639
4.803618603148355	1.137450986802122
6.845283759060861	1.410612909436295
8.870599399499873	1.598048762673253

$$\int_{-1}^{1} \widetilde{\chi}(\xi)d\xi = 0 \tag{3.28}$$

The asymptotic properties of the solution of SIE (3.27) can be determined by the solution of the corresponding characteristic equation

$$\frac{1}{\pi}\int_{0}^{1} \widehat{\chi}(\xi)\left[\frac{1}{1-\frac{x}{\xi}}+\frac{h_1}{1+\frac{x}{\xi}}+\frac{h_2\frac{x}{\xi}}{\left(1+\frac{x}{\xi}\right)^2}+\frac{h_3\frac{x}{\xi}}{\left(1+\frac{x}{\xi}\right)^3}\right]\frac{d\xi}{\xi} = f(x), \quad 0<x<1, \tag{3.29}$$

where $\widehat{\chi}(\xi) = \chi'(a_1\xi)$.

The asymptotic method [76, 77] is applied to equation (3.29). The transcendental equation for the determination of the solution's singularity factor is obtained

$$\frac{1}{\sin(\pi s)}\left[\cos(\pi s) + h_1 + h_2 s + h_3\frac{s(1-s)}{2}\right] = 0$$

This equation is reformulated as

$$\cos(\lambda\pi) + \frac{2}{3-4\mu}\lambda^2 - \frac{4}{3-4\mu}\lambda - \frac{8\mu^2-12\mu+3}{3-4\mu} = 0 \tag{3.30}$$

Its roots λ_k are found and presented in Table 3.1.

It is remarkable that equation (3.30) is equivalent to the Uflyand's equation for the problem for a wedge with the angle of openness $\pi/2$ with consideration of $p = \lambda - 1$ [149].

The special method [77] which allows to consider the fixed singularities of the unknown function is used for the solving of SIE (3.27). The convergence of this method is proved in [78]. According to this method the unknown function $\widetilde{\chi}(\xi)$ is given as the following sum:

$$\widetilde{\chi}(\xi)=\sum_{k=0}^{N-1}\left[s_k\rho_k^-(\xi)+s_{k+N}\rho_k^+(\xi)\right],\quad \xi\in[-1;1], \tag{3.31}$$

where $\rho_{2k}^-(\xi)=(1+\xi)^{\operatorname{Re}\lambda_k}\cdot\cos(\operatorname{Im}\lambda_k\ln(1+\xi)),\rho_{2k+1}^-(\xi)=(1+\xi)^{\operatorname{Re}\lambda_k}\cdot\sin(\operatorname{Im}\lambda_k\ln(1+\xi)),\ k=\overline{0,N-1},\ \rho_k^+(\xi)=\frac{T_k(\xi)}{\sqrt{1-\xi}}\ ,k=\overline{0,N-1}.$

The segment $[-1;1]$ is divided into $2N$ segments by the points $x_i: P_{2N-1}^{\lambda_0,-0.5}(x_i)=0, i=\overline{0,2N-1}$. SIE (3.27) is considered when $x=x_i, i=\overline{0,2N-1}$.

After the substitution of the expression for the unknown function (3.31) into SIE (3.27) with consideration of the condition (3.28) the system of $2N$ linear algebraic equations regarding the unknown constants $s_k, k=\overline{0,2N-1}$ is derived

$$\sum_{k=0}^{N-1}\left[s_k d_{ki}^- + s_{k+N}d_{ki}^+\right]d\xi=\widetilde{r}(x_i), i=\overline{0,2N-1}, \tag{3.32}$$

where $d_{ki}^{\pm}=\int\limits_{-1}^{1}\rho_k^{\pm}(\xi)\left[\frac{1}{\xi-x_i}+\frac{h_1}{\xi+x_i-2}+\frac{h_2(x_i-1)}{(\xi+x_i-2)^2}+\frac{h_3(\xi-1)(x_i-1)}{(\xi+x_i-2)^3}+K(x_i,\xi)\right]d\xi,$ $k=\overline{0,N-1}, i=\overline{0,2N-1}$.

3.5.3 *The Solving of SIE in the Case of the Existence of Two Fixed Singularities*

In the case when the load is applied to the whole semi-strip's short edge (3.1), the solving of the problem is reduced to the solving of SIE with two fixed singularities

$$\int\limits_0^a\chi'(\xi^*)\left[\frac{1}{\xi^*-x^*}+f(\xi^*,x^*)\right]d\xi^*=r(x^*),\quad 0<x^*<a, \tag{3.33}$$

where $r(x), f(\xi,x)$ are known regular functions. The function $f(\xi,x)$ has fixed singularities at both ends of the integration interval.

The substitution of the variables $\xi=\frac{2\xi^*-a}{a}, x=\frac{2x^*-a}{a}$ is done for the transfer to the integration interval $[-1;1]$. As the result SIE (3.33) can be rewritten in the following form:

$$\frac{1}{\pi}\int\limits_{-1}^{1}\frac{\widetilde{\chi}(\xi)}{\xi-x}d\xi+\frac{1}{\pi}\int\limits_{-1}^{1}\widetilde{\chi}(\xi)\left[h_1\left(\frac{1}{\xi+x-2}+\frac{1}{\xi+x+2}\right)+\right.$$
$$\left.+h_2\left(\frac{x-1}{(\xi+x-2)^2}+\frac{x+1}{(\xi+x+2)^2}\right)+h_3\left(\frac{(\xi-1)(x-1)}{(\xi+x-2)^3}+\frac{(\xi+1)(x+1)}{(\xi+x+2)^3}\right)\right]d\xi+$$
$$+\int\limits_{-1}^{1}\widetilde{\chi}(\xi)K(x,\xi)d\xi=\widetilde{r}(x),\qquad -1<x<1 \tag{3.34}$$

here, $\widetilde{\chi}(\xi)=\chi'\left(\frac{a(\xi+1)}{2}\right)$, $K(x,\xi),\widetilde{r}(x)$ are known regular functions, $h_1=-\frac{\kappa^2-3}{2\kappa},h_2=-\frac{2}{\kappa},h_3=\frac{4}{\kappa}$. As it can be seen, here the fixed singularities at the ends of the integration interval are present when $\xi=\pm 1$.

The transcendental equation for the determination of the solution's singularity factor (3.34) is equivalent to the transcendental equation for (3.29).

According to the scheme, presented in Sect. 3.5.2, the unknown function $\widetilde{\chi}(\xi)$ is searched as the sum

$$\widetilde{\chi}(\xi)=\sum_{k=0}^{N-1}\left[s_k\rho_k^-(\xi)+s_{k+N}\rho_k^+(\xi)\right],\quad \xi\in[-1;1], \tag{3.35}$$

where $\rho_{2k}^{\mp}(\xi)=(1\pm\xi)^{\operatorname{Re}\lambda_k}\cdot\cos(\operatorname{Im}\lambda_k\ln(1\pm\xi)),\rho_{2k+1}^{\mp}(\xi)=(1\pm\xi)^{\operatorname{Re}\lambda_k}\cdot\sin(\operatorname{Im}\lambda_k\ln(1\pm\xi)),\ k=\overline{0,N-1}$.

The segment $[-1;1]$ is divided into $2N$ segments by the points $x_i: P_{2N-1}^{\lambda_0,\lambda_0}(x_i)=0, i=\overline{0,2N-1}$. SIE (3.34) is considered when $x=x_i, i=\overline{0,2N-1}$.

The substitution of the expression for the unknown function (3.35) into SIE (3.34) with consideration of the condition (3.28) reduces to the system of $2N$ linear algebraic equations regarding the unknown coefficients $s_k, k=\overline{0,2N-1}$.

$$\sum_{k=0}^{N-1}\left[s_k d_{ki}^- + s_{k+N}d_{ki}^+\right]=\widetilde{r}(x_i), i=\overline{0,2N-1} \tag{3.36}$$

where $d_{ki}^{\pm}=\int\limits_{-1}^{1}\rho_k^{\pm}(\xi)\left[\frac{1}{\xi-x_i}+h_1\left(\frac{1}{\xi+x_i-2}+\frac{1}{\xi+x_i+2}\right)+h_2\left(\frac{x_i-1}{(\xi+x_i-2)^2}+\frac{x_i+1}{(\xi+x_i+2)^2}\right)+\right.$
$\left.+h_3\left(\frac{(\xi-1)(x_i-1)}{(\xi+x_i-2)^3}+\frac{(\xi+1)(x_i+1)}{(\xi+x_i+2)^3}\right)+K(x_i,\xi)\right]d\xi, k=\overline{0,N-1}, i=\overline{0,2N-1}$.

3.6 Conclusions to the Third Chapter

Integral transforms were applied directly to the Lame's equilibrium equations and the boundary conditions of the plane elasticity problem for a semi-strip. It allowed to avoid application of some auxiliary harmonic or biharmonic functions. The initial problem in the transform domain was reduced to the one-dimensional vector boundary problem, which was solved by the matrix differential calculation methods. For it the fundamental matrix system of solutions of the homogeneous matrix equation and Green's matrix-function for the differential vector equation of the second order were constructed. The solving was reduced to SIDE or SIE which could have one or two fixed singularities in regard to the problem's statement. The application of the special numeric method which allowed to consider the existence of fixed singularities was proposed.

Chapter 4
The Mixed Problems of Elasticity for a Semi-strip

The plane mixed problems of elasticity for a semi-strip which is fixed by its lateral semi-infinite sides and is under three configurations of the applied load by the semi-strip's short edge are investigated.

The solution of the problems for the semi-strip is reduced to the solving of the integro-differential or integral equation in regard to the configuration of the applied load. The solutions of the integral equations are obtained with consideration of the searched function's fixed singularities at the ends of the integration interval.

The numerical analysis of the stress distribution at the semi-strip's lateral sides and inside the semi-strip in regard to the applied load's configuration is conducted.

4.1 The Mixed Problem of Elasticity for a Semi-strip Under Conditions of the Second Main Problem of Elasticity

The elastic semi-strip (Fig. 4.1), which occupies the area in Cartesian coordinates $0 < x < a$, $0 < y < \infty$ is considered. The semi-strip is loaded by its short edge $y = 0$, $0 < x < a$ (3.5)–(3.6). The displacements in the semi-strip satisfy the equilibrium equations (3.4). The volume forces are neglected $X_1(x, y) = 0$, $X_2(x, y) = 0$.

Let's detail the boundary conditions at the lateral semi-infinite sides (3.2): the lateral sides $x = 0$, $x = a$, $0 < y < \infty$

$$u(0, y) = 0, \quad v(0, y) = 0, \quad 0 < y < \infty, \tag{4.1}$$

$$u(a, y) = 0, \quad v(a, y) = 0, \quad 0 < y < \infty, \tag{4.2}$$

or it can be written in a vector form with the help of boundary functionals

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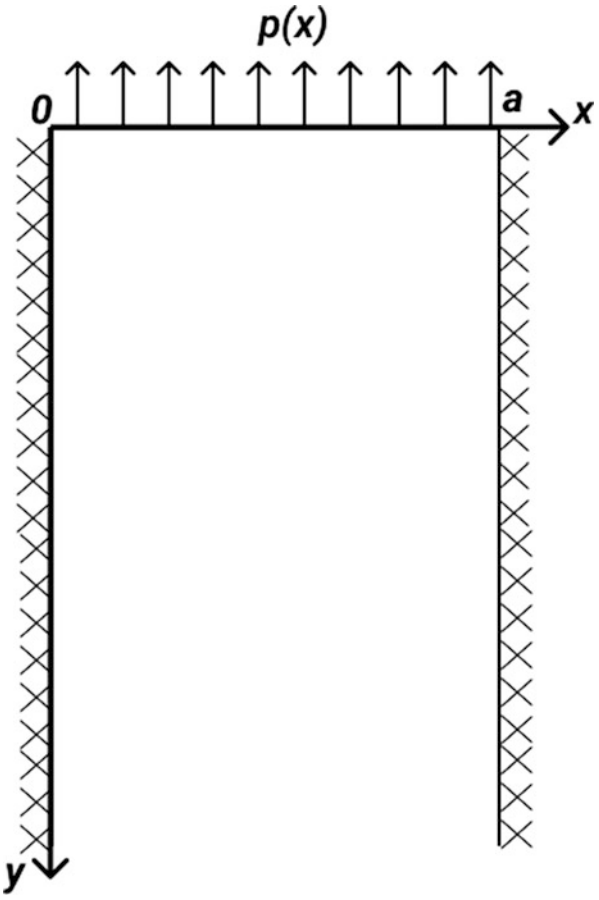


Fig. 4.1 Geometry and coordinate system of the semi-strip for the conditions of fixing

$$U_0\begin{pmatrix} u(x,y) \\ v(x,y) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} u(0,y) \\ v(0,y) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad U_1\begin{pmatrix} u(x,y) \\ v(x,y) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} u(a,y) \\ v(a,y) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{4.3}$$

The displacement and stress field inside the semi-strip and on its boundaries should be found.

According to the constructed solution (3.16) the formulae for the displacement functions have the following form:

$$u_\beta(x) = Y_1^{11}(x)c_1 + Y_1^{12}(x)c_2 + Y_2^{11}(x)c_3 + Y_2^{12}(x)c_4 + \\ + \frac{3-\kappa}{\kappa+1}\int_0^a G_{11}(x,\xi)\chi'(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int_0^a G_{12}(x,\xi)\chi(\xi)d\xi, \tag{4.4}$$

$$v_\beta(x) = Y_1^{21}(x)c_1 + Y_1^{22}(x)c_2 + Y_2^{21}(x)c_3 + Y_2^{22}(x)c_4 + \\ + \frac{3-\kappa}{\kappa+1}\int_0^a G_{21}(x,\xi)\chi'(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int_0^a G_{22}(x,\xi)\chi(\xi)d\xi. \tag{4.5}$$

The boundary conditions (4.3) (sides $x = 0$ and $x = a$) in the transformation domain are used for the finding of the coefficients $c_i, i = \overline{1,4}$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} u_\beta(0) \\ v_\beta(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} u_\beta(a) \\ v_\beta(a) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

So the next matrix system takes place

$$\begin{cases} Y_1(0)\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + Y_2(0)\begin{pmatrix} c_3 \\ c_4 \end{pmatrix} = -\vec{y}^{\,1}_\beta(0), \\ Y_1(a)\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + Y_2(a)\begin{pmatrix} c_3 \\ c_4 \end{pmatrix} = -\vec{y}^{\,1}_\beta(a). \end{cases} \tag{4.6}$$

The solution of the system (4.6) is presented in Appendix C.

4.2 The Load at the Center of the Semi-strip's Short Edge (Case 1)

The semi-strip, which is loaded at the center of its short edge $y = 0$, $a_0 < x < a_1$, $a_0 \neq 0$, $a_1 \neq a$, is considered (Fig. 4.2).

The semi-strip is loaded by the short edge $y = 0, \quad a_0 < x < a_1$

$$\sigma_y\big|_{y=0} = p(x), \quad \tau_{xy}\big|_{y=0} = 0, \quad a_0 < x < a_1 \tag{4.7}$$

At the lines $y = 0, \quad 0 < x < a_0, \ a_1 < x < a, y = 0, \quad a_1 < x < a$ the slide contact conditions are fulfilled

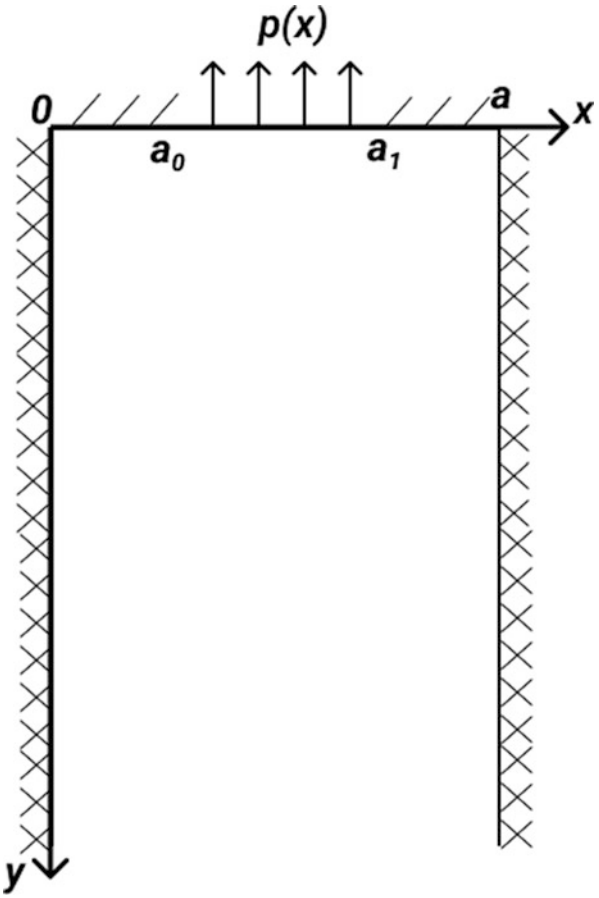


Fig. 4.2 Geometry and coordinate system of the semi-strip for the conditions of fixing. Case 1

$$v|_{y=0} = 0, \quad \tau_{xy}\big|_{y=0} = 0, \quad 0 < x < a_0, a_1 < x < a. \tag{4.8}$$

The boundary conditions (4.7) and (4.8) take the form

$$2G\mu_0\left(\mu\frac{\partial u}{\partial x}\bigg|_{y=0} + (1-\mu)\frac{\partial v}{\partial y}\bigg|_{y=0}\right) = p(x), a_0 < x < a_1, \tag{4.9}$$

$$v|_{y=0} = 0, \quad 0 < x < a_0, \quad a_1 < x < a, \tag{4.10}$$

$$\frac{\partial u}{\partial y}\bigg|_{y=0} + \frac{\partial v}{\partial x}\bigg|_{y=0} = 0, \quad 0 < x < a \tag{4.11}$$

According to the scheme presented in Chap. 2 the unknown function $\chi(x) = v|_{y=0}$, $\chi'(x) = v'|_{y=0}$ is inputted. From the boundary condition (4.11), the correspondence is obtained $\partial u/\partial y\big|_{y=0} = -\chi'(x)$ so the condition (4.11) is fulfilled automatically. The

boundary conditions at the lateral semi-infinite sides take the form (4.3). The formulae for the displacement functions are given (4.4)–(4.5).

The integral summands in (4.4) and (4.5) have the next form

$$\int_0^a G_{i1}(x,\xi)\chi'(\xi)d\xi,\ \int_0^a G_{i2}(x,\xi)\chi(\xi)d\xi, i=1,2,$$

the first of them can be transformed into the following form:

$$\int_0^a G_{i1}(x,\xi)\chi'(\xi)d\xi = \int_{+0}^{a_0-0} G_{i1}(x,\xi)\chi'(\xi)d\xi + \int_{a_0+0}^{a_1-0} G_{i1}(x,\xi)\chi'(\xi)d\xi +$$
$$+ \int_{a_1+0}^{a-0} G_{i1}(x,\xi)\chi'(\xi)d\xi = [G_{i1}(x,\xi)\chi(\xi)]|_{+0}^{a_0-0} + [G_{i1}(x,\xi)\chi(\xi)]|_{a_0+0}^{a_1-0} +$$
$$+[G_{i1}(x,\xi)\chi(\xi)]|_{a_1+0}^{a-0} - \int_0^a \frac{\partial G_{i1}}{\partial\xi}(x,\xi)\chi(\xi)d\xi = -\int_0^a \frac{\partial G_{i1}}{\partial\xi}(x,\xi)\chi(\xi)d\xi.$$

Here, $\chi(a_0-0)=0, \chi(a_1+0)=0$ from the conditions of the slide contact (4.10), $\chi(+0)=0, \chi(a-0)=0$ since the conditions of fixing (4.1)–(4.2) are fulfilled at the lateral sides of the semi-strip, $\chi(a_0+0)=0$, $\chi(a_1-0)=0$ since the solution is searched in the class of continuous functions because of the mechanical sense of the unknown function $\chi(x) = v|_{y=0}$ as the displacement function. As a result the correspondences (4.4)–(4.5) are written in the form

$$\begin{aligned} u_\beta(x) &= Y_1^{11}(x)c_1 + Y_1^{12}(x)c_2 + Y_2^{11}(x)c_3 + Y_2^{12}(x)c_4 - \\ &- \frac{3-\kappa}{\kappa+1}\int_0^a \frac{\partial}{\partial\xi}G_{11}(x,\xi)\chi(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int_0^a G_{12}(x,\xi)\chi(\xi)d\xi, \\ v_\beta(x) &= Y_1^{21}(x)c_1 + Y_1^{22}(x)c_2 + Y_2^{21}(x)c_3 + Y_2^{22}(x)c_4 - \\ &- \frac{3-\kappa}{\kappa+1}\int_0^a \frac{\partial}{\partial\xi}G_{21}(x,\xi)\chi(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int_0^a G_{22}(x,\xi)\chi(\xi)d\xi. \end{aligned} \tag{4.12}$$

In accordance with formulae (3.8), the following expression for the displacement functions are derived

$$u(x,y) = \int_0^\infty \int_0^a f(x,\xi,\beta)\cos(\beta y)\chi(\xi)d\xi d\beta, \tag{4.13}$$

$$v(x,y) = \int_0^\infty \int_0^a g(x,\xi,\beta)\sin(\beta y)\chi(\xi)d\xi d\beta. \tag{4.14}$$

Formulae (4.13)–(4.14) will define displacements completely if the function $\chi(x) = v|_{y=0}$ is known. For its finding the conditions (4.9)–(4.10) should be fulfilled. The condition (4.9) can't be fulfilled by the direct substitution of the correspondences (4.13)–(4.14) into it, because the integral in the expressions are weakly-convergent. So the method, which is presented in appendix D, is used for the separation of the weakly-convergent parts. From the condition (4.10) the following expression is derived:

$$\chi(\xi) = 0, \quad 0 < \xi < a_0, a_1 < \xi < a. \tag{4.15}$$

The following substitution of the formulae for the displacements into the boundary condition (4.9) reduces to SIDE (3.19), the solving of which is described in Sect. 3.5.1. The substitution of the found constants s_n, $n = 0, 1, \ldots$ into formulae (4.13)–(4.14) finishes the construction of the problem's solution.

The values of normal stresses $\sigma_x(0, y)$, $\sigma_y(0, y)$ at the lateral sides of the semi-strip with parameters $G = 61.2781955 \cdot 10^9$ Pa, $\mu = 0.33$, $p(x) = 1$ Pa, $a = 10$ m were analyzed when $0 \le y \le 10$ in regard to the size of the area at the short edge where the mechanical load is applied. Let's denote ϖ as the length of the slide contact zones at the short edge ($\varpi = a_0 = a - a_1$).

In figures (Figs. 4.3 and 4.4) here and further the continuous, stroke-dotted and dotted lines describe the values $\varpi = 40\ \%\ a$, $\varpi = 10\ \%\ a$ and $\varpi = 0.5\ \%\ a$ respectively.

The stresses $\sigma_x(0, y)$ are almost twice as much the stresses $\sigma_y(0, y)$.

As it can be seen from the graphics, the normal stresses at the lateral sides are compressive when $0 < y < y_1$, then they become stretching and with the further increasing of y they drop to zero. Moreover $y_1 = 2.4$ when $\varpi = 40\ \%\ a$, $y_1 = 1.2$ when $\varpi = 10\ \%\ a$, $y_1 = 1.0$ when $\varpi = 0.5\ \%\ a$. The increasing of the applied load's zone lead to the slowly decreasing of the compressing stress zones at the lateral sides.

Absolute values of the normal stresses are increasing during the increasing of the size of the area when the load is applied. When $\varpi < 0.5\ \%\ a$ the calculations lose their stability because the fixed singularities of the equation's kernel are affecting. To move the calculations near to the lateral sides the necessity of the fixed singularities' consideration appears.

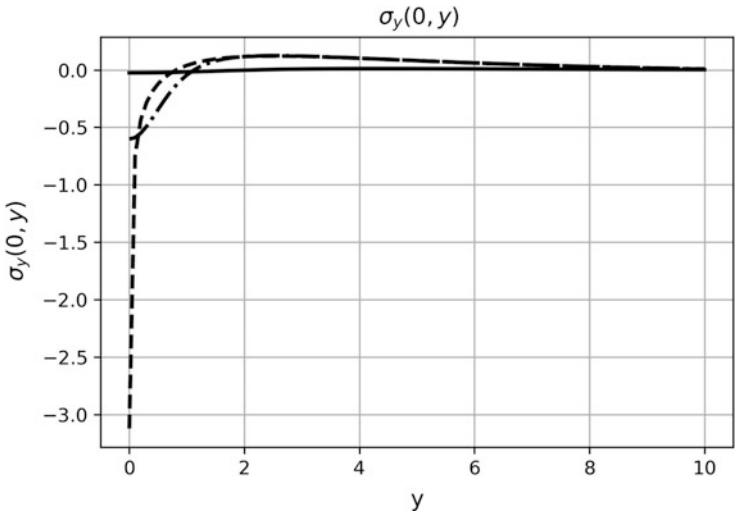


Fig. 4.3 Normal stresses $\sigma_y(0, y)$, $0 < y < a$ under the different values of a_0, a_1

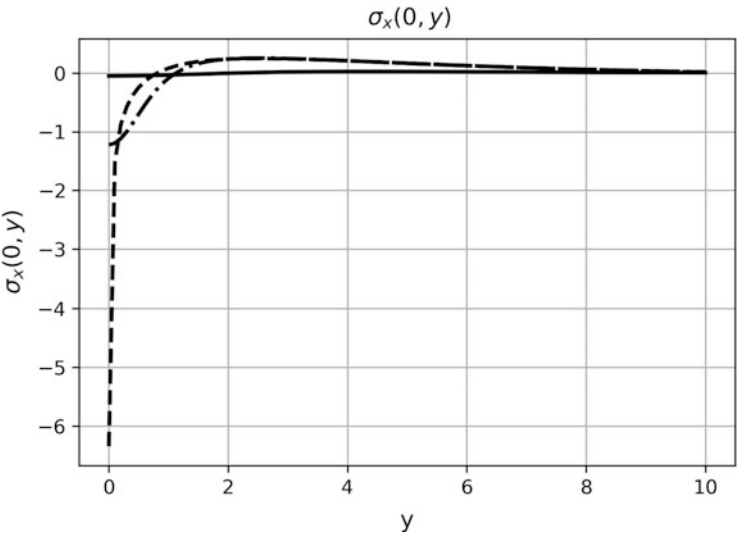


Fig. 4.4 Normal stresses $\sigma_x(0, y)$, $0 < y < a$ under the different values of a_0, a_1

4.3 The Load at the Left Part of the Semi-strip's Short Edge (Case 2)

The semi-strip, which is loaded at the left part of its short edge $y = 0, \quad 0 < x < a_1$, $a_1 \neq a$ is considered (Fig. 4.5)

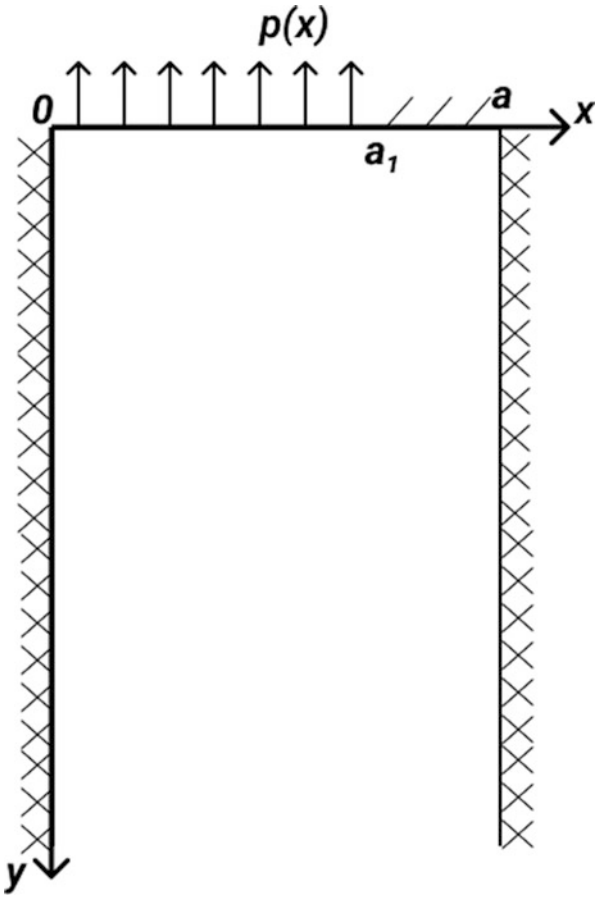


Fig. 4.5 Geometry and coordinate system of the semi-strip for the conditions of fixing. Case 2

$$\sigma_y\big|_{y=0} = p(x), \quad \tau_{xy}\big|_{y=0} = 0, \quad 0 < x < a_1 \tag{4.16}$$

At the line $y = 0, \quad a_1 < x < a$ the conditions of the slide contact are fulfilled

$$v\big|_{y=0} = 0, \quad \tau_{xy}\big|_{y=0} = 0, \quad a_1 < x < a. \tag{4.17}$$

The boundary conditions (4.7) and (4.8) can be rewritten in the displacement notation

$$2G\mu_0\left(\mu\frac{\partial u}{\partial x}\bigg|_{y=0} + (1-\mu)\frac{\partial v}{\partial y}\bigg|_{y=0}\right) = p(x), 0 < x < a_1, \tag{4.18}$$

$$v|_{y=0} = 0, \quad a_1 < x < a, \tag{4.19}$$

$$\left.\frac{\partial u}{\partial y}\right|_{y=0} + \left.\frac{\partial v}{\partial x}\right|_{y=0} = 0, \quad 0 < x < a \tag{4.20}$$

According to the scheme presented in Chap. 2, let's solve the boundary problem (3.4), (4.18)–(4.20) when $a_1 \neq a$ for the elastic semi-strip which is fixed by its lateral sides (4.3).

Formulae for the displacement functions have the form (4.4)–(4.5). Integral summands in (4.4) and (4.5) have the following form:

$$\int_0^a G^{i1}(x,\xi)\chi'(\xi)d\xi, \quad \int_0^a G^{i2}(x,\xi)\chi(\xi)d\xi, \quad i = 1,2,$$

with regard to the discontinuous properties of Green's function the second summand can be presented by the next formula

$$\int_0^a G^{i2}(x,\xi)\chi(\xi)d\xi = \int_0^{x-0} G^{i2}(x,\xi)\chi(\xi)d\xi + \int_{x+0}^a G^{i2}(x,\xi)\chi(\xi)d\xi =$$

$$= \Im^{i2}(x,x-0)\chi(x) - \Im^{i2}(x,0)\chi(0) - \int_0^{x-0} \Im^{i2}(x,\xi)\chi'(\xi)d\xi +$$

$$+ \Im^{i2}(x,a)\chi(a) - \Im^{i2}(x,x+0)\chi(x) - \int_{x+0}^a \Im^{i2}(x,\xi)\chi'(\xi)d\xi =$$

$$= \left\langle \Im^{i2}(x,x) \right\rangle \chi(x) - \int_0^a \Im^{i2}(x,\xi)\chi'(\xi)d\xi, i = 1,2,$$

where $\Im^{i2}(x,\xi)$ is the antiderivative of the function $G^{i2}(x,\xi)$, $\left\langle \Im^{12}(x,x) \right\rangle = 0$, $\left\langle \Im^{22}(x,x) \right\rangle = -\frac{\kappa-1}{\beta^2(\kappa+1)}$, $\chi(0) = \chi(a) = 0$.

As the result the correspondences (4.4)–(4.5) take the form

$$u_\beta(x) = Y_1^{11}(x)c_1 + Y_1^{12}(x)c_2 + Y_2^{11}(x)c_3 + Y_2^{12}(x)c_4 +$$
$$+\frac{3-\kappa}{\kappa+1}\int_0^a G^{11}(x,\xi)\chi'(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\int_0^a \Im^{12}(x,\xi)\chi'(\xi)d\xi,$$
$$v_\beta(x) = Y_1^{21}(x)c_1 + Y_1^{22}(x)c_2 + Y_2^{21}(x)c_3 + Y_2^{22}(x)c_4 +$$
$$+\frac{3-\kappa}{\kappa+1}\int_0^a G^{21}(x,\xi)\chi'(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\int_0^a \Im^{22}(x,\xi)\chi'(\xi)d\xi - \frac{\kappa-1}{\beta^2(\kappa+1)}\chi(x). \tag{4.21}$$

After the substitution of $Y_1(x)$, $Y_2(x)$, $c_i, i=\overline{1,4}$ obtained earlier into the formulae (4.21) and inverting of the derived transformants according to formulae (3.8), the following expressions for the displacement functions are obtained:

$$u(x,y) = \int_0^\infty \int_0^a f(x,\xi,\beta)\cos(\beta y)\chi'(\xi)d\xi d\beta, \tag{4.22}$$

$$v(x,y) = \int_0^\infty \int_0^a g(x,\xi,\beta)\sin(\beta y)\chi'(\xi)d\xi d\beta + \int_0^x \chi'(\xi)d\xi. \tag{4.23}$$

The displacements in the semi-strip will be finally determined by formulae (4.22)–(4.23), if the function $\chi'(x) = v'|_{y=0}$ is known. For its finding the conditions (4.18)–(4.19) which are not fulfilled yet should be fulfilled. The weakly-convergent integrals in the formulae for the displacements are summed up by the method presented in Appendix D.

From the condition (4.19)

$$\chi(\xi) = 0, a_1 < \xi < a \Rightarrow \chi'(\xi) = 0, a_1 < \xi < a. \tag{4.24}$$

The substitution of the displacement formulae into the boundary condition (4.18) reduces to SIE (3.26), the solving of which is described in Sect. 3.5.2. The substitution of the found coefficients $s_n, n=\overline{0,2N-1}$ into formulae (4.22)–(4.23) terminates the construction of the problem's solution.

In figures (Figs. 4.6, 4.7, 4.8 and 4.9) the plots for the normal stresses $\sigma_x(0,y)$, $\sigma_y(0,y)$, $\sigma_x(a,y)$, $\sigma_y(a,y)$ are shown when $G = 61.2781955 \cdot 10^9$ Pa, $\mu = 0.33$, $p(x) = 1$ Pa, $a = 10$ м, $\varpi = 40\ \%\ a$, $\varpi = 10\ \%\ a$ and $\varpi = 0.5\ \%\ a$.

As it can be seen, the normal stresses at the right lateral side are larger by its absolute values than the normal stresses at the left lateral side.

As it can be seen from the graphics, the normal stresses at the left lateral side are stretching when $0 < y < y_1$, then they become compressing and with the further increasing of y they decrease to zero. The normal stresses at the right lateral side are

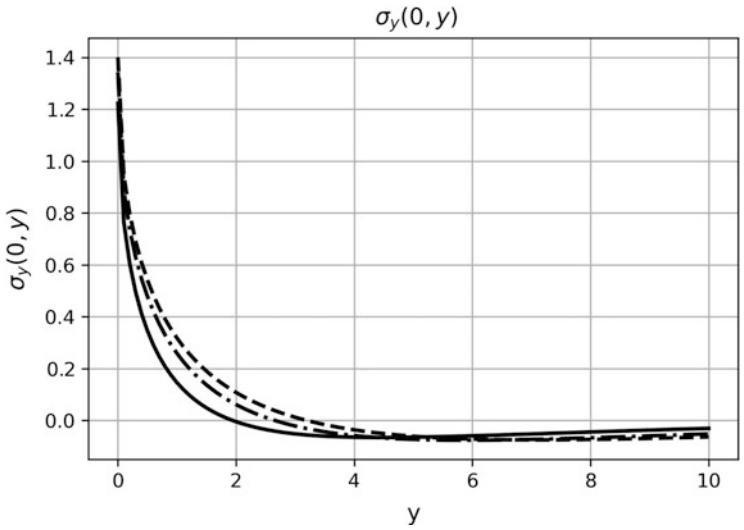


Fig. 4.6 Normal stresses $\sigma_y(0, y)$, $0 < y < a$ under the different values of a_1

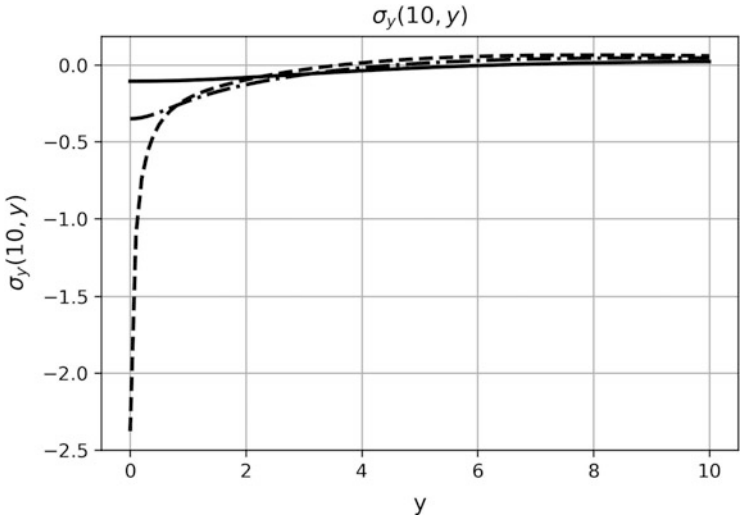


Fig. 4.7 Normal stresses $\sigma_y(a, y)$, $0 < y < a$ under the different values of a_1

compressing when $0 < y < y_2$, then they become stretching and with further increasing of y they drop to zero. Here, $y_1 = 2.1$, $y_2 = 6.4$ when $\varpi = 40\ \%\ a$, $y_1 = 3.0$, $y_2 = 4.3$ when $\varpi = 10\ \%\ a$, $y_1 = 3.3$, $y_2 = 3.8$ when $\varpi = 0.5\ \%\ a$. So, the increasing of the area when the load is applied the area of the stretching stresses at the left lateral side is slowly increasing, and the area of the compressing stresses at the right lateral side near the short edge is decreasing more

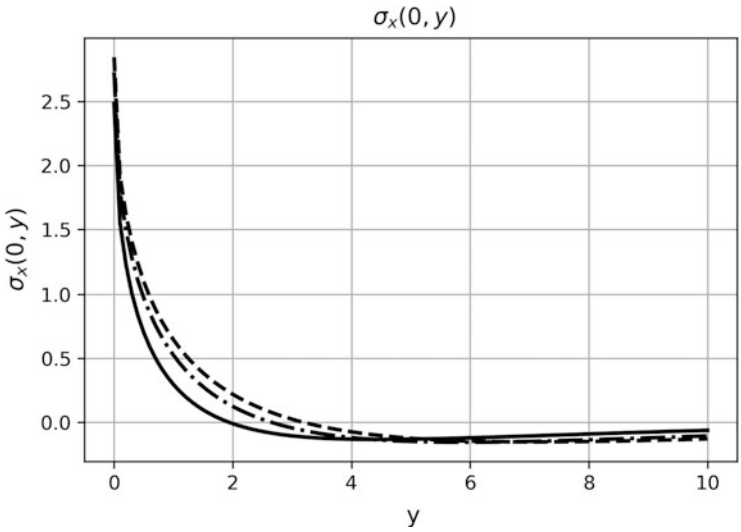


Fig. 4.8 Normal stresses $\sigma_x(0, y)$, $0 < y < a$ under the different values of a_1

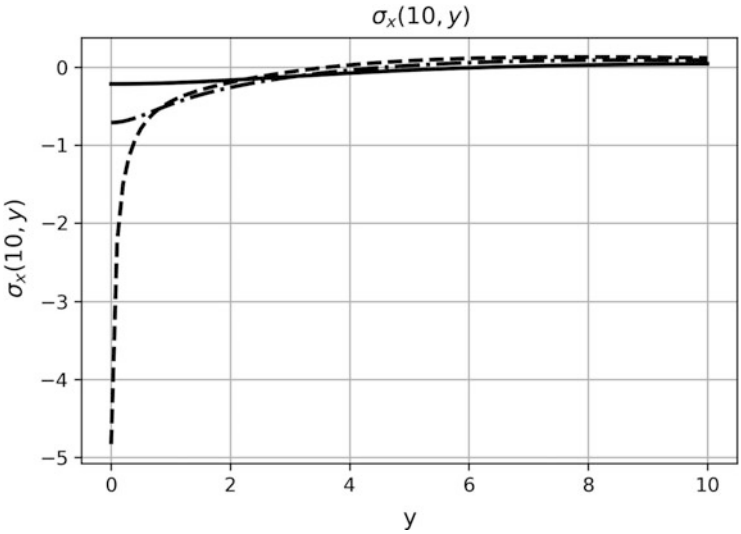


Fig. 4.9 Normal stresses $\sigma_x(a, y)$, $0 < y < a$ under the different values of a_1

than two times quickly than the area of stretching stresses is increasing at the left lateral side near the short edge.

The consideration of the fixed singularity at the point $x = 0$, $y = 0$ allowed to obtain stable calculations directly near this point. The increasing of the area when the load is applied occurs due to move near to the point $x = a$, $y = 0$ derived to the increasing of the absolute values of the normal stresses at the lateral side $x = a$

(Figs. 4.7 and 4.9). It weakly affected the change of the normal stresses at the lateral side $x = 0$ (Figs. 4.6 and 4.8). When $\varpi < 0.5\ \%\ a$ the influence of the fixed singularity in the equation's kernel starts at the point $x = a$, $y = 0$, and the calculations near this point become non-stable.

4.4 The Load at the Whole Semi-strip's Short Edge (Case 3)

The semi-strip, which is loaded at the whole short edge is considered (Fig. 4.10). Let's solve the boundary problem (3.4)–(3.6) for the elastic semi-strip with the conditions (3.2) at the lateral sides in the form (4.1)–(4.2) or (4.3).

The formulae for the displacement functions have form (4.22)–(4.23). These formulae will completely determine the displacements if the function $\chi'(\xi)$ is known. It is found by the fulfillment of the condition (3.5), which is not fulfilled

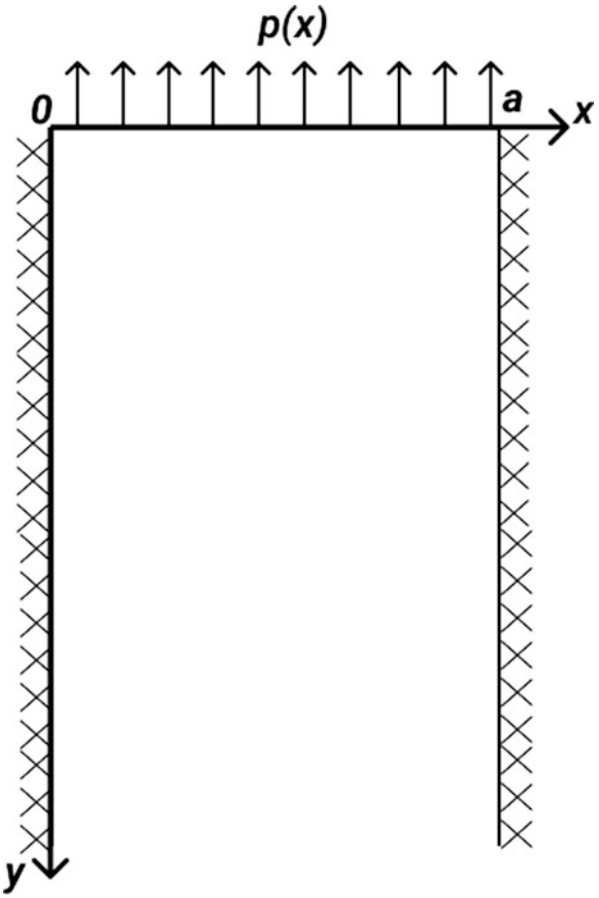


Fig. 4.10 Geometry and coordinate system of the semi-strip for the conditions of fixing. Case 3

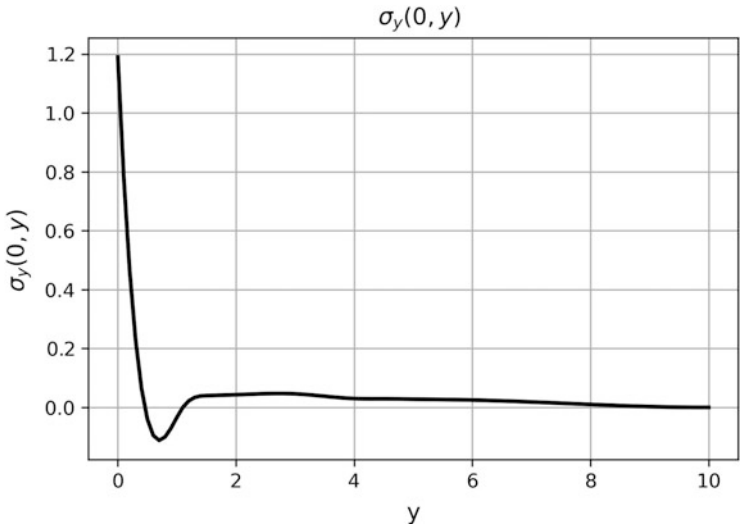


Fig. 4.11 Normal stresses $\sigma_y(0, y)$, $\sigma_y(a, y)$, $0 < y < a$

yet. This conditions cannot be fulfilled by the direct substitution of the correspondences (4.22)–(4.23) into it, because the integrals in the formulae for the displacements are weakly-convergent. So, the method presented in appendix D is used for separation of its weak-convergent parts.

The further substitution of the displacement formulae into the boundary condition (3.5) reduced to SIE (3.33). The construction of the problem's solution is completed by the substitution of the found constants $s_n, n = \overline{0, 2N-1}$ into formulae (4.22)–(4.23).

The graphics for normal stresses $\sigma_x(0, y)$, $\sigma_y(0, y)$ when $G = 61.2781955 \cdot 10^9$ Pa, $\mu = 0.33$, $p(x) = 1$ Pa, $a = 10$ m are shown in figures (Figs. 4.11 and 4.12).

As it can be seen from the plots, the normal stresses at the lateral sides are stretching when $0 < y < 1.1$, then they become compressing and with the further increasing of y they decrease to zero.

The consideration of the both fixed singularities allowed to obtain stable calculations at whole short edge of the semi-strip.

4.5 Conclusions to the Fourth Chapter

The solution of the plane problems of elasticity for the semi-strip which is fixed by two semi-infinite lateral sides is derived for three configurations of the applied load at the semi-strip's short edge.

The consideration of fixed singularities at the corner points of the semi-strip allowed to get calculations directly near these points, which is 0.5 % a closer than

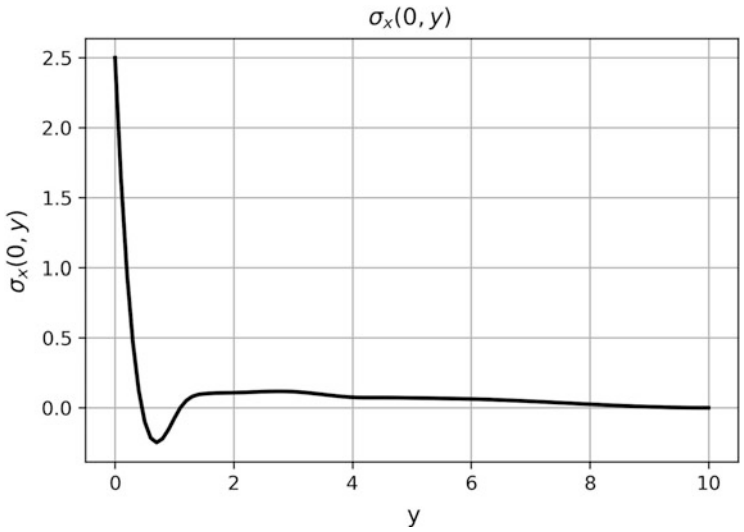


Fig. 4.12 Normal stresses $\sigma_x(0, y)$, $\sigma_x(a, y)$, $0 < y < a$

without their consideration. So the boundaries of the efficient implementations of the orthogonal polynomials method for the solving of the singular integral equation are found. Also, the maximal normal stresses at the lateral sides raise when the load is applied to the center of the semi-strip's short edge.

The areas of the stretching stresses are found for the different configurations of the applied load.

Chapter 5
The Problems of Uncoupled Thermoelasticity for the Semi-strip



The plane problems of thermoelasticity for a semi-strip with the fixed semi-infinite sides for three different configurations of the applied load by the semi-strip's short edge are investigated. The stationary thermal conductivity problem for a semi-infinite strip was solved, the found thermal field is further used during the solving of the uncoupled thermoelasticity problem for a semi-infinite strip.

The numeric analysis of the stress distribution at the lateral sides and inside the semi-strip was conducted in regard to the parameters of the stress and temperature distribution area.

5.1 The Stationary Thermal Conductivity Problem for the Semi-strip

The elastic semi-strip, the lateral sides of which $x = 0, 0 < y < \infty, x = a, 0 < y < \infty$ are thermoisolated, and the short edge $0 < x < a$, $y = 0$ is under the thermal load is considered. The semi-strip's temperature which satisfies the boundary problem should be found

$$\begin{aligned} &\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0, \qquad 0 < x < a, \;\; 0 < y < \infty \\ &T(x, 0) = f_T(x), \qquad 0 < x < a \\ &\frac{\partial T}{\partial x}(0, y) = 0, \; \frac{\partial T}{\partial x}(a, y) = 0, \qquad 0 < y < \infty \end{aligned} \tag{5.1}$$

The cosine Fourier transformation is applied over the variable x to reduce the problem (5.1) to one-dimensional boundary problem

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$$T_\alpha(y) = \int_0^a T(x,y)\cos\alpha x dx$$

with the inverse formula

$$T(x,y) = \frac{1}{a}\left[T_0(y) + 2\sum_{k=1}^{\infty} T_\alpha(y)\cos\alpha x\right], \alpha = \frac{\pi k}{a} \tag{5.2}$$

The one-dimensional boundary problem is derived

$$\begin{cases} T_\alpha''(y) - \alpha^2 T_\alpha(y) = 0, \\ T_\alpha(0) = f_\alpha, T_\alpha(y), \dfrac{dT_\alpha}{dy} \to 0, y \to \infty, \end{cases} \tag{5.3}$$

where $f_\alpha = \int_0^a f_T(x)\cos\alpha x dx$.

The semi-infinite sine Fourier transformation over the variable y is applied to the problem (5.3)

$$T_{\alpha\beta} = \int_0^\infty T_\alpha(y)\sin\beta y dy$$

with the inverse formula

$$T_\alpha(y) = \frac{2}{\pi}\int_0^\infty T_{\alpha\beta}\sin\beta y d\beta. \tag{5.4}$$

The algebraic equation in the transformation domain with regard to the displacements' transformants was obtained and solved

$$T_{\alpha\beta} = \frac{\beta}{\alpha^2+\beta^2} f_\alpha. \tag{5.5}$$

Let's invert (5.5) according to (5.4) and use formula 3.723(3) [52]. The formula $T_\alpha(y) = f_\alpha e^{-\alpha y}$ is obtained.

The found expression is inversed by formula (5.2)

$$T(x,y)=\frac{1}{a}\left[\int_0^a f_T(\xi)d\xi+2\sum_{k=1}^{\infty}\int_0^a f(\xi)\cos\alpha\xi d\xi e^{-\alpha y}\cos\alpha x\right]. \tag{5.6}$$

Consider two cases separately.

The first case $y>0$.

According to the formula for the series's summation 1.461(2) [52], the expression (5.6) can be rewritten

$$T(x,y)=\frac{1}{2a}\int_0^a f_T(\xi)\left[\frac{sh\frac{\pi y}{a}}{ch\frac{\pi y}{a}-\cos\frac{\pi(x+\xi)}{a}}+\frac{sh\frac{\pi y}{a}}{ch\frac{\pi y}{a}-\cos\frac{\pi(x-\xi)}{a}}\right]d\xi.$$

The solution of the problem (5.1) in the case $y>0$ can be written by the formula

$$T(x,y)=\frac{1-e^{-\frac{2\pi y}{a}}}{2a}\int_0^a f_T(\xi)[\delta^+(y,x,\xi)+\delta^-(y,x,\xi)]d\xi, 0\le x\le a, 0<y<\infty, \tag{5.7}$$

where $\delta^{\pm}(y,x,\xi)=\frac{1}{1+e^{-\frac{2\pi y}{a}}-2e^{-\frac{\pi y}{a}}\cos\frac{\pi(x\pm\xi)}{a}}$.

The second case $y=0$.

The expression (5.6) in this case can be rewritten

$$T(x,0)=\frac{1}{a}\int_0^a f_T(\xi)d\xi+\frac{2}{a}\sum_{k=1}^{\infty}\int_0^a f_T(\xi)\cos\frac{\pi k\xi}{a}d\xi\cos\frac{\pi kx}{a}. \tag{5.8}$$

The function $f_T(x)$ is extended in an even way for an interval $[-a;0]$ and the Fourier expansion for the function $f_T(x)$ is written

$$f_T(x)\sim\frac{1}{a}\int_0^a f_T(\xi)d\xi+\sum_{k=1}^{\infty}\frac{2}{a}\int_0^a f_T(\xi)\cos\frac{\pi k\xi}{a}d\xi\cos\frac{\pi kx}{a}, \tag{5.9}$$

where $a_0=\frac{1}{2a}\int_{-a}^{a}f_T(\xi)d\xi=\frac{1}{a}\int_0^a f_T(\xi)d\xi$, $a_n=\frac{1}{a}\int_{-a}^{a}f_T(\xi)\cos\frac{\pi k\xi}{a}d\xi=\frac{2}{a}\int_0^a f_T(\xi)\cos\frac{\pi k\xi}{a}d\xi$.

As it can be seen, the right-hand sides in (5.8) and (5.9) are equal, that is the function $T(x,0)$ represents Fourier series for the function $f_T(x)$.

As it follows from the sign of Lipschitz the Fourier series for the Helder at the point x_0 function $f_T(x)$ converges at this point to the sum $f_T(x_0)$ if the function $f_T(x)$ is continuous at this point. In this case the boundary condition $T(x,0) = f_T(x),\ 0 < x < a$ will be fulfilled on the interval $0 < x < a$ if the function $f_T(x)$ is continuous on this interval and it satisfies Helder condition.

The use of formulae (5.7) and (5.8) allows to obtain the solution of the problem (5.1) under conditions of continuity and Helder condition for the function $f_T(x)$ on the interval $0 < x < a$

$$T(x,y) = \begin{cases} \dfrac{1-e^{-\frac{2\pi y}{a}}}{2a} \displaystyle\int_0^a f_T(\xi)[\delta^+(y,x,\xi) + \delta^-(y,x,\xi)]d\xi, 0 \le x \le a, 0 < y < \infty \\ \dfrac{1}{a} \displaystyle\int_0^a f_T(\xi)d\xi + \sum_{k=1}^{\infty} \dfrac{2}{a} \int_0^a f_T(\xi) \cos\dfrac{\pi k\xi}{a} d\xi \cos\dfrac{\pi kx}{a}, 0 \le x \le a, y = 0 \end{cases} \tag{5.10}$$

5.2 The Mixed Problem of Thermoelasticity for a Semi-strip Under the Conditions of the Second Main Problem of Elasticity

The elastic semi-strip (Fig. 5.1), which occupies the area in Cartesian coordinates $0 < x < a$, $0 < y < \infty$ is considered. The semi-strip is loaded by its short edge $y = 0, \quad 0 < x < a$ by the mechanical and thermal loads (3.5)–(3.6).

The displacements in the semi-strip satisfy Lame's equilibrium equations (3.4). The generalized boundary conditions on the lateral sides (3.2) are chosen as (4.3). The displacement and stress field inside the semi-strip and on its boundaries should be found.

Lame's equilibrium equations (3.4) with consideration of the volume force's representation can be rewritten as

$$\begin{cases} \dfrac{\partial^2 u(x,y)}{\partial x^2} + \dfrac{\kappa-1}{\kappa+1}\dfrac{\partial^2 u(x,y)}{\partial y^2} + \dfrac{2}{\kappa+1}\dfrac{\partial^2 v(x,y)}{\partial x \partial y} = \dfrac{\kappa-1}{\kappa+1}\rho\dfrac{\partial T}{\partial x}, \\ \dfrac{\partial^2 v(x,y)}{\partial x^2} + \dfrac{\kappa+1}{\kappa-1}\dfrac{\partial^2 v(x,y)}{\partial y^2} + \dfrac{2}{\kappa-1}\dfrac{\partial^2 u(x,y)}{\partial x \partial y} = \rho\dfrac{\partial T}{\partial y}, \end{cases} \tag{5.11}$$

where $\rho = 2\frac{\mu+1}{1-2\mu}\alpha_t$, α_t is the coefficient of linear expansion, $T(x,y)$ is presented by formula (5.10).

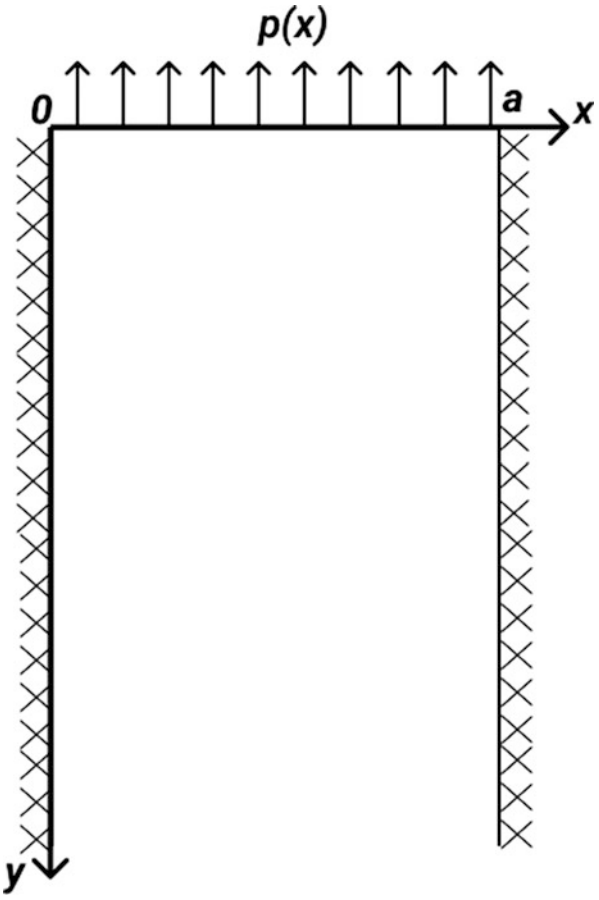


Fig. 5.1 Geometry and coordinate system of the semi-strip for conditions of fixing in the case of a temperature load

The integral Fourier transformation by the variable y is applied to (5.11), and the following system of equations in the transformation domain is derived

$$\begin{aligned}
&\frac{d^2u_\beta(x)}{dx^2} - \frac{\beta^2(\kappa\text{-}1)}{\kappa+1}u_\beta(x) + \frac{2\beta}{\kappa+1}\frac{dv_\beta(x)}{dx} = \frac{3-\kappa}{\kappa+1}\chi'(x) + \widetilde{\rho}\frac{dT_\beta}{dx}(x),\\
&\frac{d^2v_\beta(x)}{dx^2} - \frac{\beta^2(\kappa+1)}{\kappa-1}v_\beta(x) - \frac{2\beta}{\kappa-1}\frac{du_\beta(x)}{dx} = -\beta\frac{\kappa+1}{\kappa-1}\chi(x) - \beta\rho T_\beta(x),
\end{aligned} \tag{5.12}$$

here, $T_\beta(x) = \int\limits_0^\infty T(x,y)\cos\beta y dy$, $\widetilde{\rho} = \rho\frac{\kappa-1}{\kappa+1}$, $\chi(x) = v|_{y=0}$, $\chi'(x) = v'|_{y=0}$.

The vector one-dimensional boundary problem of the structure (3.10) is constructed by the scheme from the Chap. 2, where the right-hand side has the form of the vector $\vec{f}(x) = \begin{pmatrix} \frac{3-\kappa}{\kappa+1}\chi'(x) + \widetilde{\rho}\frac{dT_\beta}{dx}(x) \\ -\beta\frac{\kappa+1}{\kappa-1}\chi(x) - \beta\rho T_\beta(x) \end{pmatrix}$. According to the

constructed solution (3.16), the formulae for the displacement functions have the following representation

$$\begin{aligned} u_\beta(x) &= Y_1^{11}(x)c_1 + Y_1^{12}(x)c_2 + Y_2^{11}(x)c_3 + Y_2^{12}(x)c_4 + \\ &+ \frac{3-\kappa}{\kappa+1}\int_0^a G^{11}(x,\xi)\chi'(\xi)d\xi + \widetilde{\rho}\int_0^a G^{11}(x,\xi)\frac{dT_\beta}{d\xi}(\xi)d\xi - \\ &- \beta\frac{\kappa+1}{\kappa-1}\int_0^a G^{12}(x,\xi)\chi(\xi)d\xi - \beta\rho\int_0^a G^{12}(x,\xi)T_\beta(\xi)d\xi, \end{aligned} \tag{5.13}$$

$$\begin{aligned} v_\beta(x) &= Y_1^{21}(x)c_1 + Y_1^{22}(x)c_2 + Y_2^{21}(x)c_3 + Y_2^{22}(x)c_4 + \\ &+ \frac{3-\kappa}{\kappa+1}\int_0^a G^{21}(x,\xi)\chi'(\xi)d\xi + \widetilde{\rho}\int_0^a G^{21}(x,\xi)\frac{dT_\beta}{d\xi}(\xi)d\xi - \\ &- \beta\frac{\kappa+1}{\kappa-1}\int_0^a G^{22}(x,\xi)\chi(\xi)d\xi - \beta\rho\int_0^a G^{22}(x,\xi)T_\beta(\xi)d\xi, \end{aligned} \tag{5.14}$$

where $G^{i,j}(x,\xi)$ is the element of Green's matrix-function in the ith row and the jth column.

The coefficients $c_i, i=\overline{1,4}$ are found and presented in Appendix C. They have form (C.2).

Further solving depends on the configuration of the load applied by the semi-strip's short edge.

5.3 The Load at the Center of the Semi-strip's Short Edge (Case 1)

The semi-strip, which is loaded at the center of its short edge (Fig. 5.2) is considered. The boundary problem (3.4), (4.9)–(4.11) for the elastic semi-strip with the conditions (4.3) on the semi-infinite lateral sides should be solved when $a_0 \neq 0, \quad a_1 \neq a$. The temperature $T(x,y)$ acts as well as the mechanical load when $x \in [a_0; a_1]$, i.e. $f_T(x,0) = 0$, $0 < x < a_0$, $a_1 < x < a$.

The formulae for the displacement functions have the following form (5.13)–(5.14). The integral summands in (5.13) and (5.14) are rewritten as follows:

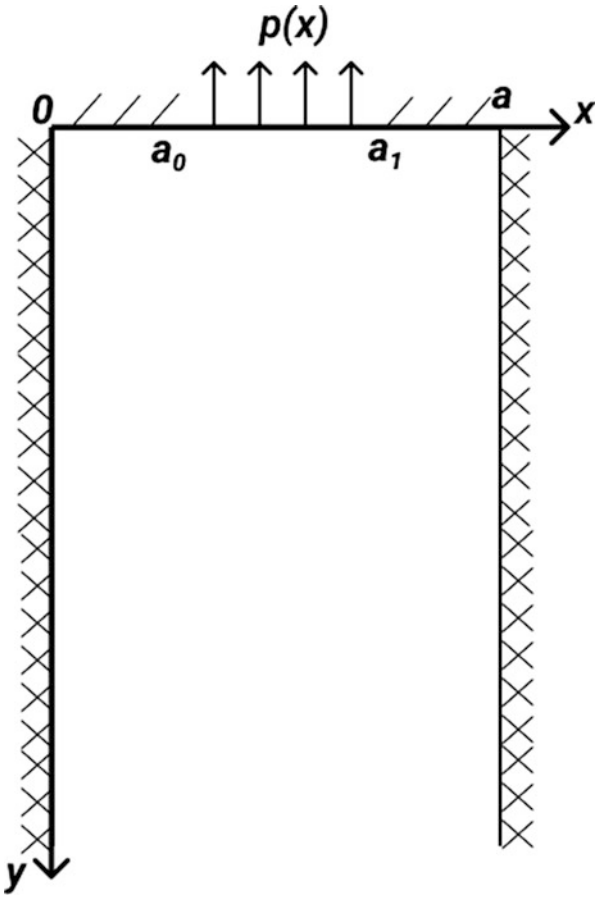


Fig. 5.2 Geometry and coordinate system of the semi-strip for conditions of fixing in the case of a temperature load. Case 1

$$\int_0^a G^{i1}(x,\xi)\chi'(\xi)d\xi,\ \int_0^a G^{i2}(x,\xi)\chi(\xi)d\xi,$$

$$\int_0^a G^{i1}(x,\xi)\frac{dT_\beta}{d\xi}(\xi)d\xi,\ \int_0^a G^{i2}(x,\xi)T_\beta(\xi)d\xi, i=1,2.$$

The third integral above can be transformed into the representation $\int_0^a G^{i1}(x,\xi)\frac{dT_\beta}{d\xi}(\xi)d\xi = -\int_0^a \frac{\partial G^{i1}}{\partial\xi}T_\beta(\xi)d\xi$ since $G^{i1}(x,a) = G^{i1}(x,0) = 0$, $i = 1, 2$.

As the result, the correspondences (5.13)–(5.14) are presented by the formulae:

$$
\begin{aligned}
u_\beta(x) &= Y_1^{11}(x)c_1 + Y_1^{12}(x)c_2 + Y_2^{11}(x)c_3 + Y_2^{12}(x)c_4 - \\
&- \frac{3-\kappa}{\kappa+1}\int\limits_0^a \frac{\partial}{\partial\xi}G^{11}(x,\xi)\chi(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int\limits_0^a G^{12}(x,\xi)\chi(\xi)d\xi - \\
&- \widetilde{\rho}\int\limits_0^a \frac{\partial G^{11}(x,\xi)}{\partial\xi}T_\beta(\xi)d\xi - \beta\rho\int\limits_0^a G^{12}(x,\xi)T_\beta(\xi)d\xi, \\
v_\beta(x) &= Y_1^{21}(x)c_1 + Y_1^{22}(x)c_2 + Y_2^{21}(x)c_3 + Y_2^{22}(x)c_4 - \\
&- \frac{3-\kappa}{\kappa+1}\int\limits_0^a \frac{\partial}{\partial\xi}G^{21}(x,\xi)\chi(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int\limits_0^a G^{22}(x,\xi)\chi(\xi)d\xi - \\
&- \widetilde{\rho}\int\limits_0^a \frac{\partial G^{21}(x,\xi)}{\partial\xi}T_\beta(\xi)d\xi - \beta\rho\int\limits_0^a G^{22}(x,\xi)T_\beta(\xi)d\xi.
\end{aligned}
\tag{5.15}
$$

The found earlier formulae for the system of fundamental matrix solutions $Y_1(x)$, $Y_2(x)$, $c_i, i=\overline{1,4}$ are substituted into the expressions (5.15), and the derived transformants are inverted by formulae (3.8). The inverse formula for the temperature has the following representation:

$$
T(x,y) = \frac{2}{\pi}\int\limits_0^\infty T_\beta(x)\cos\beta y d\beta \tag{5.16}
$$

So, the formulae for the displacements can be written as follows:

$$
u(x,y) = \int\limits_0^\infty\int\limits_0^a \left[f_1(x,\xi,\beta)\chi(\xi) + f_2(x,\xi,\beta)\int\limits_0^\infty T(\xi,\eta)\cos\beta\eta d\eta\right]\cos(\beta y)d\xi d\beta, \tag{5.17}
$$

$$
v(x,y) = \int\limits_0^\infty\int\limits_0^a \left[g_1(x,\xi,\beta)\chi(\xi) + g_2(x,\xi,\beta)\int\limits_0^\infty T(\xi,\eta)\cos\beta\eta d\eta\right]\sin(\beta y)d\xi d\beta. \tag{5.18}
$$

The expressions (5.17)–(5.18) finally determine the displacements if the function $\chi(x) = v|_{y=0}$ will be known. For its finding the unfulfilled conditions (4.9)–(4.10) by the semi-strip's short edge should be satisfied. The method described in Appendix D is used for the separation of the weak-convergent parts in the correspondences (5.17)–(5.18).

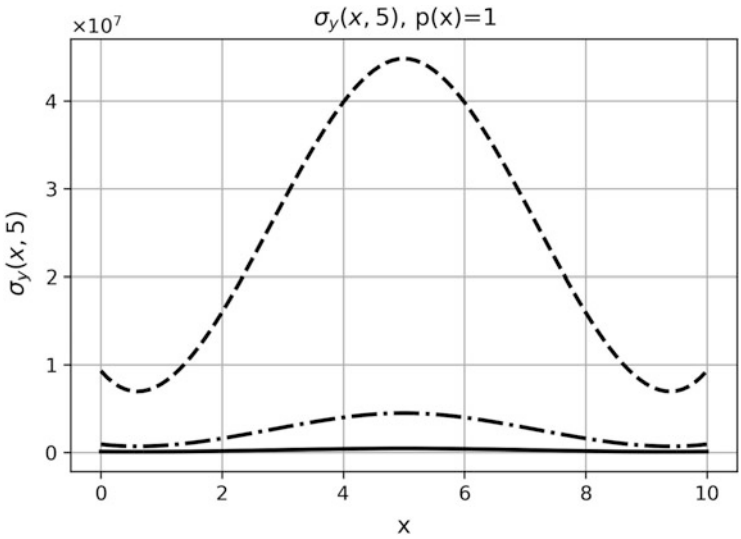


Fig. 5.3 The normal stresses $\sigma_y(x, 5)$, $0 < x < a$ under the unchanged mechanical load and the change of the temperature load

After this, the displacements' formulae were substituted into the boundary condition (4.9), which reduced to SIDE of the form (3.19). Its solving is described in Sect. 3.5.1. The substitution of the found constants s_n, $n = 0, 1, \ldots$ into formulae (5.17)–(5.18) terminates the construction of the problem's solution.

The dynamics of the stresses' $\sigma_y(x, y)$, $\sigma_x(x, y)$, $\tau_{xy}(x, y)$ changing is presented in Figs. 5.3, 5.4, 5.5, 5.6, 5.7, 5.8, 5.9 and 5.10 when $G = 61.2781955 \cdot 10^9$ Pa, $\mu = 0.33$, $\alpha_t = 1.1 \times 10^{-5}$, $a = 10$ m, $a_0 = a - a_1 = a/10$ for the different temperatures and mechanical load's values. In Figs. 5.3–5.6 the continuous, stroke-dotted, and dotted lines present the values of the temperature load $T(x, 0) = 0.1$, $T(x, 0) = 1$ and $T(x, 0) = 10$ correspondingly under the same mechanical load $p(x) = 1$. In Figs. 5.7–5.10 the continuous and stroke-dotted lines present the values of the mechanical load $p(x) = 1$ and $p(x) = 10$ accordingly under the same temperature load $T(x, 0) = 1$.

As it can be seen from the figures, the normal stresses under the fixed mechanical load and the increasement of the temperatures (Figs. 5.3–5.6) are dramatically increasing by its absolute values. Under the fixed applied temperature and the increasement of the mechanical load (Figs. 5.7–5.10), the normal stresses are far less changed. So, the influence of the temperature load is essential for the stressed state of the semi-strip.

The normal stresses at the lateral sides are compressive when $0 < y < 1.1$ both with the fixed mechanical load and the change of the temperature and with the fixed applied temperature and the increasement of the mechanical load. When $y > 1.1$, the stresses become stretching and with the further increasing of y they decrease to zero. So, it can be concluded that the value of the temperature load does not influence on

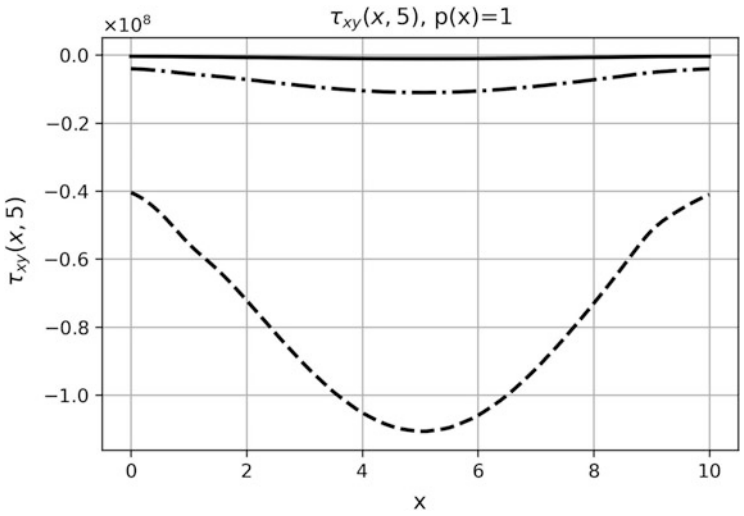


Fig. 5.4 The tangential stresses $\tau_{xy}(x, 5)$, $0 < x < a$ under the unchanged mechanical load and the change of the temperature load

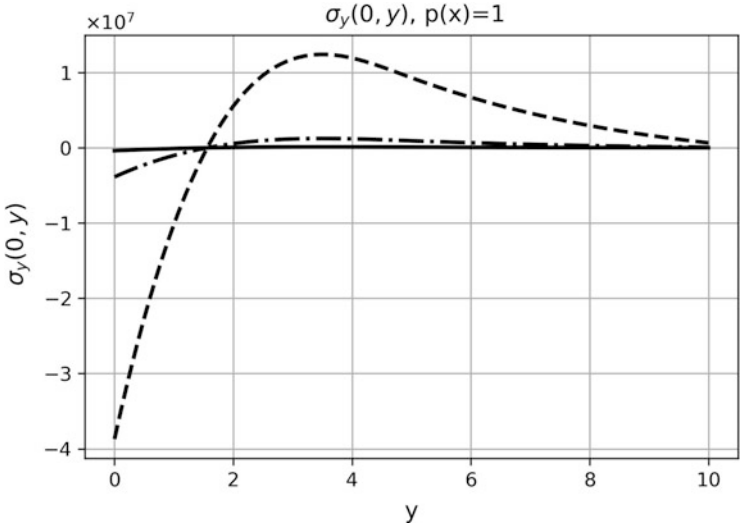


Fig. 5.5 The normal stresses $\sigma_y(0, y)$, $0 < y < 10$ under the unchanged mechanical load and the change of the temperature load

the size of the compressive stresses at the semi-strip's lateral sides near its short edge. In this case the value of the mechanical load creates a little effect on the change of the normal stress, so the size of the compressive stress does not change when the mechanical load is changing.

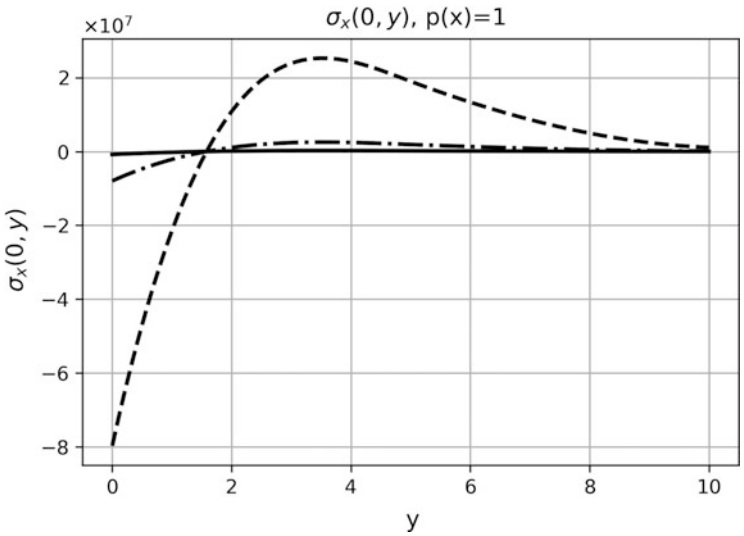


Fig. 5.6 The normal stresses $\sigma_x(0, y)$, $0 < y < 10$ under the unchanged mechanical load and the change of the temperature load

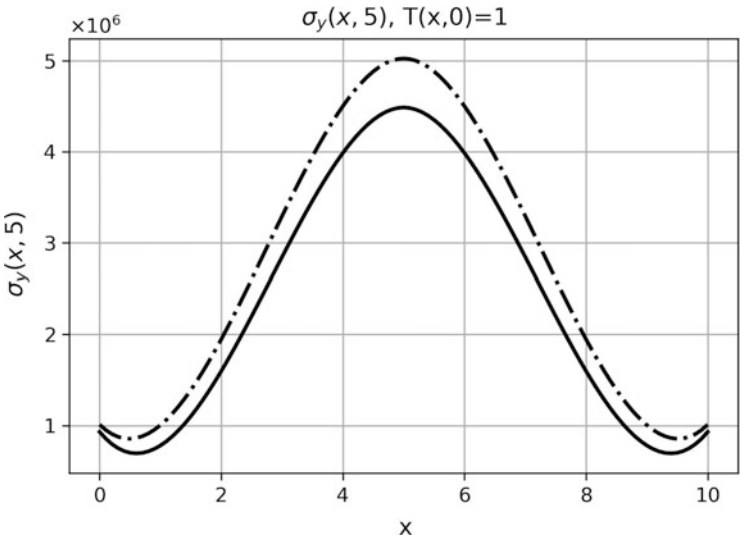


Fig. 5.7 The normal stresses $\sigma_y(x, 5)$, $0 < x < a$ under the unchanged temperature load and the change of the mechanical load

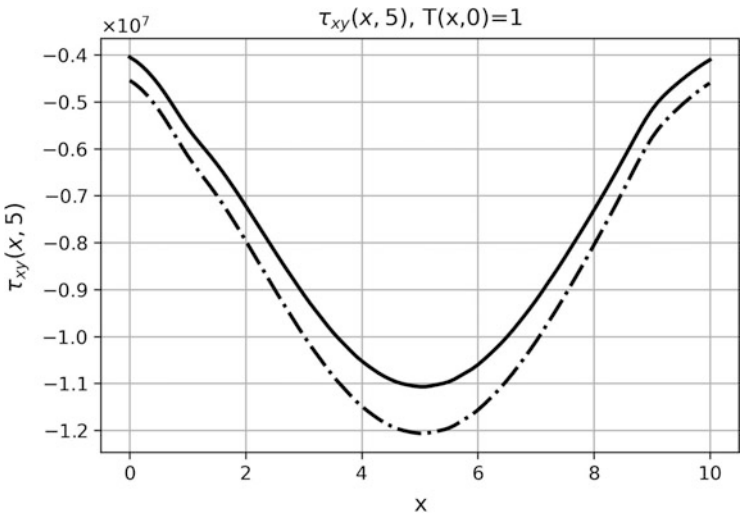


Fig. 5.8 The tangential stresses $\tau_{xy}(x, 4)$, $0 < x < a$ under the unchanged temperature load and the change of the mechanical load

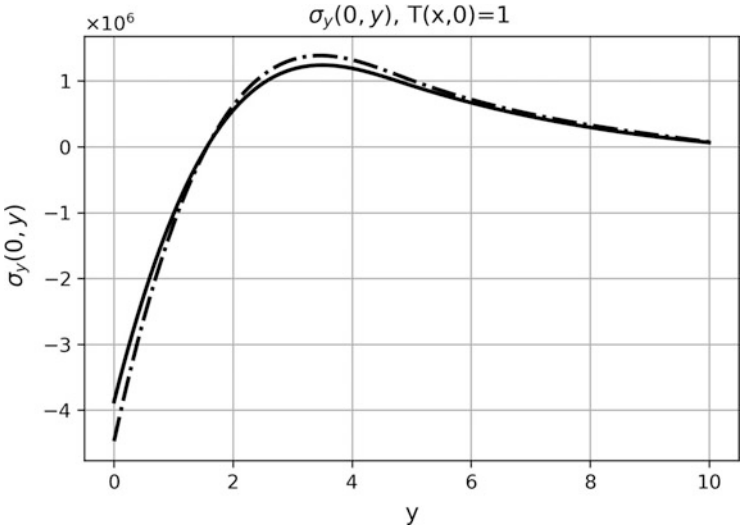


Fig. 5.9 The normal stresses $\sigma_y(0, y)$, $0 < y < 10$ under the unchanged temperature load and the change of the mechanical load

5.4 The Load at the Left Part of the Semi-strip's Short Edge (Case 2)

The semi-strip, which is loaded at the left part of its short edge (Fig. 5.11) is considered. The boundary problem (4.18)–(4.20) for the elastic semi-strip with the conditions (4.3) on the semi-infinite lateral sides should be solved when $a_1 \neq a$. The temperature $T(x, y)$ acts as well as the mechanical load when $x \in [0; a_1]$.

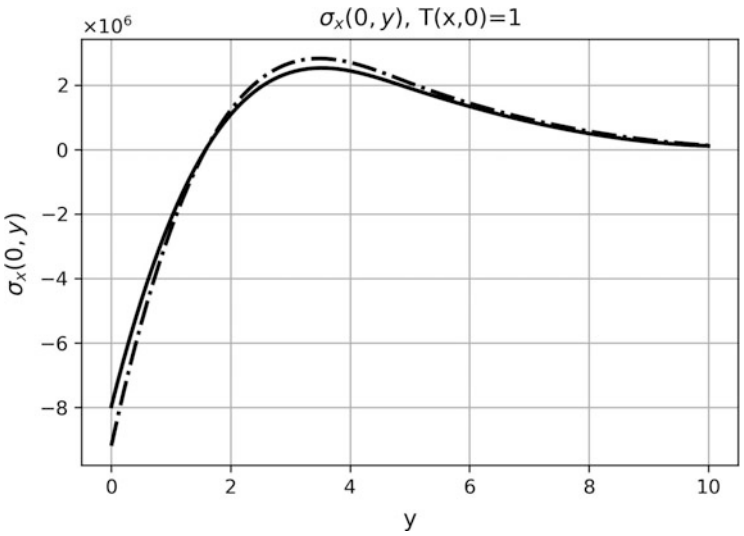


Fig. 5.10 The normal stresses $\sigma_x(0, y)$, $0 < y < 10$ under the unchanged temperature load and the change of the mechanical load

The formulae for the displacement functions have the following representations (5.13)–(5.14). According to formulae (4.21) and (5.15) the expressions for the displacement functions can be written as follows:

$$
\begin{aligned}
&u_\beta(x) = Y_1^{11}(x)c_1 + Y_1^{12}(x)c_2 + Y_2^{11}(x)c_3 + Y_2^{12}(x)c_4 + \\
&+\frac{3-\kappa}{\kappa+1}\int_0^a G^{11}(x,\xi)\chi'(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\int_0^a \mathfrak{I}^{12}(x,\xi)\chi'(\xi)d\xi - \\
&-\widetilde{\rho}\int_0^a \frac{\partial G^{11}(x,\xi)}{\partial\xi}T_\beta(\xi)d\xi - \beta\rho\int_0^a G^{12}(x,\xi)T_\beta(\xi)d\xi, \\
&v_\beta(x) = Y_1^{21}(x)c_1 + Y_1^{22}(x)c_2 + Y_2^{21}(x)c_3 + Y_2^{22}(x)c_4 + \\
&+\frac{3-\kappa}{\kappa+1}\int_0^a G^{21}(x,\xi)\chi'(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\int_0^a \mathfrak{I}^{22}(x,\xi)\chi'(\xi)d\xi - \frac{\kappa-1}{\beta^2(\kappa+1)}\chi(x) - \\
&-\widetilde{\rho}\int_0^a \frac{\partial G^{21}(x,\xi)}{\partial\xi}T_\beta(\xi)d\xi - \beta\rho\int_0^a G^{22}(x,\xi)T_\beta(\xi)d\xi.
\end{aligned}
\tag{5.19}
$$

The fundamental system of the matrix solutions is substituted into the expressions (5.19), and the obtained transformants are inverted by formulae (3.8), (5.16):

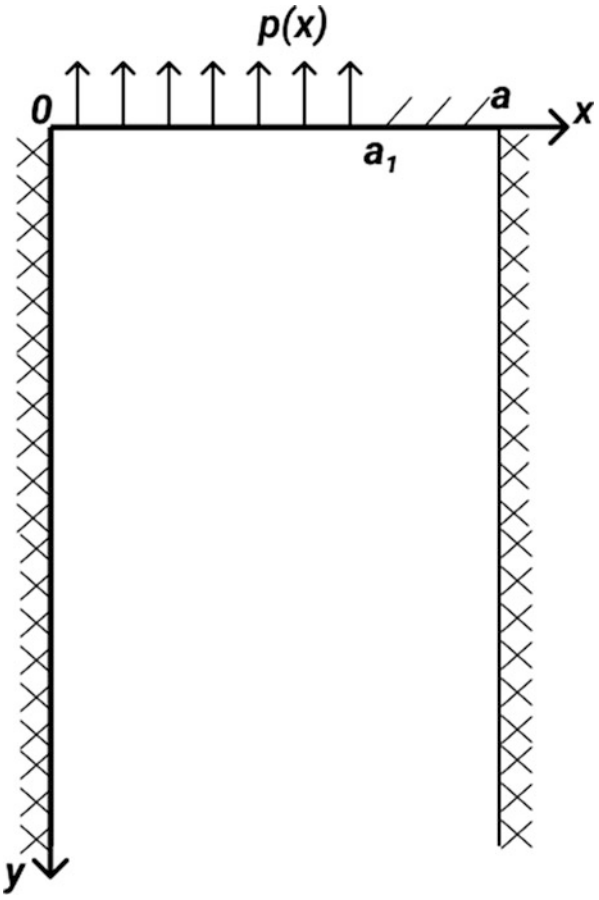


Fig. 5.11 Geometry and coordinate system of the semi-strip for conditions of fixing in the case of a temperature load. Case 2

$$u(x,y) = \int_0^{\infty} \int_0^{a} \left[f_1(x,\xi,\beta)\chi'(\xi) + f_2(x,\xi,\beta) \int_0^{\infty} T(\xi,\eta)\cos\beta\eta d\eta \right] \cos(\beta y) d\xi d\beta, \tag{5.20}$$

$$v(x,y) = \int_0^{\infty} \int_0^{a} \left[g_1(x,\xi,\beta)\chi'(\xi) + g_2(x,\xi,\beta) \int_0^{\infty} T(\xi,\eta)\cos\beta\eta d\eta \right] \sin(\beta y) d\xi d\beta + \\ + \int_0^{x} \chi'(\xi) d\xi. \tag{5.21}$$

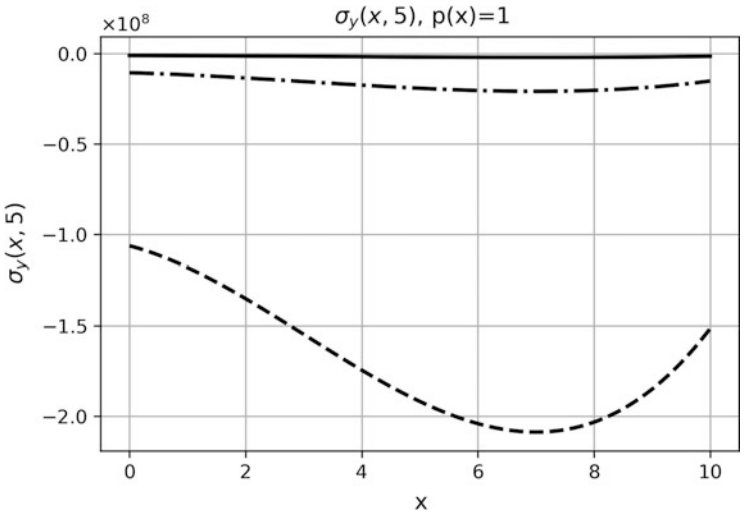


Fig. 5.12 The normal stresses $\sigma_y(x, 5)$, $0 < x < a$ under the unchanged mechanical load and the change of the temperature load

Formulae (5.20)–(5.21) finally determine the displacements if the function $\chi'(x) = v'|_{y=0}$ will be found. According to the methods presented in the previous chapter SIE is derived in the form (3.27). SIE contains the fixed singularity in the kernel. The method of this equation's solving is described in detail in Sect. 3.5.2, the solving of SIE is reduced to the solving of the system of linear algebraic equations (3.32). The substitution of the found constants $s_n, n = \overline{0, 2N-1}$ into formulae (5.20)–(5.21) finalizes the construction of the problem's solution.

In Figs. 5.12, 5.13, 5.14, 5.15, 5.16, 5.17, 5.18, 5.19, 5.20, 5.21, 5.22 and 5.23 the dynamics of the stresses' change $\sigma_y(x, y)$, $\sigma_x(x, y)$, $\tau_{xy}(x, y)$ is shown when $G = 61.2781955 \cdot 10^9$ Pa, $\mu = 0.33$, $\alpha_t = 1.1 \times 10^{-5}$, $a = 10$ m, $a - a_1 = a/10$ for the different temperatures and values of the mechanical stresses. In Figs. 5.12–5.17 the continuous, stroke-dotted, and dotted lines present the values of the temperature load $T(x, 0) = 0.1$, $T(x, 0) = 1$ and $T(x, 0) = 10$ accordingly under the unchanged mechanical load $p(x) = 1$. In Figs. 5.18–5.23 the continuous and stroke-dotted lines present the values of the mechanical load $p(x) = 1$ and $p(x) = 10$ correspondingly under the same temperature load $T(x, 0) = 1$.

The normal stresses under the fixed mechanical load and the increasing of the temperature (Figs. 5.12–5.17) are increasing significantly by its absolute values. In the case of the fixed applied temperature and the increasement of the mechanical load (Figs. 5.18–5.23) the normal stresses change far less. So, in this case the influence of the temperature load is significant as well as in the previous case.

The normal stresses on the left lateral side of the semi-strip are stretching when $0 < y < 1.1$, and on the right lateral side they are stretching when $0 < y < 1.5$ both

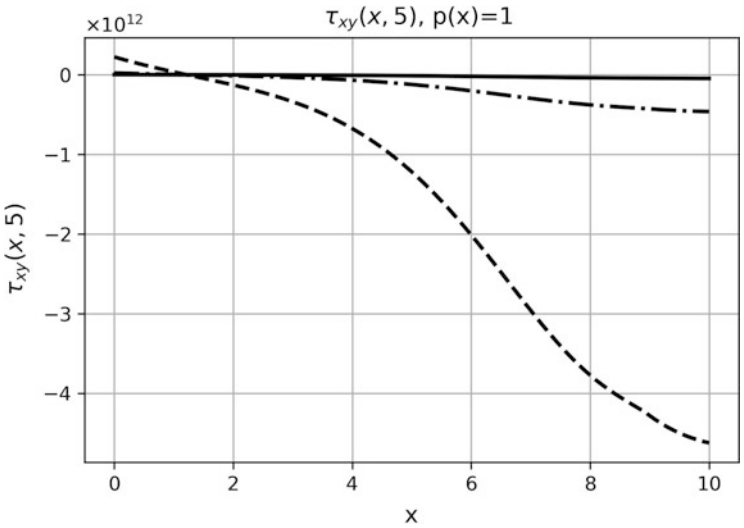


Fig. 5.13 The tangential stresses $\tau_{xy}(x, 5)$, $0 < x < a$ under the unchanged mechanical load and the change of the temperature load

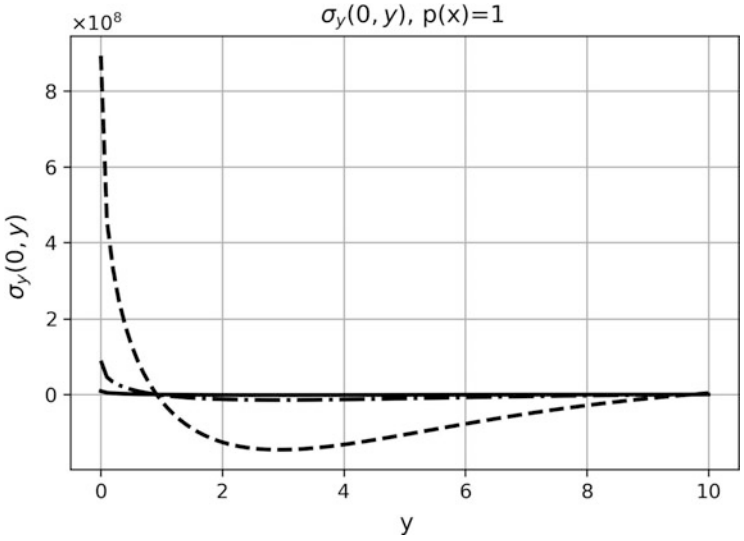


Fig. 5.14 The normal stresses $\sigma_y(0, y)$, $0 < y < 10$ under the unchanged mechanical load and the change of the temperature load

with the fixed mechanical load and the increasement of the temperature and with the fixed applied temperature and the increasing of the mechanical load, then they become compressive and with the further increasment of y they decrease to zero. So, as it can be seen, the value of the temperature load does not influence on the size

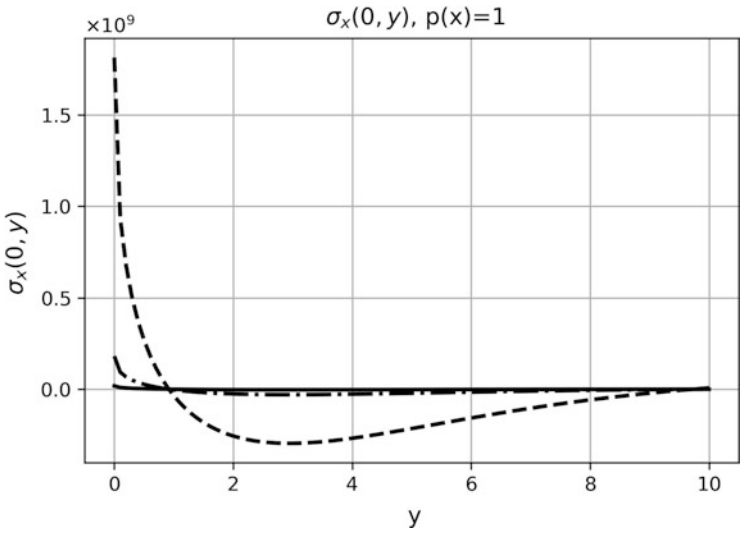


Fig. 5.15 The normal stresses $\sigma_x(0, y)$, $0 < y < 10$ under the unchanged mechanical load and the change of the temperature load

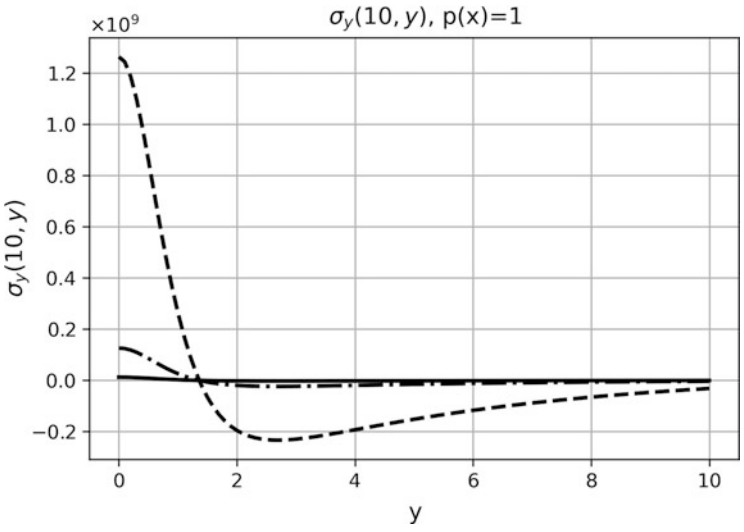


Fig. 5.16 The normal stresses $\sigma_y(a, y)$, $0 < y < 10$ under the unchanged mechanical load and the change of the temperature load

of the stretching stresses at the semi-strip's lateral sides near the short edge. In this case the value of the mechanical load slightly affects the change of the normal stresses, so the size of the stretching stresses' zone does not change when the mechanical load is changing.

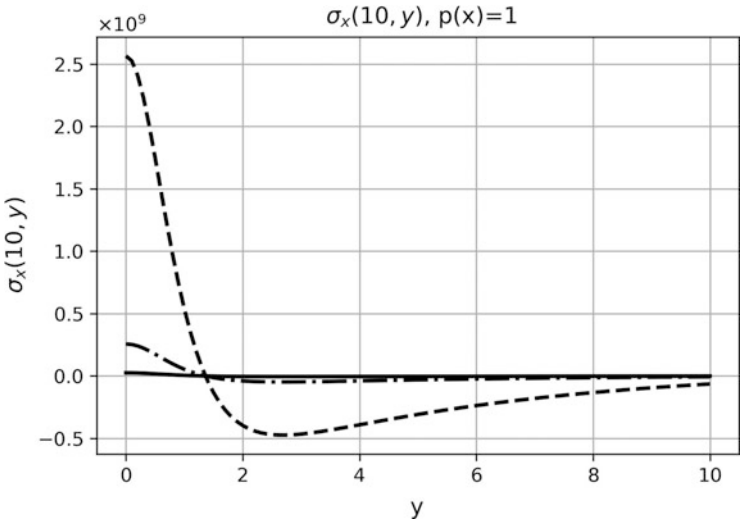


Fig. 5.17 The normal stresses $\sigma_x(a, y)$, $0 < y < 10$ under the unchanged mechanical load and the change of the temperature load

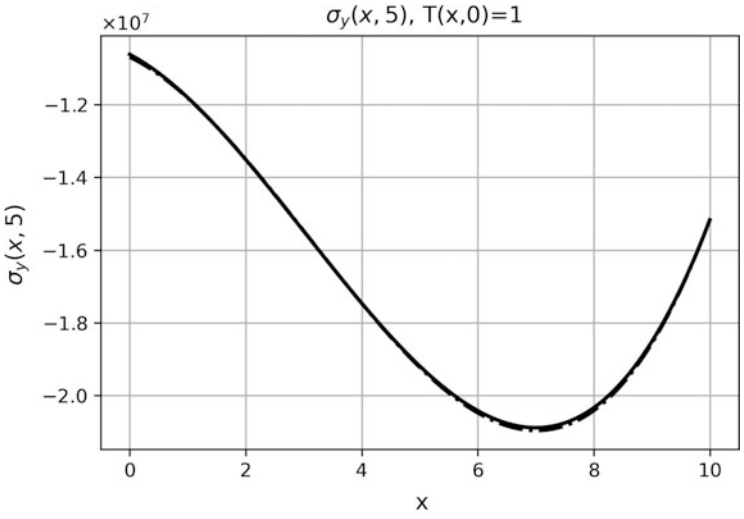


Fig. 5.18 The normal stresses $\sigma_y(x, 5)$, $0 < x < a$ under the unchanged temperature load and the change of the mechanical load

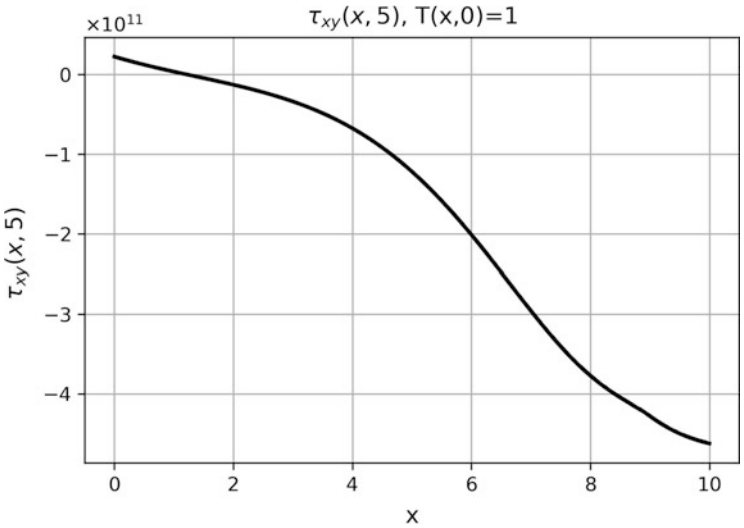


Fig. 5.19 The tangential stresses $\tau_{xy}(x, 5)$, $0 < x < a$ under the unchanged temperature load and the change of the mechanical load

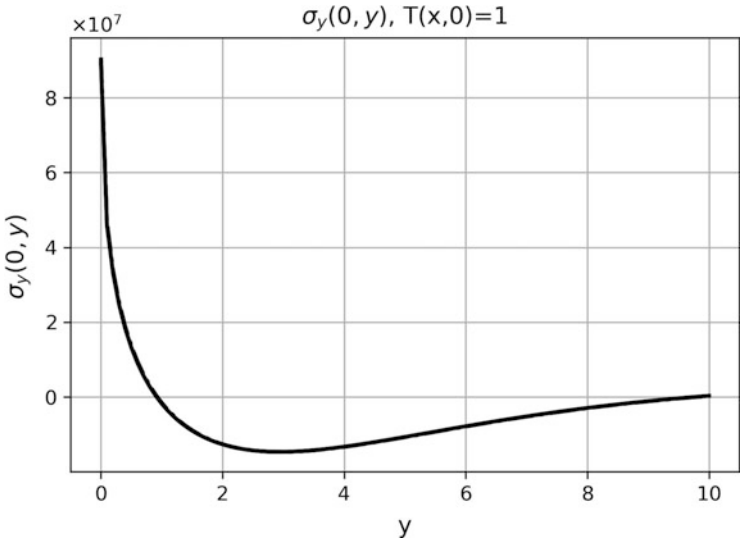


Fig. 5.20 The normal stresses $\sigma_y(0, y)$, $0 < y < 10$ under the unchanged temperature load and the change of the mechanical load

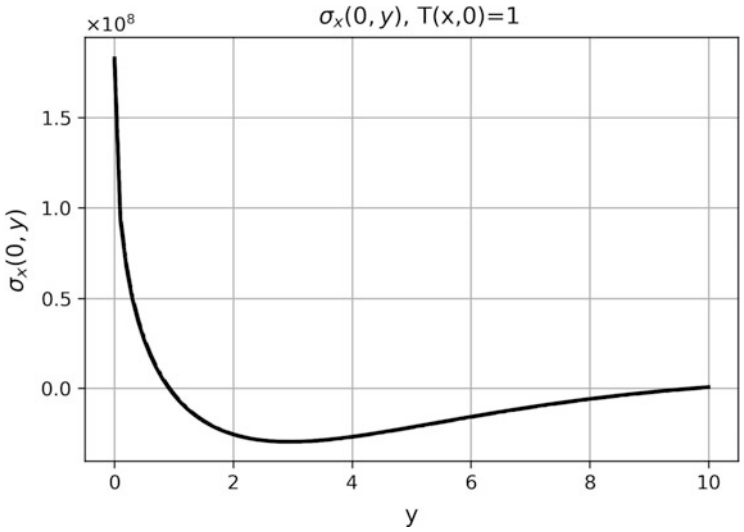


Fig. 5.21 The normal stresses $\sigma_x(0, y)$, $0 < y < 10$ under the unchanged temperature load and the change of the mechanical load

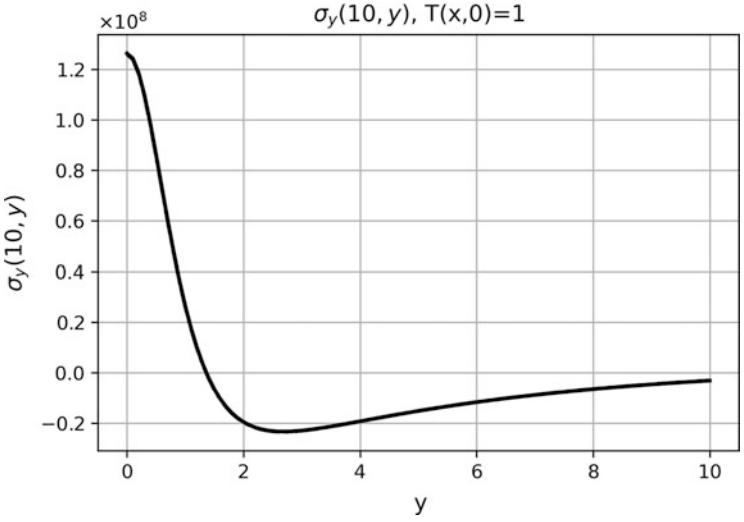


Fig. 5.22 The normal stresses $\sigma_y(a, y)$, $0 < y < 10$ under the unchanged temperature load and the change of the mechanical load

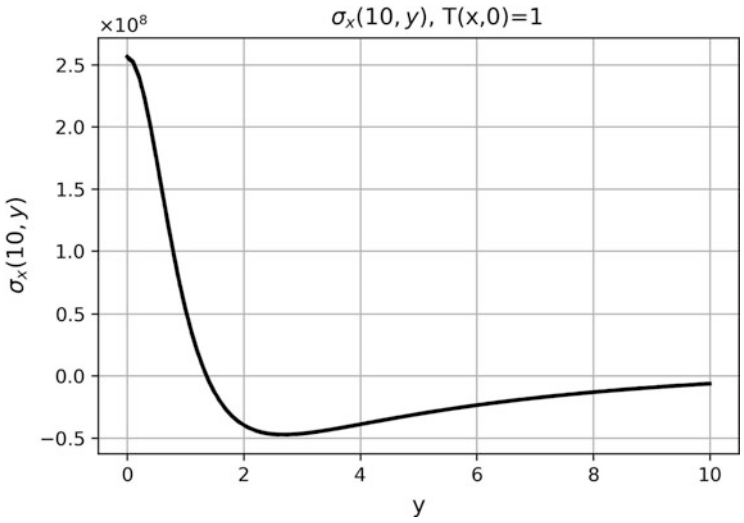


Fig. 5.23 The normal stresses $\sigma_x(a, y)$, $0 < y < 10$ under the unchanged temperature load and the change of the mechanical load

5.5 The Load at the Whole Semi-strip's Edge (Case 3)

The semi-strip, which is loaded at the whole short edge is considered (Fig. 5.24). The boundary problem (3.4)–(3.6) for the elastic semi-strip with the conditions (4.3) should be solved. The temperature $T(x, y)$ acts as well as the mechanical load, i.e. when $x \in [0; a]$.

The correspondences for the displacements have the form (5.20)–(5.21). These formulae will finally determine displacements if the function $\chi'(\xi)$ is known. According to the methods presented in the previous chapter SIE, which contains two fixed singularities, is derived in the form (3.33). Its solving is presented in Sect. 3.5.3. The substitution of the found constants $s_n, n = \overline{0, 2N-1}$ into formulae (5.20)–(5.21) terminates the construction of the problem's solution.

The dynamics of the stresses' change $\sigma_y(x, y)$, $\sigma_x(x, y)$, $\tau_{xy}(x, y)$ is shown in Figs. 5.25, 5.26, 5.27, 5.28, 5.29, 5.30, 5.31 and 5.32 when $G = 61.2781955 \cdot 10^9$ Pa, $\mu = 0.33$, $\alpha_t = 1.1 \times 10^{-5}$, $a = 10$ m for different temperatures and different values of the mechanical load. In Figs. 5.25–5.32 the continuous, stroke-dotted, and dotted lines present the values of the temperature load $T(x, 0) = 0.1$, $T(x, 0) = 1$ and $T(x, 0) = 10$ correspondingly when the mechanical load is unchanged $p(x) = 1$. In Figs. 5.29–5.32 the continuous and stroke-dotted lines present the values of the mechanical load $p(x) = 1$ and $p(x) = 10$ correspondingly when the temperature load is fixed $T(x, 0) = 1$.

The analysis of the figures shows that normal stresses under the fixed mechanical load and increasing of the temperature are dramatically increasing by its absolute values. Under the unchanged applied temperature and the mechanical load's

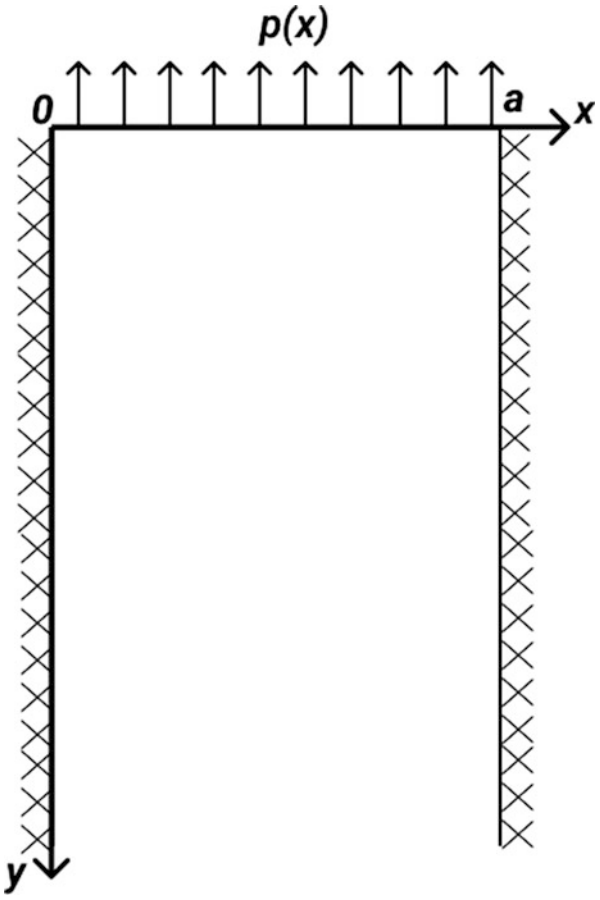


Fig. 5.24 Geometry and coordinate system of the semi-strip for conditions of fixing in the case of a temperature load. Case 3

increasment the normal stresses change slightly. So, in this case, as well as in two previous cases, the temperature influence is significant.

The normal stresses at the semi-strip's lateral sides under the fixed mechanical load and the temperature's increasing are stretching when $0 < y < 0.7$, they become compressive when $y > 0.7$, and with the further increasing of y they decrease to zero. The normal stresses at the lateral sides under the fixed applied temperature and the mechanical load's increasement are stretching when $0 < y < y_1$, then they become compressive and with the further increasing of y they decrease to zero. At that $y_1 = 0.7$ when $p(x) = 1$, $y_1 = 0.4$ when $p(x) = 10$. So, it can be concluded that the values of the temperature load, unlikely the mechanical load, does not affect the size of the compressive stresses' zone at the semi-strip's lateral sides near the short edge.

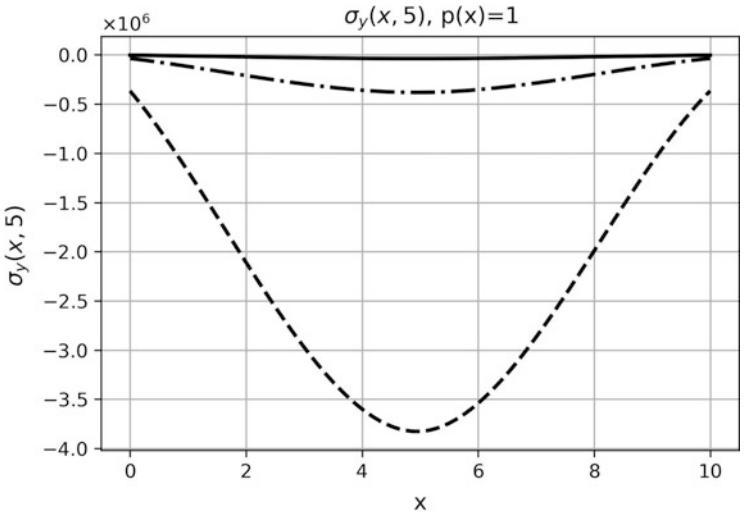


Fig. 5.25 The normal stresses $\sigma_y(x, 5)$, $0 < x < a$ under the unchanged mechanical load and the change of the temperature load

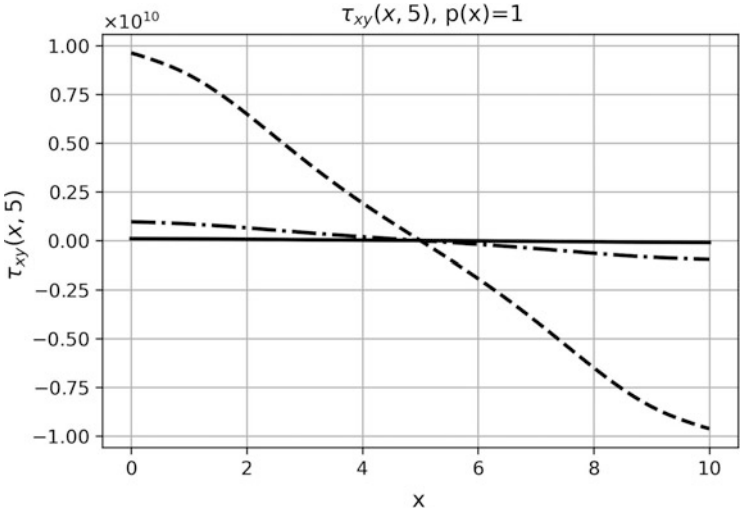


Fig. 5.26 The tangential stresses $\tau_{xy}(x, 5)$, $0 < x < a$ under the unchanged mechanical load and the change of the temperature load

5.6 Conclusions to the Fifth Chapter

The exact solution of the stationary thermal conductivity problem for the semi-strip is found. It was used during the uncoupled thermoelasticity problem's solving.

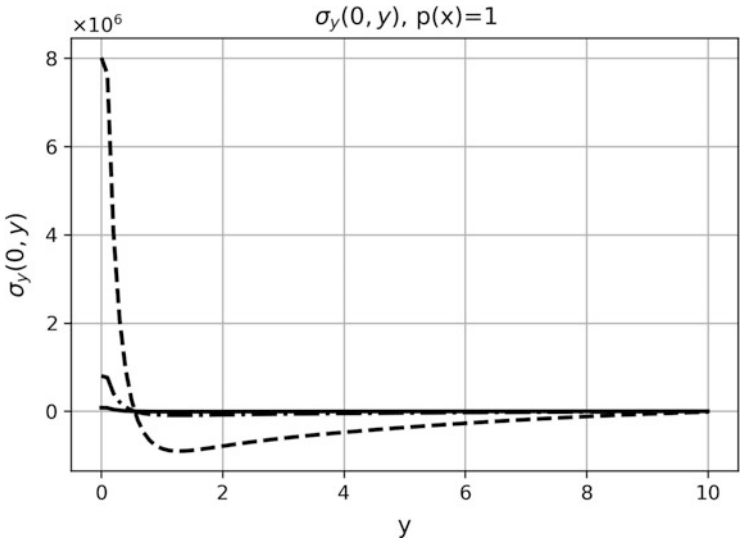


Fig. 5.27 The normal stresses $\sigma_y(0, y)$, $0 < y < 10$ under the unchanged mechanical load and the change of the temperature load

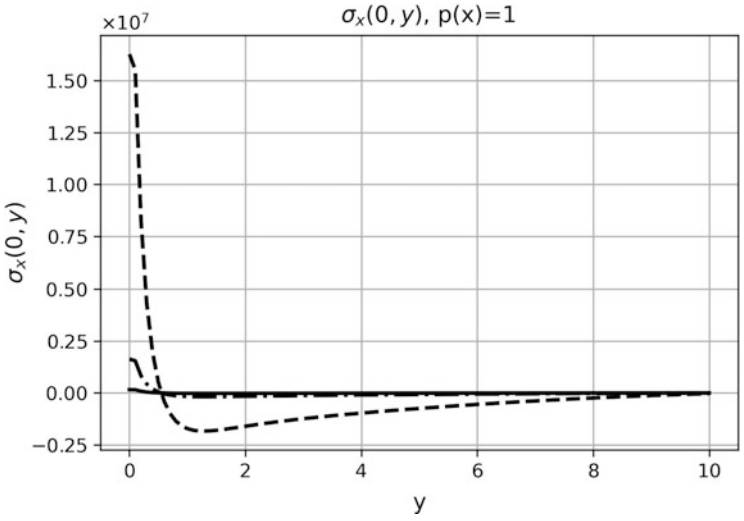


Fig. 5.28 The normal stresses $\sigma_x(0, y)$, $0 < y < 10$ under the unchanged mechanical load and the change of the temperature load

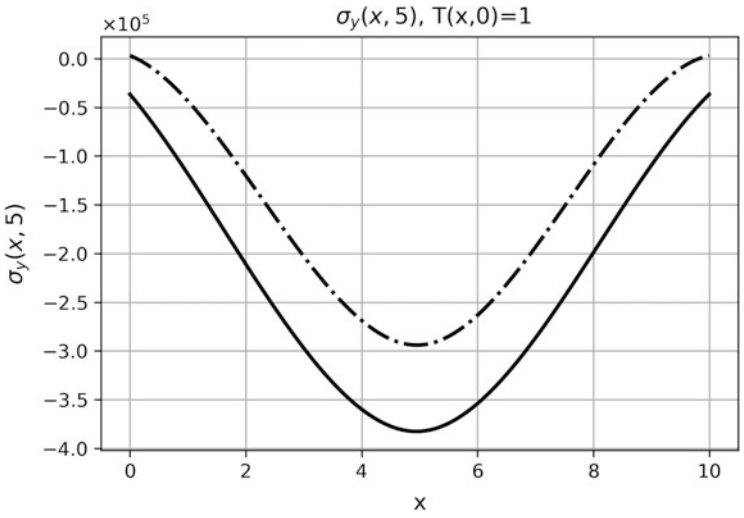


Fig. 5.29 The normal stresses $\sigma_y(x, 5)$, $0 < x < a$ under the unchanged temperature load and the change of the mechanical load

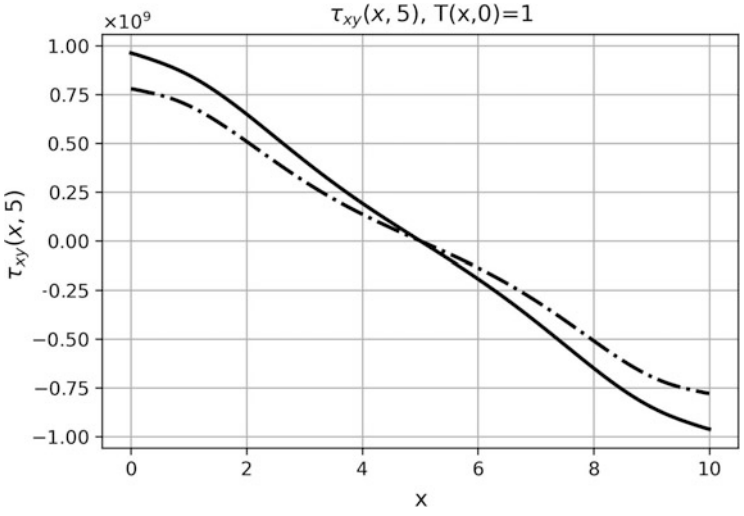


Fig. 5.30 The tangential stresses $\tau_{xy}(x, 5)$, $0 < x < a$ under the unchanged temperature load and the change of the mechanical load

The nature of the temperature influence regarding the value of the normal stresses at the semi-strip's lateral sides and inside the semi-strip is established. The most essentially the semi-strip's stress state is affected by the thermal load. The biggest stresses are observed at the semi-strip's lateral sides in the case when the load is

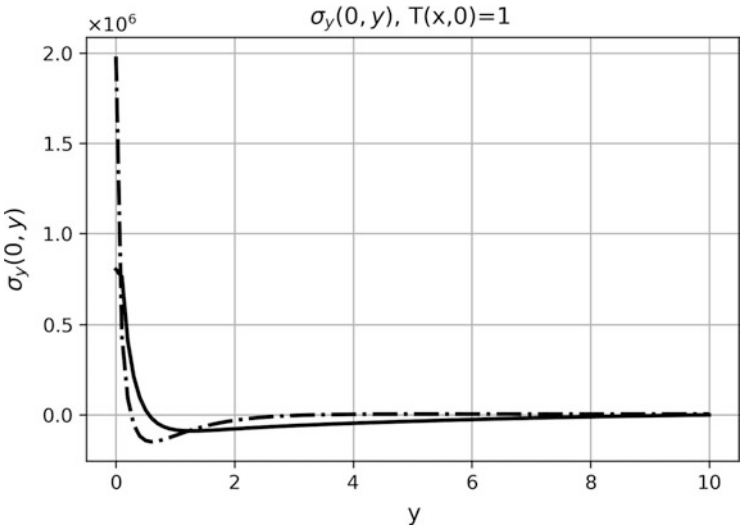


Fig. 5.31 The normal stresses $\sigma_y(0, y)$, $0 < y < 10$ under the unchanged temperature load and the change of the mechanical load

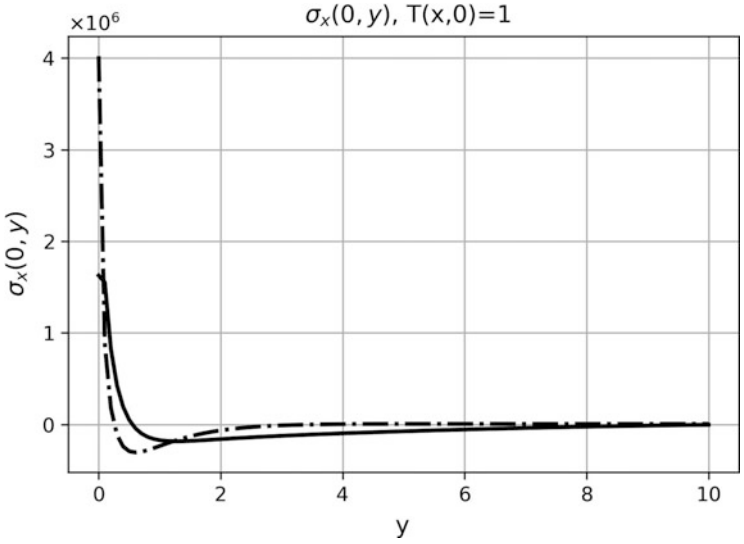


Fig. 5.32 The normal stresses $\sigma_x(0, y)$, $0 < y < 10$ under the unchanged temperature load and the change of the mechanical load

applied at the center of the semi-strip's short edge. The values of the thermal load unlikely the values of the mechanical load does not affect the size of the compressive stresses' zone at the semi-strip's lateral sides near the short edge.

Chapter 6
The Stress Concentration Problems for the Semi-strip with a Transverse Crack



The plane mixed problems of elasticity for a semi-strip which is fixed by its lateral sides and contains a transverse crack are investigated in the chapter for three configurations of the load applied by the semi-strip's short edge.

The solving of the problem for the semi-strip with a transverse crack is reduced to the solving of the system of singular integro-differential or integral equations depending on the configuration of the load applied by the semi-strip's short edge. The solutions of the integral equations are obtained with consideration of the fixed singularities at the ends of the integration segment.

The numeric analysis of the stress intensity coefficients (SIF) is conducted depending on the crack's length.

6.1 The Stress Concentration Problem for a Semi-strip with a Transverse Crack

The elastic semi-strip (Fig. 6.1), which occupies the area in Cartesian coordinates $0 < x < a,\ 0 < y < \infty$ is considered. The semi-strip is loaded by its short edge $y = 0, \quad 0 < x < a$ (3.5)–(3.6) by the normal load $p(x)$.

The displacements in the semi-strip satisfy the equilibrium equations (3.4). The boundary conditions on the semi-infinite lateral sides are detailed in the form (4.3). SIF should be found when no volume forces act on the semi-strip.

The transverse crack is located inside the semi-strip by the line $c_0 < x < c_1, y = B$

$$\begin{aligned} u(x, B-0) - u(x, B+0) &= \langle u(x, B) \rangle = \psi_1(x) \neq 0, \quad c_0 < x < c_1, \\ v(x, B-0) - v(x, B+0) &= \langle v(x, B) \rangle = \psi_2(x) \neq 0, \quad c_0 < x < c_1, \end{aligned} \tag{6.1}$$

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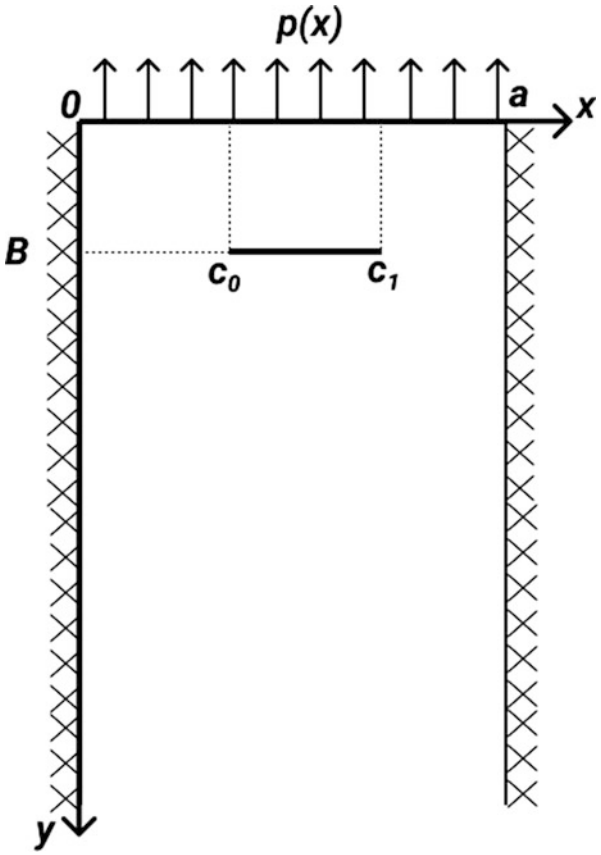


Fig. 6.1 Geometry and coordinate system of the semi-strip with a transverse crack

$$\begin{aligned}\tau_{xy}(x,B-0)-\tau_{xy}(x,B+0)&=\langle\tau_{xy}(x,B)\rangle=0,\quad c_0<x<c_1,\\ \sigma_y(x,B-0)-\sigma_y(x,B+0)&=\langle\sigma_y(x,B)\rangle=0,\quad c_0<x<c_1,\end{aligned}\tag{6.2}$$

$$\tau_{xy}\big|_{y=B\pm0}=0,\quad \sigma_y\big|_{y=B\pm0}=p_\sigma(x),\quad c_0<x<c_1.\tag{6.3}$$

From the conditions (6.2) the jumps of the normal displacements' derivatives can be expressed

$$\langle\tau_{xy}(x,B)\rangle=0\Rightarrow\left\langle\frac{\partial u}{\partial y}(x,B)\right\rangle=-\psi'_2(x),c_0<x<c_1,$$
$$\langle\sigma_y(x,B)\rangle=0\Rightarrow\left\langle\frac{\partial v}{\partial y}(x,B)\right\rangle=-\frac{\mu}{1-\mu}\psi'_1(x)=\frac{\kappa-3}{\kappa+1}\psi'_1(x),c_0<x<c_1.$$

The problem (3.4)–(3.6), (4.1)–(4.2), (6.1)–(6.3) is reduced to the one-dimensional boundary problem with the help of the integral sin-cos Fourier transformation applied by the variable y (3.7) by generalized scheme [112]. The step-

by-step obtaining of the equations in the transformation domain is given in Appendix A.

As the result the continuous one-dimensional boundary problem in the transformation domain is derived

$$\begin{cases} u''_\beta(x) - \dfrac{\beta^2(\kappa\text{-}1)}{\kappa+1}u_\beta(x) + \dfrac{2\beta}{\kappa+1}v'_\beta(x) = \dfrac{3-\kappa}{\kappa+1}\chi'(x) - \\ \qquad - \beta\sin\beta B\dfrac{\kappa-1}{\kappa+1}\psi_1(x) + \cos\beta B\dfrac{\kappa-3}{\kappa+1}\psi'_2(x), \\ v''_\beta(x) - \dfrac{\beta^2(\kappa+1)}{\kappa-1}v_\beta(x) - \dfrac{2\beta}{\kappa-1}u'_\beta(x) = -\beta\dfrac{\kappa+1}{\kappa-1}\chi(x) - \\ \qquad - \sin\beta B\psi'_1(x) + \beta\dfrac{\kappa+1}{\kappa-1}\cos\beta B\psi_2(x), \\ u_\beta(0) = 0, u_\beta(a) = 0, \\ v_\beta(0) = 0, v_\beta(a) = 0. \end{cases} \tag{6.4}$$

This problem is reformulated as the continuous vector boundary problem by the scheme shown in Chap. 2. The right-hand side of the vector equation (3.10) can be written in the form $\vec{f}(x) = \begin{pmatrix} \dfrac{3-\kappa}{\kappa+1}\chi'(x) - \beta\sin\beta B\dfrac{\kappa-1}{\kappa+1}\psi_1(x) + \cos\beta B\dfrac{\kappa-3}{\kappa+1}\psi'_2(x) \\ -\beta\dfrac{\kappa+1}{\kappa-1}\chi(x) - \sin\beta B\psi'_1(x) + \beta\dfrac{\kappa+1}{\kappa-1}\cos\beta B\psi_2(x) \end{pmatrix}$, where $\chi(x) = v|_{y=0}, \chi'(x) = v'|_{y=0}$. According to the constructed solution (3.16) formulae for the displacement functions have the following form:

$$\begin{aligned} u_\beta(x) &= Y_1^{11}(x)c_1 + Y_1^{12}(x)c_2 + Y_2^{11}(x)c_3 + Y_2^{12}(x)c_4 + \\ &+ \frac{3-\kappa}{\kappa+1}\int_0^a G^{11}(x,\xi)\chi'(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int_0^a G^{12}(x,\xi)\chi(\xi)d\xi - \\ &- \frac{\kappa-1}{\kappa+1}\beta\sin\beta B\int_0^a G^{11}(x,\xi)\psi_1(\xi)d\xi + \cos\beta B\frac{\kappa-3}{\kappa+1}\int_0^a G^{11}(x,\xi)\psi'_2(\xi)d\xi - \\ &- \sin\beta B\int_0^a G^{12}(x,\xi)\psi'_1(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\cos\beta B\int_0^a G^{12}(x,\xi)\psi_2(\xi)d\xi, \end{aligned} \tag{6.5}$$

$$v_\beta(x) = Y_1^{21}(x)c_1 + Y_1^{22}(x)c_2 + Y_2^{21}(x)c_3 + Y_2^{22}(x)c_4 +$$
$$+\frac{3-\kappa}{\kappa+1}\int_0^a G^{21}(x,\xi)\chi'(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int_0^a G^{22}(x,\xi)\chi(\xi)d\xi -$$
$$-\frac{\kappa-1}{\kappa+1}\beta\sin\beta B\int_0^a G^{21}(x,\xi)\psi_1(\xi)d\xi + \cos\beta B\frac{\kappa-3}{\kappa+1}\int_0^a G^{21}(x,\xi)\psi'_2(\xi)d\xi -$$
$$-\sin\beta B\int_0^a G^{22}(x,\xi)\psi'_1(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\cos\beta B\int_0^a G^{22}(x,\xi)\psi_2(\xi)d\xi. \tag{6.6}$$

The coefficients $c_i, i=\overline{1,4}$ are given in Appendix C, they have the form (C.2).

Further solving of the problem depends on the configuration of the normal load applied by the semi-strip's short edge.

6.2 The Load at the Center of the Semi-strip's Short Edge (Case 1)

The semi-strip, which is loaded at the center of its short edge and the crack is located inside the semi-strip far from its lateral sides is considered (Fig. 6.2). The boundary value problem (3.4), (4.9)–(4.11), (6.1)–(6.3) when $a_0 \neq 0$, $a_1 \neq a$ for the elastic semi-strip with conditions at the lateral sides (4.3) should be solved.

Integral summands in (6.5) and (6.6) have the following forms:

$$\int_0^a G^{i1}(x,\xi)\chi'(\xi)d\xi,\ \int_0^a G^{i2}(x,\xi)\chi(\xi)d\xi,\ \int_0^a G^{i1}(x,\xi)\psi_1(\xi)d\xi,$$
$$\int_0^a G^{i1}(x,\xi)\psi'_2(\xi)d\xi,\ \int_0^a G^{i2}(x,\xi)\psi'_1(\xi)d\xi,\ \int_0^a G^{i2}(x,\xi)\psi_2(\xi)d\xi,$$

moreover,

$$\int_0^a G^{ij}(x,\xi)\psi'_k(\xi)d\xi = -\int_0^a \frac{\partial G^{ij}}{\partial\xi}(x,\xi)\psi_k(\xi)d\xi, \quad i,j,k=1,2.$$

As the result the correspondences (6.5)–(6.6) take the form:

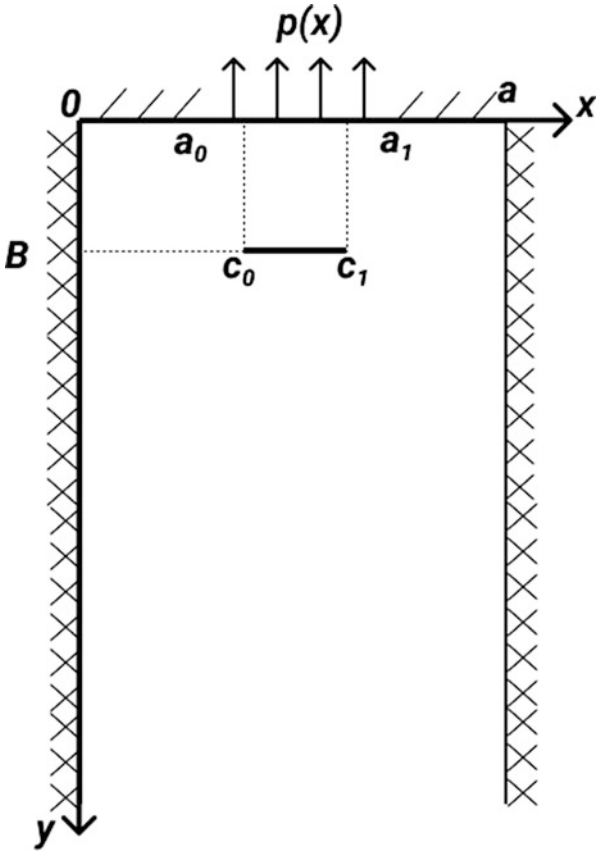


Fig. 6.2 Geometry and coordinate system of the semi-strip with a transverse crack, which is located far from the semi-strip's lateral sides. Case 1

$$
\begin{aligned}
u_\beta(x) &= Y_1^{11}(x)c_1 + Y_1^{12}(x)c_2 + Y_2^{11}(x)c_3 + Y_2^{12}(x)c_4 - \\
&- \frac{3-\kappa}{\kappa+1}\int_0^a \frac{\partial}{\partial\xi}G^{11}(x,\xi)\chi(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int_0^a G^{12}(x,\xi)\chi(\xi)d\xi - \\
&- \frac{\kappa-1}{\kappa+1}\beta\sin\beta B\int_{c_0}^{c_1} G^{11}(x,\xi)\psi_1(\xi)d\xi + \sin\beta B\int_{c_0}^{c_1}\frac{\partial G^{12}}{\partial\xi}(x,\xi)\psi_1(\xi)d\xi - \\
&- \cos\beta B\frac{\kappa-3}{\kappa+1}\int_{c_0}^{c_1}\frac{\partial G^{11}}{\partial\xi}(x,\xi)\psi_2(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\cos\beta B\int_{c_0}^{c_1} G^{12}(x,\xi)\psi_2(\xi)d\xi,
\end{aligned}
\tag{6.7}
$$

$$v_\beta(x) = Y_1^{21}(x)c_1 + Y_1^{22}(x)c_2 + Y_2^{21}(x)c_3 + Y_2^{22}(x)c_4 -$$
$$- \frac{3-\kappa}{\kappa+1} \int_0^a \frac{\partial}{\partial\xi} G^{21}(x,\xi)\chi(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1} \int_0^a G^{22}(x,\xi)\chi(\xi)d\xi -$$
$$- \frac{\kappa-1}{\kappa+1}\beta\sin\beta B \int_{c_0}^{c_1} G^{21}(x,\xi)\psi_1(\xi)d\xi + \sin\beta B \int_{c_0}^{c_1} \frac{\partial G^{22}}{\partial\xi}(x,\xi)\psi_1(\xi)d\xi -$$
$$- \cos\beta B\frac{\kappa-3}{\kappa+1} \int_{c_0}^{c_1} \frac{\partial G^{21}}{\partial\xi}(x,\xi)\psi_2(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\cos\beta B \int_{c_0}^{c_1} G^{22}(x,\xi)\psi_2(\xi)d\xi, \tag{6.8}$$

since $\psi_1(x), \psi_2(x) = 0, x \notin [c_0; c_1]$.

The obtained earlier $Y_1(x)$, $Y_2(x)$, $c_i, i = \overline{1,4}$ are substituted into formulae (6.7)–(6.8) and the derived transformants are inverted according to formulae (3.8). Then the following expressions for the displacement functions are obtained:

$$u(x,y) = \int_0^a \chi(\xi)f_u(x,y,\xi)d\xi + \int_{c_0}^{c_1} \psi_1(\xi)g_{u1}(x,y,\xi)d\xi + \int_{c_0}^{c_1} \psi_2(\xi)g_{u2}(x,y,\xi)d\xi, \tag{6.9}$$

$$v(x,y) = \int_0^a \chi(\xi)f_v(x,y,\xi)d\xi + \int_{c_0}^{c_1} \psi_1(\xi)g_{v1}(x,y,\xi)d\xi + \int_{c_0}^{c_1} \psi_2(\xi)g_{v2}(x,y,\xi)d\xi. \tag{6.10}$$

Formulae (6.9)–(6.10) will finally determine displacements if the functions $\chi(x)$, $\psi_1(x)$, $\psi_2(x)$ are known. For its finding the unfulfilled conditions at the semi-strip's short edge (4.9)–(4.10) and crack conditions (6.3) should be satisfied.

Conditions (4.9), (6.3) can't be fulfilled by direct substitution of (6.9)–(6.10) in it, because integrals in expressions for the displacements are weakly-convergent. So, the method given in Appendix D is used for its weakly-convergent parts' separation.

Formula (4.15) follows from the condition (4.10). The substitution of the formulae for the displacements into the boundary condition (4.9) and crack conditions (6.3) reduces to the system of singular integro-differential equations (SIDE) of the following form:

$$\begin{cases} \dfrac{d^2}{dx^2}\displaystyle\int_{a_0}^{a_1}\chi(\xi^*)\ln\frac{1}{|\xi^*-x^*|}d\xi^*+K_0(x^*)=r(x^*), \quad a_0<x^*<a_1 \\ \dfrac{d^2}{dx^2}\displaystyle\int_{c_0}^{c_1}\psi_1(\xi^*)\ln\frac{1}{|\xi^*-x^*|}d\xi^*+K_1(x^*)=0, \quad c_0<x^*<c_1 \\ \dfrac{d^2}{dx^2}\displaystyle\int_{c_0}^{c_1}\psi_2(\xi^*)\ln\frac{1}{|\xi^*-x^*|}d\xi^*+K_2(x^*)=q(x^*), \quad c_0<x^*<c_1, \end{cases} \tag{6.11}$$

where

$K_i(x)=\int_{a_0}^{a_1}\chi(\xi)f_i(\xi,x)d\xi+\int_{c_0}^{c_1}\psi_1(\xi)R_{i,1}(x,\xi)d\xi+\int_{c_0}^{c_1}\psi_2(\xi)R_{i,2}(x,\xi)d\xi, i=0,1,2$

$f_i(\xi,x)$, $R_{i,\,1}(x,\xi)$, $R_{i,\,2}(x,\xi)$, $r(x)$, $q(x)$, $i=0,\,1,\,2$ are known regular functions when $a_0\neq 0$, $a_1\neq a$.

The following substitution $\xi=\frac{2\xi^*-(\vartheta_0+\vartheta_1)}{\vartheta_1-\vartheta_0}, x=\frac{2x^*-(\vartheta_0+\vartheta_1)}{\vartheta_1-\vartheta_0}$ is done, here $\vartheta\in\{a, c\}$, for the transfer to the integration interval $[-1;1]$. As the result SIDE (6.11) takes the following form:

$$\begin{cases} \dfrac{d^2}{dx^2}\displaystyle\int_{-1}^{1}\widetilde{\chi}(\xi)\ln\frac{1}{|\xi-x|}d\xi+\widetilde{K}_0(x)=\widetilde{r}(x), \quad -1<x<1 \\ \dfrac{d^2}{dx^2}\displaystyle\int_{-1}^{1}\widetilde{\psi}_1(\xi)\ln\frac{1}{|\xi-x|}d\xi+\widetilde{K}_1(x)=0, \quad -1<x<1 \\ \dfrac{d^2}{dx^2}\displaystyle\int_{-1}^{1}\widetilde{\psi}_2(\xi)\ln\frac{1}{|\xi-x|}d\xi+\widetilde{K}_2(x)=\widetilde{q}(x), \quad -1<x<1, \end{cases} \tag{6.12}$$

where $\widetilde{\chi}(\xi)=\chi\left(\frac{(a_1-a_0)\xi+(a_0+a_1)}{2}\right)$, $\widetilde{\psi}_i(\xi)=\psi_i\left(\frac{(c_1-c_0)\xi+(c_0+c_1)}{2}\right), i=1,2,$

$\widetilde{r}(x)=\frac{(a_1-a_0)}{2}r\left(\frac{(a_1-a_0)\xi+(a_0+a_1)}{2}\right)$, $\widetilde{q}(x)=\frac{(c_1-c_0)}{2}q\left(\frac{(c_1-c_0)\xi+(c_0+c_1)}{2}\right)$,

$K_i(x)=\int_{-1}^{1}\widetilde{\chi}(\xi)\widetilde{f}_i(\xi,x)d\xi+\int_{-1}^{1}\widetilde{\psi}_1(\xi)\widetilde{R}_{i,1}(x,\xi)d\xi+\int_{-1}^{1}\widetilde{\psi}_2(\xi)\widetilde{R}_{i,2}(x,\xi)d\xi, i=0,1,2$

$\widetilde{f}_i(\xi,x)=\frac{(a_1-a_0)(\vartheta_1-\vartheta_0)}{4}f_i\left(\frac{(a_1-a_0)\xi+(a_0+a_1)}{2}, \frac{(\vartheta_1-\vartheta_0)x+(\vartheta_0+\vartheta_1)}{2}\right)$,

$\widetilde{R}_{i,j}(x,\xi)=\frac{(c_1-c_0)(\vartheta_1-\vartheta_0)}{4}R_{i,j}\left(\frac{(\vartheta_1-\vartheta_0)x+(\vartheta_0+\vartheta_1)}{2}, \frac{(c_1-c_0)\xi+(c_0+c_1)}{2}\right), j=1,2,$

$\vartheta\in\begin{cases} a, i=0 \\ c, i=1,2 \end{cases}$.

SIDE (6.12) is solved approximately by the orthogonal polynomials method [112] analogously to the scheme, presented in Sect. 3.5.1. According to this method,

functions $\widetilde{\chi}(\xi), \widetilde{\psi}_i(\xi), i = 1, 2$ are expanded into series by the Chebyshev polynomials of the second kind

$$\widetilde{\chi}(\xi) = \sum_{n=0}^{\infty} s_n^0 \sqrt{1-\xi^2} U_n(\xi), \quad \xi \in [-1; 1], \tag{6.13}$$

$$\widetilde{\psi}_i(\xi) = \sum_{n=0}^{\infty} s_n^i \sqrt{1-\xi^2} U_n(\xi), \quad \xi \in [-1; 1], i = 1, 2, \tag{6.14}$$

here, $U_n(x)$ are Chebyshev polynomials of the second kind. Expressions (6.13)–(6.14) are substituted into SIDE (6.12), the integration and summation orders are changed. The spectral correspondence (3.22) is used and the standard scheme of orthogonal polynomials method is applied. As the result the infinite system of linear algebraic equations with regard to unknown coefficients $s_n^i, i = 0, 1, 2, n = 0, 1, \ldots$ is derived

$$\vec{S}_m + \sum_{n=0}^{\infty} D_{mn} \vec{S}_n = \vec{f}_m, m = 0, 1, 2 \ldots, \tag{6.15}$$

where $\vec{S}_m = \begin{pmatrix} s_m^0 \\ s_m^1 \\ s_m^2 \end{pmatrix}$, components of $D_{mn} = \left\{ d_{mn}^{ij} \right\}, i, j = 0, 1, 2, \vec{f}_m = \begin{pmatrix} f_m^0 \\ f_m^1 \\ f_m^2 \end{pmatrix}$ are known constants.

The system (6.15) is solved by the reduction method. The substitution of the found constants $s_n^i, i = 0, 1, 2, n = 0, 1, \ldots$ into the expressions (6.13)–(6.14) and its further use in formulae (6.9)–(6.10) complete the construction of the problem's solution.

For SIF calculation the stress functions at the line $c_0 < x < c_1, y = B$ are written as follows:

$$\begin{aligned} \sigma_y(x^*, B) &= \frac{1}{\pi} \int_{c_0}^{c_1} \psi_2(\xi^*) \frac{1}{(\xi^* - x^*)^2} d\xi^* + \sigma_y^0(x^*, B), \\ \tau_{xy}(x^*, B) &= \frac{1}{\pi} \int_{c_0}^{c_1} \psi_1(\xi^*) \frac{1}{(\xi^* - x^*)^2} d\xi^* + \tau_{xy}^0(x^*, B). \end{aligned} \tag{6.16}$$

After the substitution of the variables $\xi = \frac{2\xi^* - (c_0 + c_1)}{c_1 - c_0}, x = \frac{2x^* - (c_0 + c_1)}{c_1 - c_0}$ the expressions for the stress functions (6.16) take the following form:

$$\sigma_y(x,B)=\frac{1}{\pi}\frac{2}{c_1-c_0}\frac{d^2}{dx^2}\int\limits_{-1}^{1}\ln\frac{1}{|\xi-x|}\widetilde{\psi}_2(\xi)d\xi+\widetilde{\sigma}_y^0(x,B),$$
$$\tau_{xy}(x,B)=\frac{1}{\pi}\frac{2}{c_1-c_0}\frac{d^2}{dx^2}\int\limits_{-1}^{1}\ln\frac{1}{|\xi-x|}\widetilde{\psi}_1(\xi)d\xi+\widetilde{\tau}_{xy}^0(x,B). \tag{6.17}$$

where $\widetilde{\sigma}_y^0(x,B),\widetilde{\tau}_{xy}^0(x,B)$ are known regular functions.

The expressions for the functions $\widetilde{\psi}_i(\xi)$ (6.14) are substituted into formulae (6.17) and the order of summation and integration is changed

$$\sigma_y(x,B)=\frac{1}{\pi}\frac{2}{c_1-c_0}\sum_{n=0}^{\infty}s_n^2\frac{d^2}{dx^2}\int\limits_{-1}^{1}\ln\frac{1}{|\xi-x|}\sqrt{1-\xi^2}U_n(\xi)d\xi+\widetilde{\sigma}_y^0(x,B),$$
$$\tau_{xy}(x,B)=\frac{1}{\pi}\frac{2}{c_1-c_0}\sum_{n=0}^{\infty}s_n^1\frac{d^2}{dx^2}\int\limits_{-1}^{1}\ln\frac{1}{|\xi-x|}\sqrt{1-\xi^2}U_n(\xi)d\xi+\widetilde{\tau}_{xy}^0(x,B). \tag{6.18}$$

SIF are calculated by formula [112]:

$$K_{\pm}=\lim_{x\to\pm1\pm0}\sqrt{\pi(c_1-c_0)(\pm x-1)}\begin{bmatrix}\sigma_y(x,B)\\ \tau_{xy}(x,B)\end{bmatrix} \tag{6.19}$$

The expressions (6.18) are substituted into formula (6.19) and the following spectral correspondence is used

$$\frac{1}{\pi}\frac{d^2}{dx^2}\int\limits_{-1}^{1}\ln\frac{1}{|x-s|}\sqrt{1-s^2}U_m(s)ds=\frac{|x|U_n(x)}{\sqrt{x^2-1}}+\sqrt{x^2-1}U'_n(x)\operatorname{sgn}x-$$
$$-0.5(n+1)U_n(x),\quad |x|>1.$$

Then

$$\lim_{x\to\pm1\pm0}\sqrt{\pi(c_1-c_0)(\pm x-1)}\left(\frac{1}{\pi}\frac{d^2}{dx^2}\int\limits_{-1}^{1}\ln\frac{1}{|\xi-x|}\sqrt{1-\xi^2}U_n(\xi)d\xi\right)=$$
$$=\sqrt{\pi(c_1-c_0)}(\pm1)^n(n+1)\frac{1}{\sqrt{2}}.$$

As the result the calculation formulae for the SIF calculation are obtained

$$K_{I-} = \sum_{k=0}^{\infty} s_k^2 \frac{\sqrt{\pi(c_1 - c_0)}(n+1)(-1)^k}{\sqrt{2}}, K_{I+} = \sum_{k=0}^{\infty} s_k^2 \frac{\sqrt{\pi(c_1 - c_0)}(n+1)}{\sqrt{2}}, \tag{6.20}$$

$$K_{II-} = \sum_{k=0}^{\infty} s_k^1 \frac{\sqrt{\pi(c_1 - c_0)}(n+1)(-1)^k}{\sqrt{2}}, K_{II+} = \sum_{k=0}^{\infty} s_k^1 \frac{\sqrt{\pi(c_1 - c_0)}(n+1)}{\sqrt{2}}. \tag{6.21}$$

Hereinafter, the length of distances between left and right semi-strip's lateral sides and left and right crack tips are denoted as γ_0, γ_1, respectively.

In figures (Figs. 6.3 and 6.4) the numeric analysis of SIF under the following values of parameters $G = 61.2781955 \cdot 10^9$ Pa, $\mu = 0.33$, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa, $a = 10$ m, $a_0 = a - a_1 = 1$ and the symmetric location of the crack inside the semi-strip was conducted.

When the crack length is decreasing the SIF values K_I, K_{II} are decreasing to zero. When the parameters take values $\gamma_0 = \gamma_1 < 15\ \%\ a$ calculations become unstable, because of the influence of the fixed singularities at the crack tips. To bring the calculations when the crack is close to the semi-strip's lateral sides the necessity of the fixed singularities' consideration occurs.

In figures (Figs. 6.5 and 6.6) the graphics for SIF K_I, K_{II} are presented when $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa.

As it can be seen from plot, SIF values K_I, K_{II} are decreasing to zero when the crack length is decreasing. The fixed singularities at the crack tips should be taken into consideration when $\gamma_0 = \gamma_1 < 5\ \%\ a$.

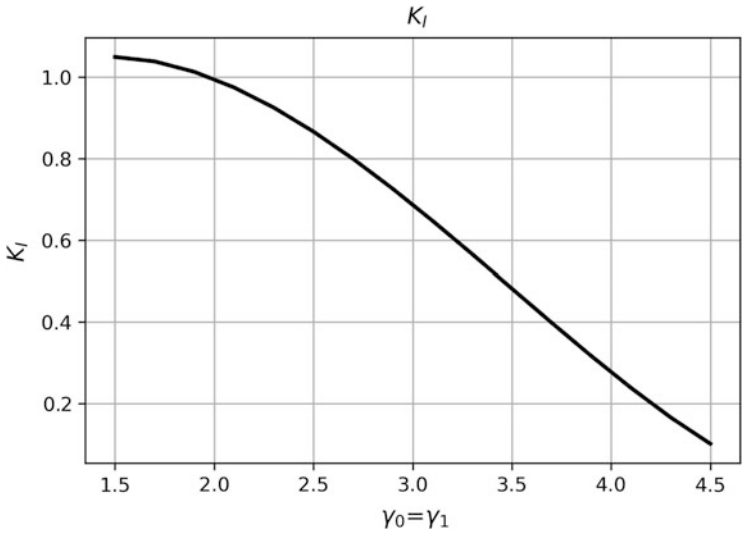


Fig. 6.3 SIF K_I under symmetric change of the crack's length, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa

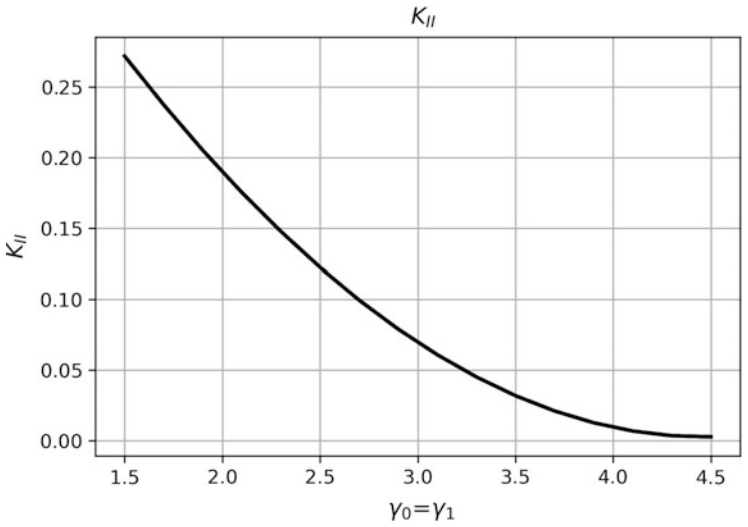


Fig. 6.4 SIF K_{II} under symmetric change of the crack's length, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa

The values of SIF K_I are significantly bigger than the values of SIF K_{II}.

6.3 The Load at the Whole Semi-strip's Short Edge (Case 3)

The semi-strip, which is loaded at the whole short edge and the crack is located inside the semi-strip far from its lateral sides is considered (Fig. 6.7). The boundary problem (3.4)–(3.6), (6.1)–(6.3) for the elastic semi-strip with

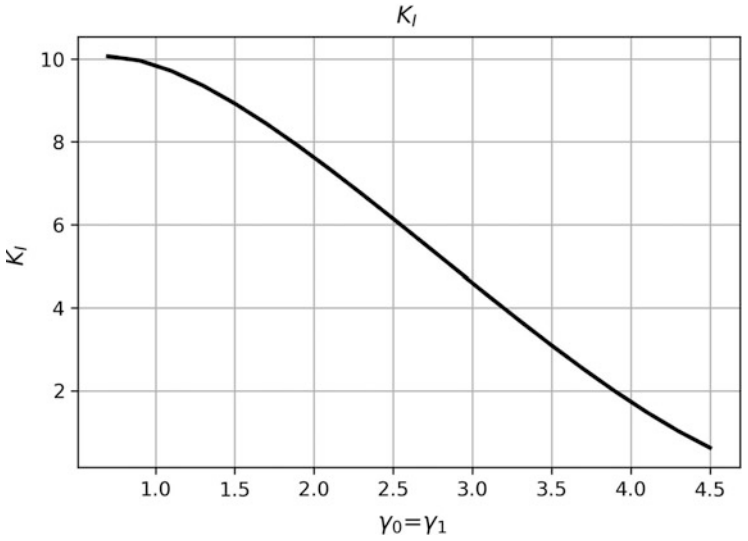


Fig. 6.5 SIF K_I under symmetric change of the crack's length, $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa

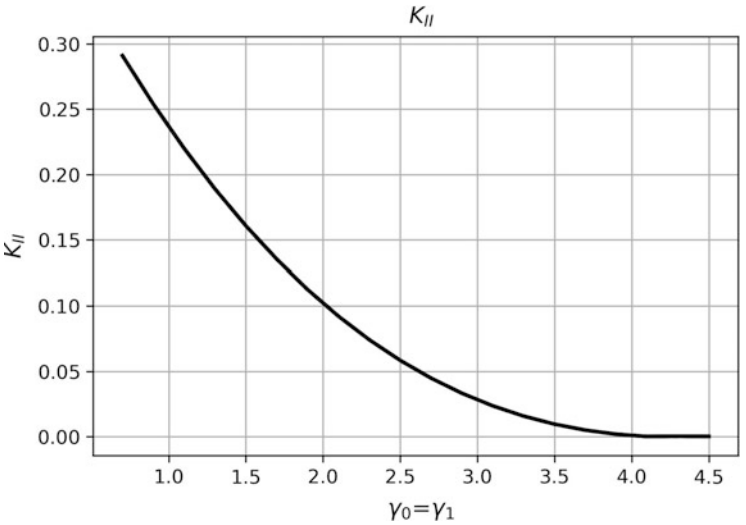


Fig. 6.6 SIF K_{II} under symmetric change of the crack's length, $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa

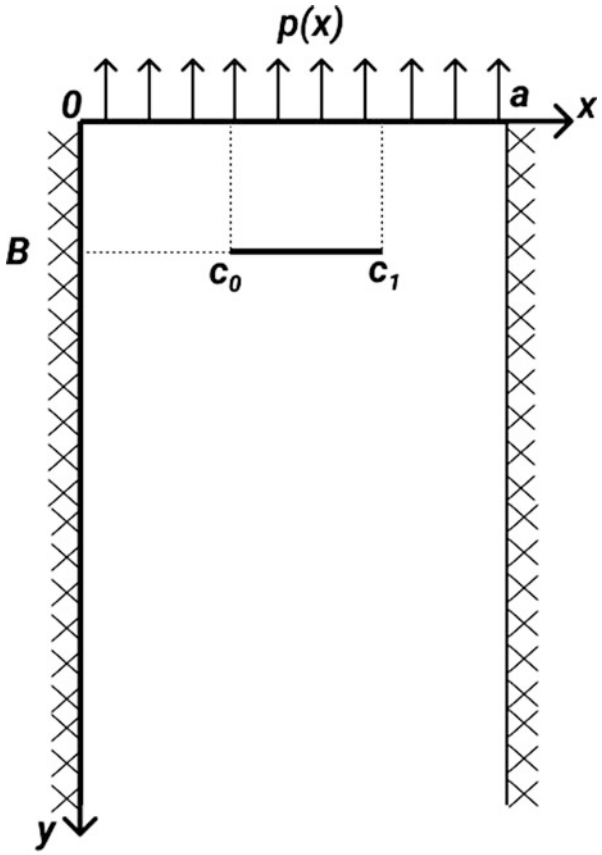


Fig. 6.7 Geometry and coordinate system of the semi-strip with a transverse crack, which is located far from the semi-strip's lateral sides. Case 3

conditions (3.2) at the semi-strip's lateral sides taken in the form (4.1)–(4.2) or (4.3) should be solved.

The correspondences (6.5)–(6.6) take the following form:

$$u_\beta(x) = Y_1^{11}(x)c_1 + Y_1^{12}(x)c_2 + Y_2^{11}(x)c_3 + Y_2^{12}(x)c_4 + \\ + \frac{3-\kappa}{\kappa+1}\int_0^a G^{11}(x,\xi)\chi'(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\int_0^a \mathfrak{I}^{12}(x,\xi)\chi'(\xi)d\xi - \\ - \frac{\kappa-1}{\kappa+1}\beta\sin\beta B\int_{c_0}^{c_1} G^{11}(x,\xi)\psi_1(\xi)d\xi + \sin\beta B\int_{c_0}^{c_1}\frac{\partial G^{12}}{\partial\xi}(x,\xi)\psi_1(\xi)d\xi - \\ - \cos\beta B\frac{\kappa-3}{\kappa+1}\int_{c_0}^{c_1}\frac{\partial G^{11}}{\partial\xi}(x,\xi)\psi_2(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\cos\beta B\int_{c_0}^{c_1} G^{12}(x,\xi)\psi_2(\xi)d\xi, \tag{6.22}$$

$$v_\beta(x) = Y_1^{21}(x)c_1 + Y_1^{22}(x)c_2 + Y_2^{21}(x)c_3 + Y_2^{22}(x)c_4 + \\ + \frac{3-\kappa}{\kappa+1}\int_0^a G^{21}(x,\xi)\chi'(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\int_0^a \mathfrak{I}^{22}(x,\xi)\chi'(\xi)d\xi - \frac{\kappa-1}{\beta^2(\kappa+1)}\chi(x) - \\ - \frac{\kappa-1}{\kappa+1}\beta\sin\beta B\int_{c_0}^{c_1} G^{21}(x,\xi)\psi_1(\xi)d\xi + \sin\beta B\int_{c_0}^{c_1}\frac{\partial G^{22}}{\partial\xi}(x,\xi)\psi_1(\xi)d\xi - \\ - \cos\beta B\frac{\kappa-3}{\kappa+1}\int_{c_0}^{c_1}\frac{\partial G^{21}}{\partial\xi}(x,\xi)\psi_2(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\cos\beta B\int_{c_0}^{c_1} G^{22}(x,\xi)\psi_2(\xi)d\xi, \tag{6.23}$$

since $\psi_1(x), \psi_2(x) = 0,\ x \notin [c_0; c_1]$.

After the substitution of the derived earlier $Y_1(x),\ Y_2(x),\ c_i, i = \overline{1,4}$ into formulae (6.22)–(6.23) and inversion of the obtained transformants, respectively, to the presentation (3.8) the following expressions for the displacement functions are derived:

$$u(x,y)=\int_0^a \chi'(\xi)f_u(x,y,\xi)d\xi+\int_{c_0}^{c_1}\psi_1(\xi)g_{u1}(x,y,\xi)d\xi+ \\ +\int_{c_0}^{c_1}\psi_2(\xi)g_{u2}(x,y,\xi)d\xi, \tag{6.24}$$

$$v(x,y)=\int_0^a \chi'(\xi)f_v(x,y,\xi)d\xi+\int_{c_0}^{c_1}\psi_1(\xi)g_{v1}(x,y,\xi)d\xi+ \\ +\int_{c_0}^{c_1}\psi_2(\xi)g_{v2}(x,y,\xi)d\xi+\int_0^x \chi'(\xi)d\xi. \tag{6.25}$$

Formulae (6.24)–(6.25) will finally determine displacements if the functions $\chi'(x)$, $\psi_1(x)$, $\psi_2(x)$ are found. For its finding the unfulfilled yet condition (3.5) and crack conditions (6.3) should be satisfied. Since the integrals in the expressions for the displacements are weakly-convergent, the method presented in Appendix D is used for the separation of its weakly-convergent parts.

The substitution of the formulae for the displacement functions into the boundary condition (3.5) and into the crack conditions (6.3) leads to the system of singular integral equations (SSIE) of the following form:

$$\begin{cases} \displaystyle\int_0^a \chi'(\xi^*)\left[\frac{1}{\xi^*-x^*}+Z(x^*,\xi^*)\right]d\xi^*+K_0(x^*)=r(x^*), \quad 0<x^*<a \\ \displaystyle\frac{d^2}{dx^2}\int_{c_0}^{c_1}\psi_1(\xi^*)\ln\frac{1}{|\xi^*-x^*|}d\xi^*+K_1(x^*)=0, \quad c_0<x^*<c_1 \\ \displaystyle\frac{d^2}{dx^2}\int_{c_0}^{c_1}\psi_2(\xi^*)\ln\frac{1}{|\xi^*-x^*|}d\xi^*+K_2(x^*)=0, \quad c_0<x^*<c_1, \end{cases} \tag{6.26}$$

where

$K_i(x)=\int_0^a \chi'(\xi)f_i(\xi,x)d\xi+\int_{c_0}^{c_1}\psi_1(\xi)R_{i,1}(x,\xi)d\xi+\int_{c_0}^{c_1}\psi_2(\xi)R_{i,2}(x,\xi)d\xi, i=0,1,2$,

$f_i(\xi,x)$, $R_{i,\ 1}(x,\xi)$, $R_{i,\ 2}(x,\xi)$, $r(x)$, $i=0,\ 1,\ 2$ are known regular functions,

$Z(x,\xi)=h_1\left(\frac{1}{\xi+x}+\frac{1}{\xi+x-2a}\right)+h_2\left(\frac{x}{(\xi+x)^2}+\frac{x-a}{(\xi+x-2a)^2}\right)+h_3\left(\frac{x\xi}{(\xi+x)^3}+\frac{(x-a)(\xi-a)}{(\xi+x-2a)^3}\right)$,

$h_1=-\frac{\kappa^2-3}{2\kappa}, h_2=-\frac{2}{\kappa}, h_3=\frac{4}{\kappa}$.

For the transfer to the integration intervals $[-1;1]$ the following variable substitution $\xi = \frac{2\xi^* - (\vartheta_0+\vartheta_1)}{\vartheta_1 - \vartheta_0}, x = \frac{2x^* - (\vartheta_0+\vartheta_1)}{\vartheta_1 - \vartheta_0}$ is used, where $\vartheta \in \{a, c\}$. As the result SSIE (6.26) takes the following form:

$$\begin{cases} \int\limits_{-1}^{1} \widetilde{\chi}(\xi)\left[\dfrac{1}{\xi - x} + \widetilde{Z}(x,\xi)\right]d\xi + \widetilde{K}_0(x) = \widetilde{r}(x), \quad -1 < x < 1 \\ \dfrac{d^2}{dx^2} \int\limits_{-1}^{1} \widetilde{\psi}_1(\xi) \ln \dfrac{1}{|\xi - x|} d\xi + \widetilde{K}_1(x) = 0, \quad -1 < x < 1 \\ \dfrac{d^2}{dx^2} \int\limits_{-1}^{1} \widetilde{\psi}_2(\xi) \ln \dfrac{1}{|\xi - x|} d\xi + \widetilde{K}_2(x) = \widetilde{q}(x), \quad -1 < x < 1, \end{cases} \tag{6.27}$$

here, $\widetilde{\chi}(\xi) = \chi'\left(\frac{a(\xi+1)}{2}\right)$, $\widetilde{\psi}_i(\xi) = \psi_i\left(\frac{(c_1 - c_0)\xi + (c_0 + c_1)}{2}\right), i = 1, 2$, $\widetilde{r}(x) = r\left(\frac{a\xi + a}{2}\right)$, $\widetilde{q}(x) = q\left(\frac{(c_1 - c_0)\xi + (c_0 + c_1)}{2}\right)$, $K_i(x) = \int\limits_{-1}^{1} \widetilde{\chi}(\xi)\widetilde{f}_i(\xi, x)d\xi + \int\limits_{-1}^{1} \widetilde{\psi}_1(\xi)\widetilde{R}_{i,1}(x,\xi)d\xi + \int\limits_{-1}^{1} \widetilde{\psi}_2(\xi)\widetilde{R}_{i,2}(x,\xi)d\xi, i = 0, 1, 2, \widetilde{f}_0(\xi, x) = \frac{a}{2} f_0\left(\frac{a\xi + a}{2}, \frac{(c_1 - c_0)x + (c_0 + c_1)}{2}\right)$, $\widetilde{f}_i(\xi, x) = \frac{a}{2}\frac{(c_1 - c_0)}{2} f_i\left(\frac{a\xi + a}{2}, \frac{(c_1 - c_0)x + (c_0 + c_1)}{2}\right), i = 1, 2, \widetilde{R}_{0,j}(x,\xi) = \frac{(c_1 - c_0)}{2} R_{i,j}\left(\frac{ax + a}{2}, \frac{(c_1 - c_0)\xi + (c_0 + c_1)}{2}\right), j = 1, 2,$ $\widetilde{R}_{i,j}(x,\xi) = \frac{(c_1 - c_0)}{2}\frac{a}{2} R_{i,j}\left(\frac{ax + a}{2}, \frac{(c_1 - c_0)\xi + (c_0 + c_1)}{2}\right), i, j = 1, 2,$ $\widetilde{Z}(x,\xi) = h_1\left(\frac{1}{\xi + x + 2} + \frac{1}{\xi + x - 2}\right) + + h_2\left(\frac{x+1}{(\xi + x + 1)^2} + \frac{x - 1}{(\xi + x - 2)^2}\right) + h_3\left(\frac{(x+1)(\xi+1)}{(\xi + x + 2)^3} + \frac{(x-1)(\xi - 1)}{(\xi + x - 2)^3}\right)$.

The transcendental equation for the determination of the singularity order of the first equation's solution in (6.27) is equivalent to the transcendental equation for (3.29).

According to the scheme, presented in Sect. 3.5.2, the unknown functions $\widetilde{\chi}(\xi)$ are expanded into sum at each interval

$$\widetilde{\chi}(\xi) = \sum_{k=0}^{N-1} \left[s_k^0 \rho_k^-(\xi) + s_{k+N}^0 \rho_k^+(\xi)\right], \quad \xi \in [-1; 1], \tag{6.28}$$

where $\rho_{2k}^{\mp}(\xi) = (1 \pm \xi)^{\operatorname{Re}\lambda_k} \cdot \cos(\operatorname{Im}\lambda_k \ln(1 \pm \xi)), \rho_{2k+1}^{\mp}(\xi) = (1 \pm \xi)^{\operatorname{Re}\lambda_k} \cdot \sin(\operatorname{Im}\lambda_k \ln(1 \pm \xi)), \quad k = \overline{0, N-1}$.

The functions $\widetilde{\psi}_i(\xi)$ are expanded in series by the Chebyshev polynomials of the second kind:

$$\widetilde{\psi}_i(\xi) = \sum_{k=0}^{2N-1} s_k^i \sqrt{1-\xi^2} U_k(\xi), \quad \xi \in [-1;1], i = 1,2. \tag{6.29}$$

The segment $[-1;1]$ is divided into $2N$ intervals by the points $x_i : P_{2N-1}^{\lambda_0,\lambda_0}(x_i) = 0, i = \overline{0,2N-1}$. SSIE (6.27) is considered when $x = x_i$.

After the substitution of the expressions for the unknown functions (6.28), (6.29) into SSIE (6.27) and consideration of the condition (3.28) the system of $6N$ linear algebraic equations, respectively, unknown constants $s_k^i, i = 0,1,2, k = \overline{0,2N-1}$ is derived:

$$\sum_{n=0}^{2N-1} D_{mn} \vec{S}_n = \vec{f}_m, m = \overline{0,2N-1}, \tag{6.30}$$

where $\vec{S}_m = \begin{pmatrix} s_m^0 \\ s_m^1 \\ s_m^2 \end{pmatrix}$, components of $D_{mn} = \{d_{mn}^{ij}\}, i,j = 0,1,2, \vec{f}_m = \begin{pmatrix} f_m^0 \\ f_m^1 \\ f_m^2 \end{pmatrix}$ are known constants.

The substitution of the found constants $s_k^i, i = 0,1,2, k = \overline{0,2N-1}$ into formulae (6.24)–(6.25) completes the constriction of the problem's solution.

SIF are calculated by the formulae

$$K_{I-} = \sum_{k=0}^{2N-1} s_k^2 \frac{\sqrt{\pi(c_1-c_0)}(n+1)(-1)^k}{\sqrt{2}}, K_{I+} = \sum_{k=0}^{2N-1} s_k^2 \frac{\sqrt{\pi(c_1-c_0)}(n+1)}{\sqrt{2}}, \tag{6.31}$$

$$K_{II-} = \sum_{k=0}^{2N-1} s_k^1 \frac{\sqrt{\pi(c_1-c_0)}(n+1)(-1)^k}{\sqrt{2}}, K_{II+} = \sum_{k=0}^{2N-1} s_k^1 \frac{\sqrt{\pi(c_1-c_0)}(n+1)}{\sqrt{2}}. \tag{6.32}$$

In figures (Figs. 6.8 and 6.9) the plots of SIF K_I, K_{II} are depicted when при $G = 61.2781955 \cdot 10^9$ Pa, $\mu = 0.33$, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa, $a = 10$ m.

When the crack's length is decreasing the values of SIF K_I, K_{II} are decreasing to zero. The stable calculations are obtained when the parameters values are $\gamma_0 = \gamma_1 \geq 5\ \%\ a$, and when $\gamma_0 = \gamma_1 < 5\ \%\ a$ the fixed singularities at the crack tips should be considered.

The graphics of SIF K_I, K_{II} are presented in figures (Figs. 6.10 and 6.11) when $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa.

As it can be seen from the plots, the SIF values K_I, K_{II} are decreasing to zero when crack length is decreasing. Under the values of parameters $\gamma_0 = \gamma_1 < 1\ \%\ a$ the fixed singularities at the crack tips should be taken into account.

In this case the SIF values K_I are significantly larger than the values of SIF K_{II}.

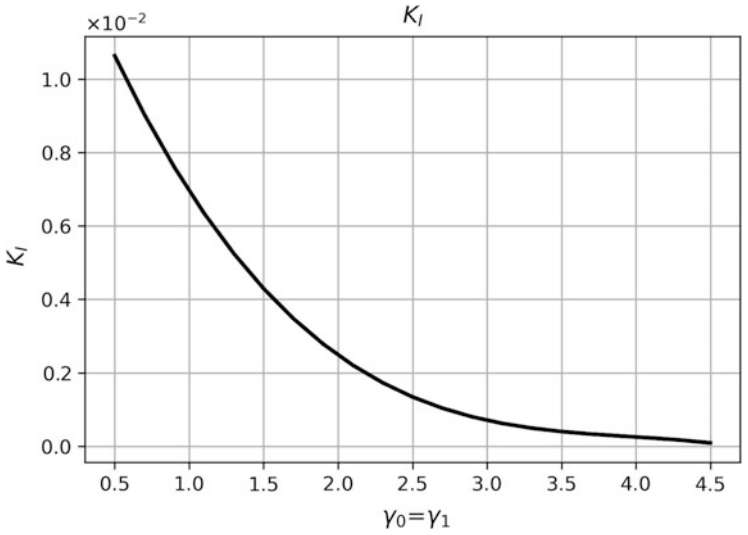


Fig. 6.8 SIF K_I under symmetric change of the crack's length, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa

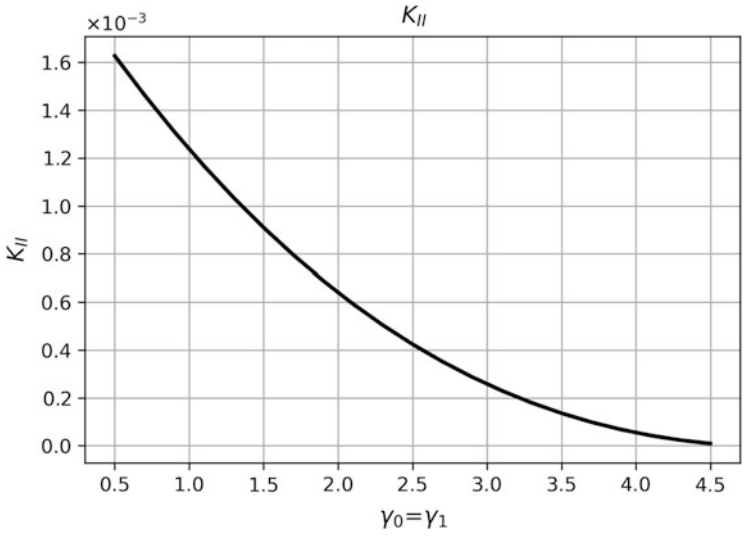


Fig. 6.9 SIF K_{II} under symmetric change of the crack's length, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa

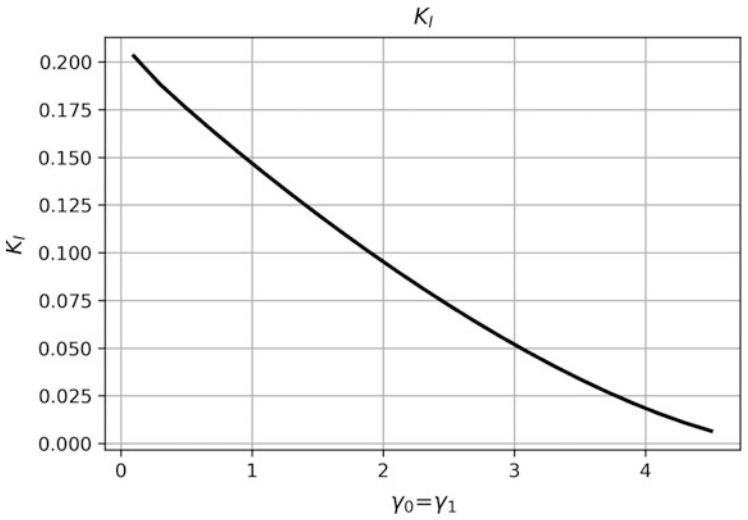


Fig. 6.10 SIF K_I under symmetric change of the crack's length, $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa

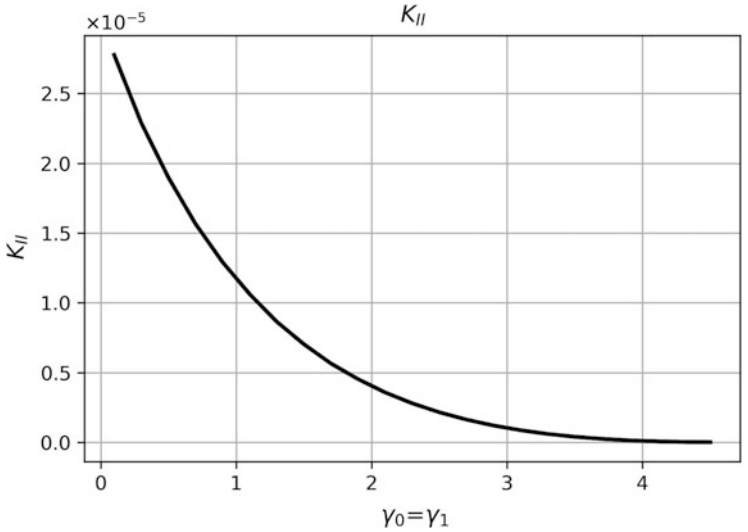


Fig. 6.11 SIF K_{II} under symmetric change of the crack's length, $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa

6.4 The Load at the Left Part of the Semi-strip's Short Edge (Case 2)

The semi-strip, which is loaded at the left part of its short edge and the crack is located inside the semi-strip far from its lateral sides is considered (Fig. 6.12). Let's solve the boundary problem (3.4), (4.18)–(4.20), (6.1)–(6.3) when $a_1 \neq a$ for the elastic semi-strip with the conditions (4.3) at the lateral sides.

The correspondences (6.5)–(6.6) take the form (6.22)–(6.23). After the substitution of the found earlier $Y_1(x), Y_2(x), c_i, i = \overline{1,4}$ into the expressions (6.22)–(6.23) and inverting of the derived transformants according to formulae (3.8) the expressions for the displacement functions are obtained in the form (6.24)–(6.25).

Formulae (6.24)–(6.25) will completely determine the displacements if the functions $\chi'(x)$, $\psi_1(x)$, $\psi_2(x)$ are found. For its establishment the conditions by the semi-strip's short edge (4.18)–(4.19) and the crack conditions (6.3) should be fulfilled. For the summation of the weakly-convergent integrals in the formulae for the displacement functions the method presented in Appendix D is used.

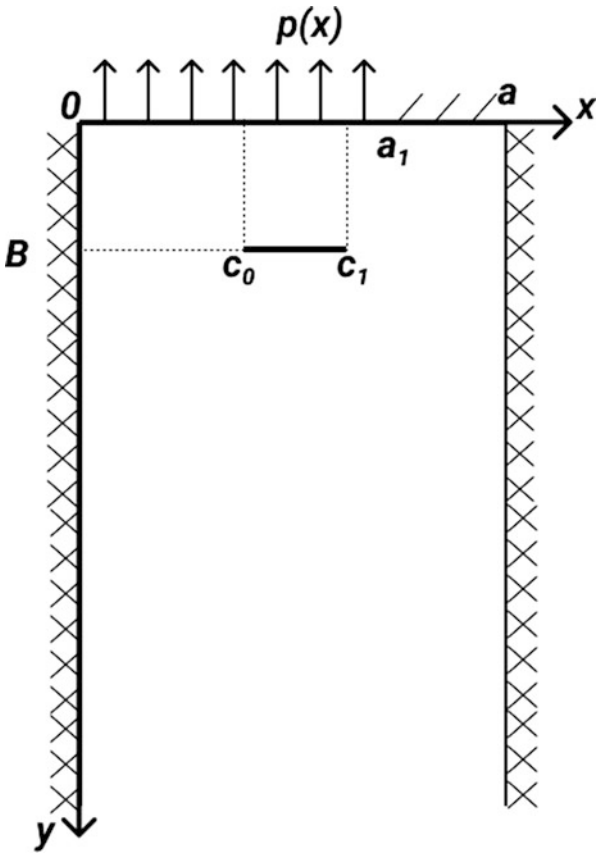


Fig. 6.12 Geometry and coordinate system of the semi-strip with a transverse crack, which is located far from the semi-strip's lateral sides. Case 2

From the condition (4.19) formula (4.24) is obtained. The substitution of the expressions for the displacement functions into the boundary condition at the semi-strip's short edge (4.18) and crack conditions (6.3) reduces to SSIE

$$\begin{cases} \int\limits_{-1}^{1} \widetilde{\chi}(\xi)\left[\dfrac{1}{\xi-x}+\widetilde{Z}(x,\xi)\right]d\xi+\widetilde{K}_0(x)=\widetilde{r}(x), \quad -1<x<1 \\ \dfrac{d^2}{dx^2}\int\limits_{-1}^{1}\widetilde{\psi}_1(\xi)\ln\dfrac{1}{|\xi-x|}d\xi+\widetilde{K}_1(x)=0, \quad -1<x<1 \\ \dfrac{d^2}{dx^2}\int\limits_{-1}^{1}\widetilde{\psi}_2(\xi)\ln\dfrac{1}{|\xi-x|}d\xi+\widetilde{K}_2(x)=\widetilde{q}(x), \quad -1<x<1 \end{cases} \tag{6.33}$$

where $\widetilde{\chi}(\xi)=\chi'\left(\frac{a_1(\xi+1)}{2}\right)$, $\widetilde{\psi}_i(\xi)=\psi_i\left(\frac{(c_1-c_0)\xi+(c_0+c_1)}{2}\right), i=1,2,$

$\widetilde{K}_i(x)=\int\limits_{-1}^{1}\widetilde{\chi}(\xi)\widetilde{f}_i(\xi,x)d\xi+\int\limits_{-1}^{1}\widetilde{\psi}_1(\xi)\widetilde{R}_{i,1}(x,\xi)d\xi+\int\limits_{-1}^{1}\widetilde{\psi}_2(\xi)\widetilde{R}_{i,2}(x,\xi)d\xi, i=0,1,2$

$\widetilde{f}_i(\xi,x),\widetilde{R}_{i,1}(x,\xi),\widetilde{R}_{i,2}(x,\xi),\widetilde{r}(x),\widetilde{q}(x), i=0,1,2$ are known regular functions when $a_1\neq a$, $\widetilde{Z}(x,\xi)=\frac{h_1}{\xi+x+2}+\frac{h_2(x+1)}{(\xi+x+2)^2}+\frac{h_3(\xi+1)(x+1)}{(\xi+x+2)^3}$, $h_1=-\frac{\kappa^2-3}{2\kappa}, h_2=-\frac{2}{\kappa}, h_3=\frac{4}{\kappa}$.

The transcendental equation for the determination of the singularity order of the first equation's in (6.33) solution is equivalent to the transcendental equation for (3.29).

The method presented in Sect. 2.6.2 is applied for the solving of SSIE (6.33). According to it the unknown function $\widetilde{\chi}(\xi)$ is expanded into sum at each interval

$$\widetilde{\chi}(\xi)=\sum_{k=0}^{N-1}\left[s_k^0\rho_k^-(\xi)+s_{k+N}^0\rho_k^+(\xi)\right], \quad \xi\in[-1;1], \tag{6.34}$$

here, $\rho_{2k}^-(\xi)=(1+\xi)^{\operatorname{Re}\lambda_k}\cdot\cos(\operatorname{Im}\lambda_k\ln(1+\xi)), \rho_{2k+1}^-(\xi)=(1+\xi)^{\operatorname{Re}\lambda_k}\cdot\sin(\operatorname{Im}\lambda_k\ln(1+\xi)), \quad k=\overline{0,N-1}, \quad \rho_k^+(\xi)=\frac{T_k(\xi)}{\sqrt{1-\xi}}, k=\overline{0,N-1}, \quad T_k(x)$ are Chebyshev polynomials of the first kind.

The functions $\widetilde{\psi}_i(\xi)$ are expanded in sums by the Chebyshev polynomials of the second kind (6.29).

The segment $[-1;1]$ is divided into $2N$ intervals by the points $x_i: P_{2N-1}^{\lambda_0,-0.5}(x_i)=0, i=\overline{0,2N-1}$. SSIE (6.33) is considered when $x=x_i$.

After the substitution of the expressions for the unknown functions (6.34), (6.29) into SSIE (6.33) with consideration of the condition (3.28) the system of $6N$ linear algebraic equation regarding unknown constants $s_k^i, i=0,1,2, k=\overline{0,2N-1}$ is derived in the form (6.30). The substitution of the found constants $s_k^i, i=0,1,2, k=\overline{0,2N-1}$ into formulae (6.24)–(6.25) completes the construction of the problem's solution. SIF are calculated by formulae (6.31)–(6.32).

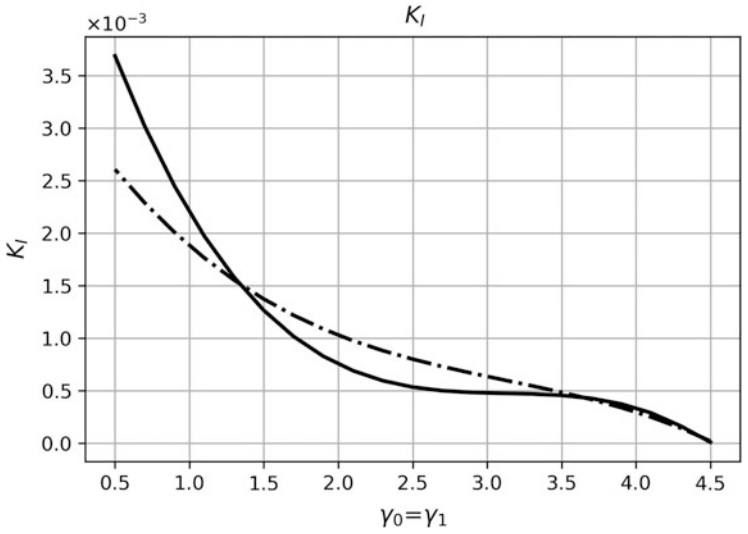


Fig. 6.13 SIF K_I under symmetric change of the crack's length, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa

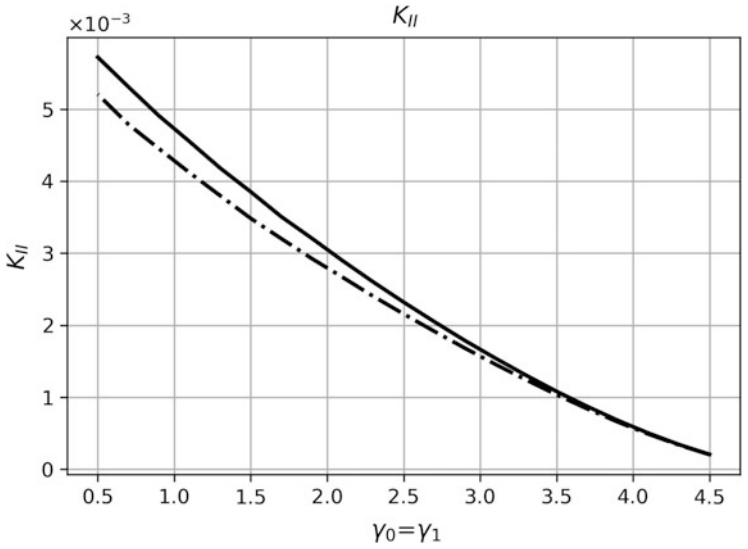


Fig. 6.14 SIF K_{II} under symmetric change of the crack's length, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa

In Figs. 6.13, 6.14, 6.15 and 6.16 the continuous line describes SIF at the left crack tip, and stroke-dotted line describe SIF at the right crack tip.

In figures (Figs. 6.13 and 6.14) graphics for SIF K_I, K_{II} are presented when $G = 61.2781955 \cdot 10^9$ Pa, $\mu = 0.33$, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa, $a = 10$ m.

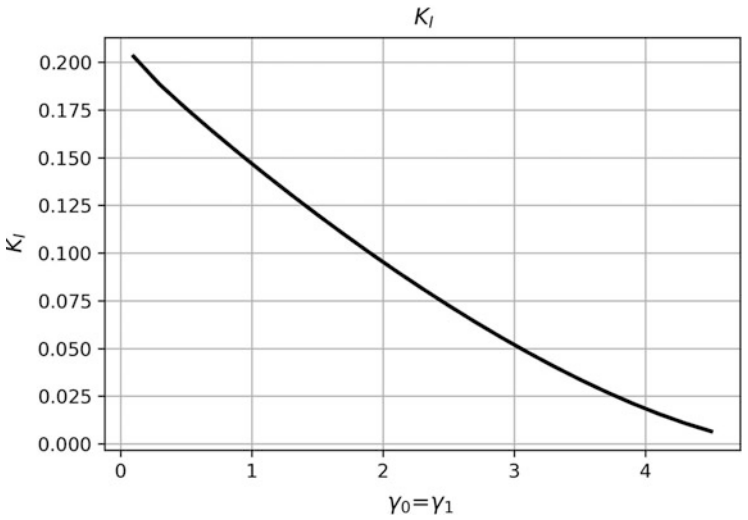


Fig. 6.15 SIF K_I under symmetric change of the crack's length, $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa

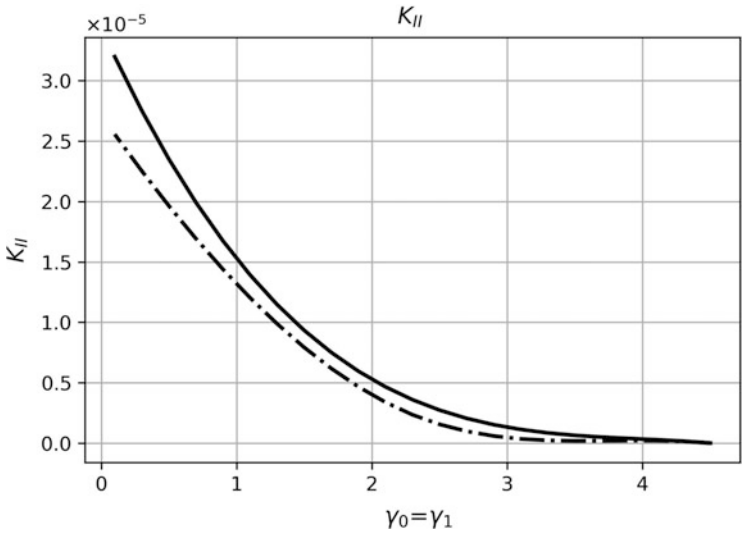


Fig. 6.16 SIF K_{II} under symmetric change of the crack's length, $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa

As it can be seen, it is necessary to consider the fixed singularity at the left crack tip when $\gamma_0 < 5\ \%\ a$ and at the right crack tip when $\gamma_1 < 11\ \%\ a$.

In this case SIF K_I and K_{II} have the same order, and the values of SIF K_{II} are larger than the correspondent values of SIF K_I.

Plots for SIF K_I, K_{II} are depicted in figures (Figs. 6.15 and 6.16) when $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa.

The SIF K_{I-}, K_{I+} coincide by its values correspondingly and SIF K_{II-}, K_{II+} are different.

In this case stable results are obtained under the parameters' values $\gamma_0 = \gamma_1 \geq 1\ \% a$, and when $\gamma_0 = \gamma_1 < 1\ \% a$ the fixed singularities at the crack tips should be taken into account.

In this case of loading SIF K_I are significantly bigger than SIF K_{II}.

6.5 The Stress Concentration Problem for a Fixed Semi-strip with a Transverse Crack. Symmetric Case

The elastic semi-strip (Fig. 6.17), which occupies the area in Cartesian coordinates $0 < x < a$, $0 < y < \infty$ is considered. The semi-strip is loaded by its short edge $y = 0$, $0 < x < a$ (3.5)–(3.6) by the normal load $p(x)$.

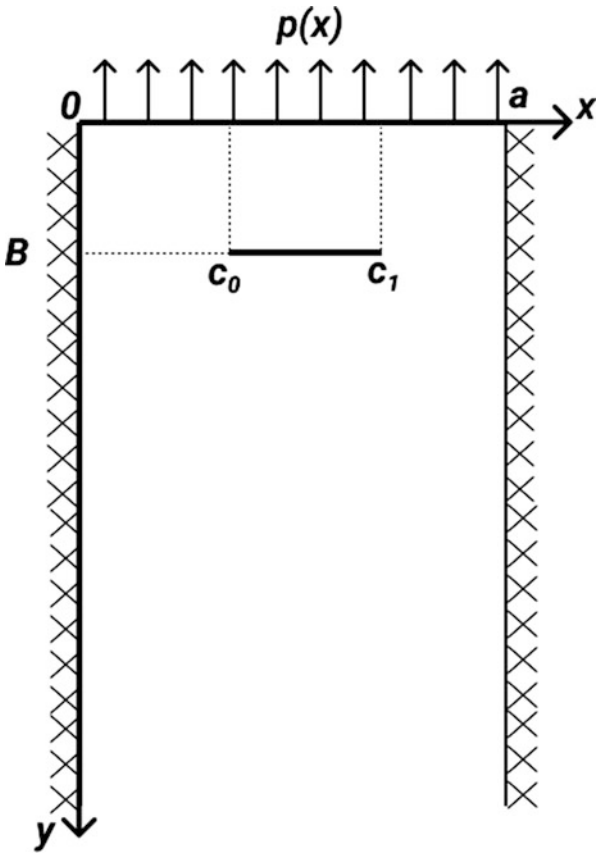


Fig. 6.17 Geometry and coordinate system of the semi-strip with a transverse crack in the symmetric case

The displacements in the semi-strip satisfy the equilibrium equations (3.4). The generalized boundary conditions on the semi-infinite lateral sides are detailed in the form (4.3). SIF should be found when no volume forces act on the semi-strip.

The transverse crack is located inside the semi-strip by the line $c_0 < x < c_1, y = B$. Consider the case when the crack is located far from the semi-strip's lateral sides and it is symmetric with respect to the semi-strip's center, i.e. $c_0 = a - c_1$. Also it is done that $a_0 = a - a_1$ and the load $p(x)$ is a symmetric function with regard to the semi-strip's center, i.e.

$$p\left(\frac{a}{2} - x\right) = p\left(\frac{a}{2} + x\right), 0 < x < \frac{a}{2}. \tag{6.35}$$

Then conditions (6.1)–(6.3) take the form

$$\begin{aligned} &u(x, B-0) - u(x, B+0) = \langle u(x, B)\rangle = 0, \quad c_0 < x < c_1, \\ &v(x, B-0) - v(x, B+0) = \langle v(x, B)\rangle = \psi_2(x) \neq 0, \quad c_0 < x < c_1, \end{aligned} \tag{6.36}$$

$$\sigma_y(x, B-0) - \sigma_y(x, B+0) = \langle \sigma_y(x, B)\rangle = 0, \quad c_0 < x < c_1, \tag{6.37}$$

$$\sigma_y\big|_{y=B\pm 0} = p_\sigma(x), \quad c_0 < x < c_1. \tag{6.38}$$

The condition $\langle \tau_{xy}(x, B)\rangle = 0$ is fulfilled automatically in this case.

The one-dimensional boundary problem in the transformation domain (6.4) takes the following form:

$$\begin{cases} u''_\beta(x) - \dfrac{\beta^2(\kappa\text{-}1)}{\kappa+1} u_\beta(x) + \dfrac{2\beta}{\kappa+1} v'_\beta(x) = \dfrac{3-\kappa}{\kappa+1}\chi'(x) + \\ \qquad + \cos\beta B \dfrac{\kappa-3}{\kappa+1}\psi'_2(x), \\ v''_\beta(x) - \dfrac{\beta^2(\kappa+1)}{\kappa-1} v_\beta(x) - \dfrac{2\beta}{\kappa-1} u'_\beta(x) = -\beta\dfrac{\kappa+1}{\kappa-1}\chi(x) + \\ \qquad + \beta\dfrac{\kappa+1}{\kappa-1}\cos\beta B\psi_2(x), \\ u_\beta(0) = 0, u_\beta(a) = 0, \\ v_\beta(0) = 0, v_\beta(a) = 0. \end{cases} \tag{6.39}$$

The problem (6.39) is reformulated as the continuous boundary problem by the scheme from Chap. 2. The right-hand side of the vector equation (3.10) is presented by formula $\vec{f}(x) = \begin{pmatrix} \dfrac{3-\kappa}{\kappa+1}\chi'(x) + \cos\beta b \dfrac{\kappa-3}{\kappa+1}\psi'_2(x) \\ -\beta\dfrac{\kappa+1}{\kappa-1}\chi(x) + \beta\dfrac{\kappa+1}{\kappa-1}\cos\beta b\psi_2(x) \end{pmatrix}$. The correspondences for the displacement functions (6.5)–(6.6) are simplified

$$u_\beta(x) = Y_1^{11}(x)c_1 + Y_1^{12}(x)c_2 + Y_2^{11}(x)c_3 + Y_2^{12}(x)c_4 +$$
$$+\frac{3-\kappa}{\kappa+1}\int_0^a G^{11}(x,\xi)\chi'(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int_0^a G^{12}(x,\xi)\chi(\xi)d\xi +$$
$$+\cos\beta B\frac{\kappa-3}{\kappa+1}\int_0^a G^{11}(x,\xi)\psi'_2(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\cos\beta B\int_0^a G^{12}(x,\xi)\psi_2(\xi)d\xi, \tag{6.40}$$

$$v_\beta(x) = Y_1^{21}(x)c_1 + Y_1^{22}(x)c_2 + Y_2^{21}(x)c_3 + Y_2^{22}(x)c_4 +$$
$$+\frac{3-\kappa}{\kappa+1}\int_0^a G^{21}(x,\xi)\chi'(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int_0^a G^{22}(x,\xi)\chi(\xi)d\xi +$$
$$+\cos\beta B\frac{\kappa-3}{\kappa+1}\int_0^a G^{21}(x,\xi)\psi'_2(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\cos\beta B\int_0^a G^{22}(x,\xi)\psi_2(\xi)d\xi. \tag{6.41}$$

The coefficients $c_i, i=\overline{1,4}$ take the form (C.2).

6.6 The Load at the Center of the Semi-strip's Short Edge in the Symmetric Statement (Case 1)

The semi-strip, which is loaded at the center of its short edge and the crack is located inside the semi-strip far from its lateral sides is considered in the symmetric statement (Fig. 6.18). The boundary problem (3.4), (4.9)–(4.11), (6.36)–(6.38) when $a_0 \neq 0$, $a_1 \neq a$ should be solved for the elastic semi-strip with the conditions at the semi-strip's lateral sides (4.3).

The correspondences (6.7)–(6.8) are simplified

$$u_\beta(x) = Y_1^{11}(x)c_1 + Y_1^{12}(x)c_2 + Y_2^{11}(x)c_3 + Y_2^{12}(x)c_4 -$$
$$-\frac{3-\kappa}{\kappa+1}\int_0^a \frac{\partial}{\partial\xi}G^{11}(x,\xi)\chi(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int_0^a G^{12}(x,\xi)\chi(\xi)d\xi -$$
$$-\cos\beta B\frac{\kappa-3}{\kappa+1}\int_{c_0}^{c_1}\frac{\partial G^{11}}{\partial\xi}(x,\xi)\psi_2(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\cos\beta B\int_{c_0}^{c_1} G^{12}(x,\xi)\psi_2(\xi)d\xi, \tag{6.42}$$

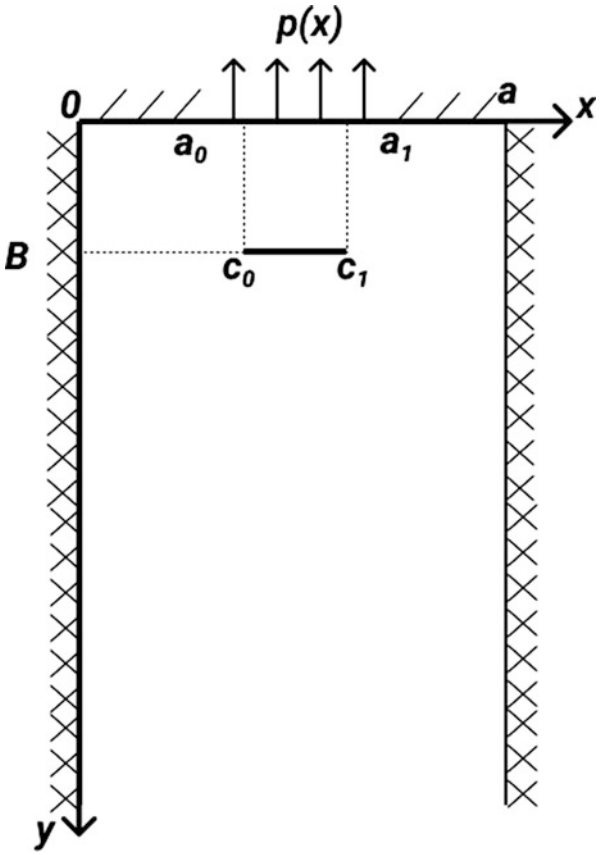


Fig. 6.18 Geometry and coordinate system of the semi-strip with a transverse crack, which is located far from the semi-strip's lateral sides in the symmetric statement. Case 1

$$v_\beta(x) = Y_1^{21}(x)c_1 + Y_1^{22}(x)c_2 + Y_2^{21}(x)c_3 + Y_2^{22}(x)c_4 -$$
$$- \frac{3-\kappa}{\kappa+1}\int_0^a \frac{\partial}{\partial\xi}G^{21}(x,\xi)\chi(\xi)d\xi - \beta\frac{\kappa+1}{\kappa-1}\int_0^a G^{22}(x,\xi)\chi(\xi)d\xi -$$
$$- \cos\beta B\frac{\kappa-3}{\kappa+1}\int_{c_0}^{c_1}\frac{\partial G^{21}}{\partial\xi}(x,\xi)\psi_2(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\cos\beta B\int_{c_0}^{c_1}G^{22}(x,\xi)\psi_2(\xi)d\xi, \tag{6.43}$$

After the substitution of the constructed earlier $Y_1(x), Y_2(x), c_i, i=\overline{1,4}$ into formulae (6.42)–(6.43) and the inversion of the derived transformants according to the correspondence (3.8) the following expressions for the displacement functions are obtained:

$$u(x,y)=\int\limits_0^a \chi(\xi)f_u(x,y,\xi)d\xi+\int\limits_{c_0}^{c_1}\psi_2(\xi)g_{u2}(x,y,\xi)d\xi, \tag{6.44}$$

$$v(x,y)=\int\limits_0^a \chi(\xi)f_v(x,y,\xi)d\xi+\int\limits_{c_0}^{c_1}\psi_2(\xi)g_{v2}(x,y,\xi)d\xi. \tag{6.45}$$

Formulae (6.44)–(6.45) will finally determine the displacements if the functions $\chi(x)$, $\psi_2(x)$ are found. For its finding the conditions at the semi-strip's short edge (4.9) –(4.10), which are not fulfilled yet, and the following crack conditions should be satisfied

$$\sigma_y(x,B+0)=0 \tag{6.46}$$

The conditions (4.9), (6.46) can't be fulfilled by the direct substitution of the correspondences (6.44)–(6.45) into them, because the integrals in the expressions for the displacements are weakly-convergent. So the method, presented in Appendix D, is used for the separation of its weakly-convergent parts.

From the condition (4.10) the expression (4.15) is derived. The substitution of the formulae for the displacements into the boundary condition (3.5) and the crack condition (6.46) reduces to SSIDE of the form

$$\begin{cases} \dfrac{d^2}{dx^2}\displaystyle\int\limits_{-1}^{1}\widetilde{\chi}(\xi)\ln\frac{1}{|\xi-x|}d\xi+\widetilde{K}_0(x)=\widetilde{r}(x), \quad -1<x<1 \\ \dfrac{d^2}{dx^2}\displaystyle\int\limits_{-1}^{1}\widetilde{\psi}_2(\xi)\ln\frac{1}{|\xi-x|}d\xi+\widetilde{K}_2(x)=\widetilde{q}(x), \quad -1<x<1, \end{cases} \tag{6.47}$$

where $\widetilde{\chi}(\xi)=\chi\left(\frac{(a_1-a_0)\xi+(a_0+a_1)}{2}\right)$, $\widetilde{\psi}_2(\xi)=\psi_2\left(\frac{(c_1-c_0)\xi+(c_0+c_1)}{2}\right)$,

$\widetilde{K}_i(x)=\int\limits_{-1}^{1}\widetilde{\chi}(\xi)\widetilde{f}_i(\xi,x)d\xi+\int\limits_{-1}^{1}\widetilde{\psi}_2(\xi)\widetilde{R}_{i,2}(x,\xi)d\xi, i=0,2$,

$\widetilde{f}_i(\xi,x),\widetilde{R}_{i,2}(x,\xi),\widetilde{r}(x),\widetilde{q}(x), i=0,1,2$ are known regular functions when $a_0\neq 0$, $a_1\neq a$.

SSIDE (6.47) is solved approximately with the help of the orthogonal polynomials method applied by the scheme presented in Sect. 5.2. The functions $\widetilde{\chi}(\xi),\widetilde{\psi}_2(\xi), i=1,2$ are expanded in series by the Chebyshev polynomials of the second kind (6.13)–(6.14). The infinity system of linear algebraic equations regarding unknown coefficients $s_n^i, i=0,2, n=0,1,\ldots$ is derived

$$\vec{S}_m + \sum_{n=0}^{\infty} D_{mn}\vec{S}_n = \vec{f}_m, \quad m = 0, 1, 2 \ldots \tag{6.48}$$

here, $\vec{S}_m = \begin{pmatrix} s_m^0 \\ s_m^2 \end{pmatrix}$, components of $D_{mn} = \{d_{mn}^{ij}\}, i,j = 0, 2$, $\vec{f}_m = \begin{pmatrix} f_m^0 \\ f_m^2 \end{pmatrix}$ are known constants.

The system (6.48) is solved by the reduction method. The substitution of the found constants $s_n^i, i = 0, 2, n = 0, 1, \ldots$ into formulae (6.13)–(6.14) and its further usage in formulae (6.44)–(6.45) completes the construction of the problem's solution. SIF are calculated by formulae (6.20).

In Fig. 6.19 the graphic of SIF K_I is presented when $G = 61.2781955 \cdot 10^9$ Pa, $\mu = 0.33$, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa, $a = 10$ m, $a_0 = a - a_1 = 1$.

The SIF values are decreasing to zero when the crack length is decreasing. From the comparison of SIF K_I in Fig. 6.3 for the general case and SIF K_I in Fig. 6.19 for the symmetric case it can be concluded that corresponding values of K_I are equal for these two cases. Under the parameters' values $\gamma_0 = \gamma_1 < 15\ \%\ a$ the calculations become unstable, and the fixed singularities at crack tips should be considered.

In Fig. 6.20 the plot for SIF K_I is depicted when $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa.

As it can be seen from graphics, the decreasing of the crack length leads to the decreasing to zero of the SIF values K_I. The values of SIF K_I coincide with the corresponding SIF values K_I for the general case (Fig. 6.5). Stable calculations are derived under the following values of the parameters $\gamma_0 = \gamma_1 \geq 5\ \%\ a$, and when $\gamma_0 = \gamma_1 < 5\ \%\ a$ the fixed singularities at crack tips should be taken into account.

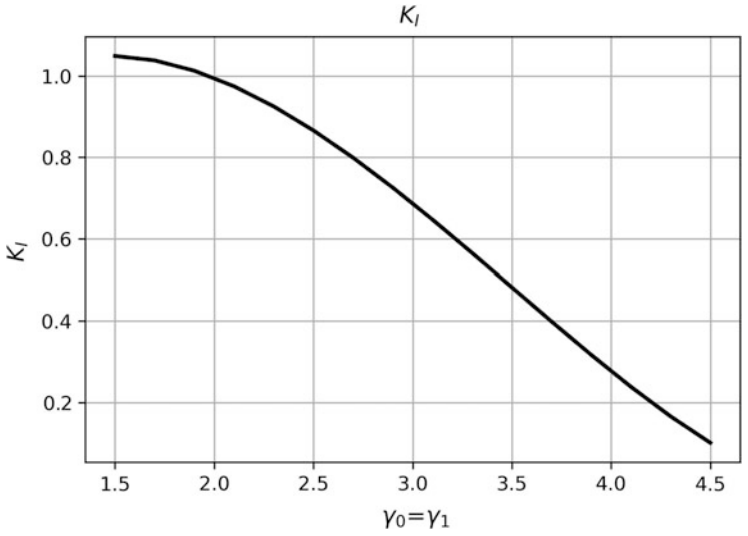


Fig. 6.19 SIF K_I under the symmetric change of the crack length, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa

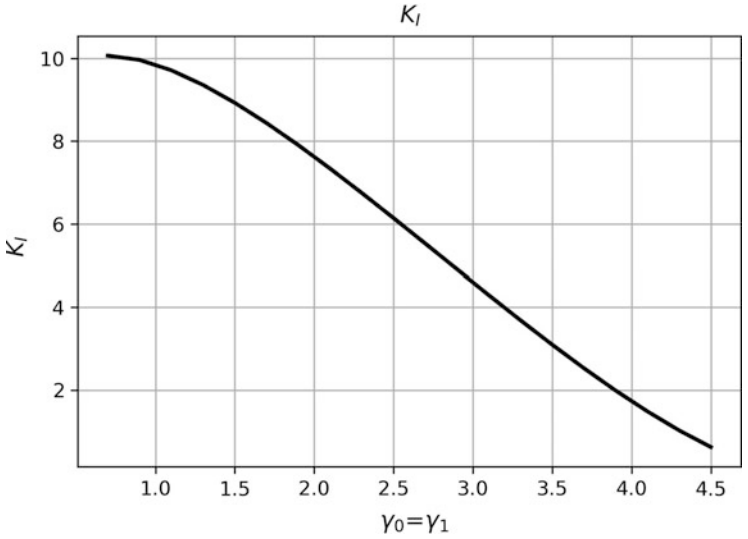


Fig. 6.20 SIF K_I under the symmetric change of the crack length, $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa

6.7 The Load at the Whole Semi-strip's Short Edge in the Symmetric Statement (Case 3)

The semi-strip, which is loaded at the whole short edge and the crack is located inside the semi-strip far from its lateral sides is considered (Fig. 6.21). The boundary problem (3.4)–(3.6), (6.36)–(6.38) for the elastic semi-strip with conditions (4.3) at the semi-strip's lateral sides should be solved.

The correspondences (6.22)–(6.23) are simplified

$$u_\beta(x) = Y_1^{11}(x)c_1 + Y_1^{12}(x)c_2 + Y_2^{11}(x)c_3 + Y_2^{12}(x)c_4 +$$
$$+\frac{3-\kappa}{\kappa+1}\int_0^a G^{11}(x,\xi)\chi'(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\int_0^a \Im^{12}(x,\xi)\chi'(\xi)d\xi -$$
$$-\cos\beta B\frac{\kappa-3}{\kappa+1}\int_{c_0}^{c_1}\frac{\partial G^{11}}{\partial\xi}(x,\xi)\psi_2(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\cos\beta B\int_{c_0}^{c_1} G^{12}(x,\xi)\psi_2(\xi)d\xi, \tag{6.49}$$

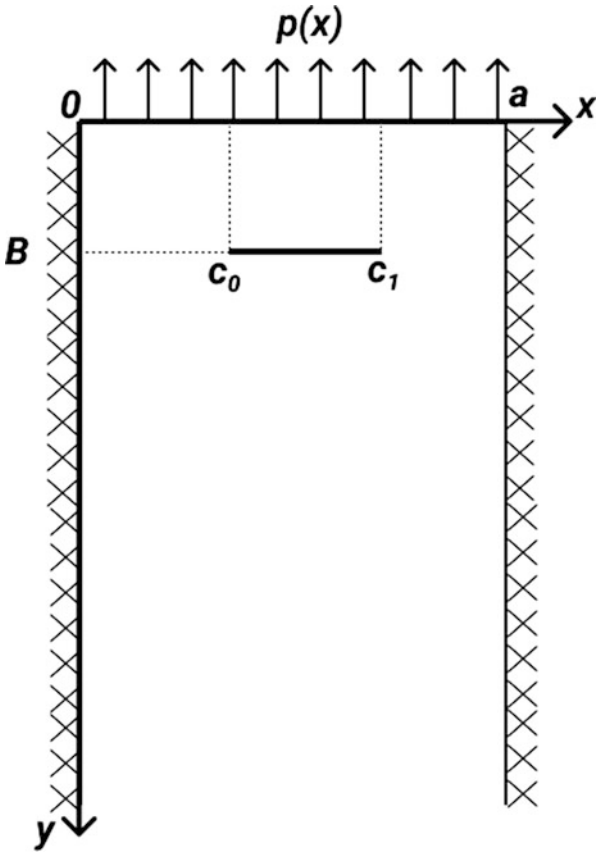


Fig. 6.21 Geometry and coordinate system of the semi-strip with a transverse crack, which is located far from the semi-strip's lateral sides in the symmetric statement. Case 3

$$v_\beta(x) = Y_1^{21}(x)c_1 + Y_1^{22}(x)c_2 + Y_2^{21}(x)c_3 + Y_2^{22}(x)c_4 +$$
$$+\frac{3-\kappa}{\kappa+1}\int_0^a G^{21}(x,\xi)\chi'(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\int_0^a \Im^{22}(x,\xi)\chi'(\xi)d\xi - \frac{\kappa-1}{\beta^2(\kappa+1)}\chi(x) -$$
$$-\cos\beta B\frac{\kappa-3}{\kappa+1}\int_{c_0}^{c_1}\frac{\partial G^{21}}{\partial\xi}(x,\xi)\psi_2(\xi)d\xi + \beta\frac{\kappa+1}{\kappa-1}\cos\beta B\int_{c_0}^{c_1} G^{22}(x,\xi)\psi_2(\xi)d\xi, \tag{6.50}$$

The substitution of the fundamental matrix solving system $Y_1(x), Y_2(x), c_i, i = \overline{1,4}$ into formulae (6.49)–(6.50) and inversion of the obtained transformants according to formulae (3.8) reduce to the following correspondences for the displacement functions:

$$u(x,y)=\int\limits_0^a \chi'(\xi)f_u(x,y,\xi)d\xi+\int\limits_{c_0}^{c_1}\psi_2(\xi)g_{u2}(x,y,\xi)d\xi, \tag{6.51}$$

$$v(x,y)=\int\limits_0^a \chi'(\xi)f_v(x,y,\xi)d\xi+\int\limits_{c_0}^{c_1}\psi_2(\xi)g_{v2}(x,y,\xi)d\xi+\int\limits_0^x \chi'(\xi)d\xi \tag{6.52}$$

Formulae (6.51)–(6.52) will completely determine the displacements if the functions $\chi'(x)$, $\psi_2(x)$ are known. For their finding the unfulfilled condition at the semi-strip's short edge (3.5) and crack condition (6.46) should be satisfied. The weakly-convergent integrals in the expressions for the displacement functions are summed up by the method, which is presented in Appendix D.

The substitution of the formulae for the displacements into the boundary condition (3.5) and into the crack condition (6.46) leads to SSIE of the following form:

$$\begin{cases} \int\limits_{-1}^{1}\widetilde{\chi}(\xi)\left[\dfrac{1}{\xi-x}+\widetilde{Z}(x,\xi)\right]d\xi+\widetilde{K}_0(x)=\widetilde{r}(x), \quad -1<x<1 \\ \dfrac{d^2}{dx^2}\int\limits_{-1}^{1}\widetilde{\psi}_2(\xi)\ln\dfrac{1}{|\xi-x|}d\xi+\widetilde{K}_2(x)=\widetilde{q}(x), \quad -1<x<1, \end{cases} \tag{6.53}$$

here, $\widetilde{\chi}(\xi)=\chi'\left(\frac{a(\xi+1)}{2}\right)$, $\widetilde{\psi}_2(\xi)=\psi_2\left(\frac{(c_1-c_0)\xi+(c_0+c_1)}{2}\right)$,

$\widetilde{K}_i(x)=\int\limits_{-1}^{1}\widetilde{\chi}(\xi)\widetilde{f}_i(\xi,x)d\xi+\int\limits_{-1}^{1}\widetilde{\psi}_2(\xi)\widetilde{R}_{i,2}(x,\xi)d\xi, i=0,2$

$\widetilde{f}_i(\xi,x),\widetilde{R}_{i,2}(x,\xi),\widetilde{r}(x),\widetilde{q}(x), i=0,1,2$ are known regular functions when $a_0\neq 0$, $a_1\neq a$, $\widetilde{Z}(x,\xi)=h_1\left(\frac{1}{\xi+x+2}+\frac{1}{\xi+x-2}\right)+h_2\left(\frac{x+1}{(\xi+x+1)^2}+\frac{x-1}{(\xi+x-2)^2}\right)+$

$+h_3\left(\frac{(x+1)(\xi+1)}{(\xi+x+2)^3}+\frac{(x-1)(\xi-1)}{(\xi+x-2)^3}\right)$.

The transcendental equation for the determination of the singular order of the first equation's in (6.53) solution coincides with the transcendental equation for (3.29).

The method presented in Sect. 2.6.2 is used for the solving of SSIE (6.53). According to it, the unknown function $\widetilde{\chi}(\xi)$ is expanded into sums at each interval (6.28). The function $\widetilde{\psi}_2(\xi)$ is expanded into sum by the Chebyshev polynomial of the second kind (6.29). Segment $[-1;1]$ is divided into $2N$ intervals by the points $x_i: P_{2N-1}^{\lambda_0,\lambda_0}(x_i)=0, i=\overline{0,2N-1}$. SSIE (6.53) is considered when $x=x_i$.

After the substitution of the expressions for the unknown functions (6.28), (6.29) into SSIE (6.53) with consideration of the condition (3.28) the system of $4N$ linear algebraic equations regarding unknown constants $s_k^i, i=0,2, k=\overline{0,2N-1}$ is derived.

$$\sum_{n=0}^{2N-1} D_{mn} \vec{S}_n = \vec{f}_m, \quad m = \overline{0, 2N-1} \tag{6.54}$$

where $\vec{S}_m = \begin{pmatrix} s_m^0 \\ s_m^2 \end{pmatrix}$, components of $D_{mn} = \{d_{mn}^{ij}\}, i,j = 0,2, \vec{f}_m = \begin{pmatrix} f_m^0 \\ f_m^2 \end{pmatrix}$ are known constants.

The substitution of the found constants $s_k^i, i = 0,2, k = \overline{0, 2N-1}$ into formulae (6.51)–(6.52) completes the construction of the problem's solution. SIF are calculated by formulae (6.31).

In Fig. 6.22 the graphic for SIF K_I is presented when $G = 61.2781955 \cdot 10^9$ Па, $\mu = 0.33$, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa, $a = 10$ m, $a_0 = a - a_1 = 1$.

As it can be seen from the plot, SIF values K_I, K_{II} are decreasing to zero when the crack length is decreasing. The values of SIF K_I are equal to the corresponding SIF values K_I for the general case (Fig. 6.8). The stable calculations are obtained when $\gamma_0 = \gamma_1 \geq 5\ \%\ a$ and when $\gamma_0 = \gamma_1 < 5\ \%\ a$ the fixed singularities at the crack tips should be considered.

In figure (Fig. 6.23) the graphic for SIF K_I is depicted when $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa.

When the crack length is decreasing the SIF values K_I are decreasing to zero. The comparison with the general case (Fig. 6.10) shows that the SIF values K_I for this case are equal to the corresponding SIF values K_I for the general case. The fixed singularities at the crack tips should be taken into account when $\gamma_0 = \gamma_1 < 1\ \%\ a$.

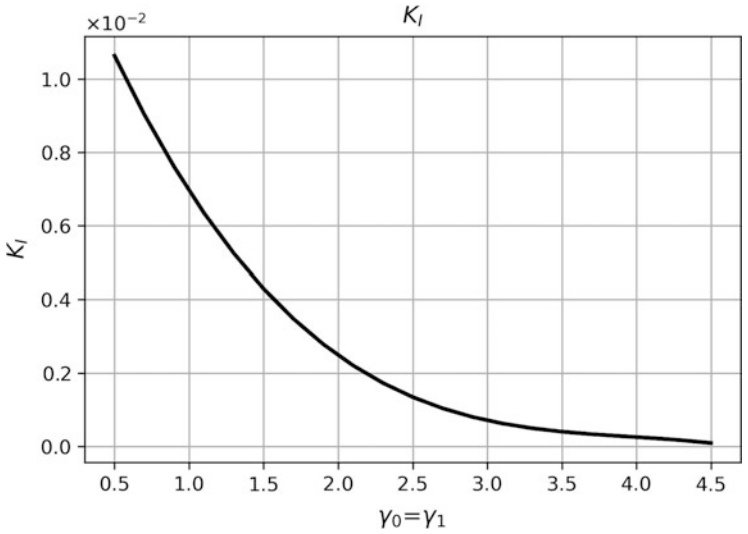


Fig. 6.22 SIF K_I under the symmetric change of the crack length, $p(x) = 1$ Pa, $p_\sigma(x) = 0$ Pa

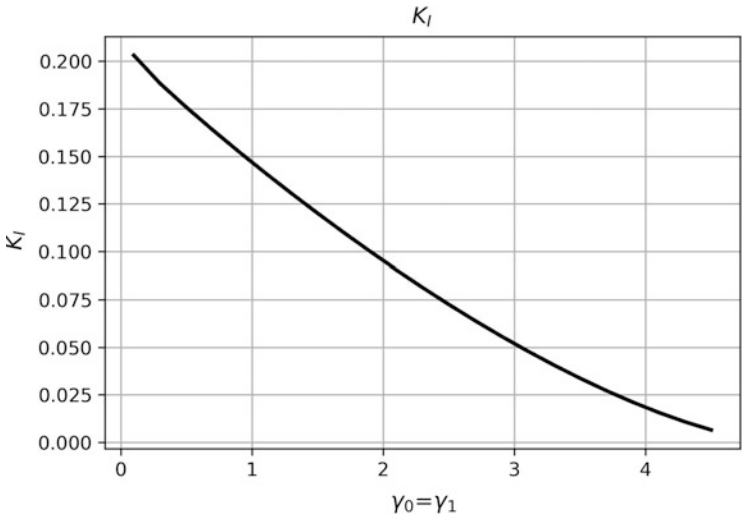


Fig. 6.23 SIF K_I under the symmetric change of the crack length, $p(x) = 0$ Pa, $p_\sigma(x) = 1$ Pa

6.8 Conclusions to the Sixth Chapter

The solutions of the plane elasticity problems for a semi-strip with a transverse crack are derived for three configurations of the applied load by the semi-strip's short edge. The distances between crack tips and the semi-strip's lateral sides when the fixed singularities of the unknown functions at the ends of the integration intervals should be considered are established.

Stress intensity coefficients are investigated for three cases of the applied load. The comparative analysis is carried out. The maximal SIF are observed for the case of mechanical load's application at the center of the semi-strip's short edge.

The symmetric case is considered, the conditions of its use are established.

Conclusions

The material of this book shows the importance of the theory of boundary mixed problems of mathematical physics in the application to the modeling of quite complicated engineering problems. The first chapter of the monograph contains all the necessary theoretical concepts used in the future to solve problems on the stress state of a semi-strip. On the basis of this theoretical basis, in the second chapter, a new analytical method for solving mixed problems of elasticity for a rectangular region is developed and proposed. In the following chapters, the mixed plane problem for a semi-infinite elastic strip under the load of a different nature applied by the semi-strip's short edge is investigated. The problem for the case when the semi-strip is weakened by the transverse crack is considered. The following main results are derived:

1. The new method is proposed. Its unlikely traditional approaches for the solving of the mixed plane problem for a semi-strip allow to avoid additional harmonic, biharmonic functions and to operate directly with the equilibrium equations. The solutions obtained by this method are exact presentation of the searched semi-strip's mechanical characteristics.
2. Green's function is constructed as the bilinear form for the problems of a stress state for the semi-strip. It allows to essentially simplificate the numerical calculations. The solving is reduced to one singular integral equation which contains fixed singularities. The transcendental equation is constructed to consider the influence of the fixed singularities. The roots of transcendental equation are found, and the efficient numeric-analytical method for the solving of the singular integral equation is applied.
3. The efficient approximate solution of the problem for a semi-strip's stress state weakened by the transverse crack is constructed. The dependence of the stress intensity factor from the crack length and the configuration of the load applied by the semi-strip's short edge are obtained.

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These results allow to set the following mechanical features of the stress and displacement distribution in the semi-strip:

– The maximal normal stresses in the case of mechanical load are observed at the semi-strip's lateral sides when the load is applied by the center of the semi-strip's short edge.
– The influence of the thermal load on the stress field of the semi-strip and the dependence of the stress values from the ratio of the thermal and mechanical load are established.
– The boundaries of application of the proposed method are established in dependence of the transverse crack's length for three configurations of the mechanical load by the semi-strip's short edge. Namely, the crack length should be not bigger than 90 % of a. When these sizes are exceeded the fixed singularities which appear in corresponding singular equations should be taken into consideration. The maximal SIF values are observed when the load is concentrated at the center of the semi-strip's short edge.

Appendix A: Step-by-Step Integration of the Lame's Equation by the Variable *y*

The first equation in (3.4) is multiplied by $\cos\beta y$ and integrated by the variable y on the segment from 0 to infinity. The second equation in (3.4) is multiplied by $\sin\beta y$ and integrated by the variable y on the segment from 0 to infinity. The decreasing of the displacement functions $u(x, y)$ and $v(x, y)$ on the infinity, and the introduced change of variables $v(x, 0) = \chi(x)$, $v'(x, 0) = \chi'(x)$, $u^{\bullet}(x, 0) = -\chi'(x)$ are used.

$$\int_0^\infty \frac{\partial^2 u(x,y)}{\partial x^2}\cos\beta y dy + \frac{\kappa-1}{\kappa+1}\int_0^\infty \frac{\partial^2 u(x,y)}{\partial y^2}\cos\beta y dy + \frac{2}{\kappa+1}\int_0^\infty \frac{\partial^2 v(x,y)}{\partial x\partial y}\cos\beta y dy +$$

$$+\frac{\kappa-1}{\kappa+1}\int_0^\infty X_1(x,y)\cos\beta y dy = 0;$$

$$\int_0^\infty \frac{\partial^2 u(x,y)}{\partial x^2}\cos\beta y dy = \frac{d^2}{dx^2}\int_0^\infty u(x,y)\cos\beta y dy = \frac{d^2}{dx^2}u_\beta(x);$$

$$\int_0^\infty \frac{\partial^2 u(x,y)}{\partial y^2}\cos\beta y dy = \left[\frac{\partial u(x,y)}{\partial y}\cos\beta y\right]\Big|_{y=0}^{y=\infty} + \beta\int_0^\infty \frac{\partial u(x,y)}{\partial y}\sin\beta y dy = \chi'(x) +$$

$$+\beta[u(x,y)\sin\beta y]\Big|_{y=0}^{y=\infty} - \beta^2\int_0^\infty u(x,y)\cos\beta y dy = \chi'(x) - \beta^2 u_\beta(x);$$

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$$\int_0^\infty \frac{\partial^2 v(x,y)}{\partial x \partial y}\cos\beta y dy = \frac{d}{dx}\int_0^\infty \frac{\partial v(x,y)}{\partial y}\cos\beta y dy =$$

$$= \frac{d}{dx}\left([v(x,y)\cos\beta y]\Big|_{y=0}^{y=\infty} + \beta\int_0^\infty v(x,y)\sin\beta y dy \right)$$

$$= \frac{d}{dx}\big(-\chi(x) + \beta v_\beta(x)\big) = -\chi'(x) + \beta v'_\beta(x)$$

Then, the first equation in the transformation domain takes the form:

$$u_\beta''(x) - \frac{\beta^2(\kappa-1)}{\kappa+1}u_\beta(x) + \frac{2\beta}{\kappa+1}v_\beta'(x) = \frac{3-\kappa}{\kappa+1}\chi'(x) - \frac{\kappa-1}{\kappa+1}X_{1\beta}(x).$$

The integral transformation is applied to the second equation

$$\int_0^\infty \frac{\partial^2 v}{\partial x^2}(x,y)\sin\beta y dy + \frac{\kappa+1}{\kappa-1}\int_0^\infty \frac{\partial^2 v}{\partial y^2}(x,y)\sin\beta y dy - \frac{2}{\kappa-1}\int_0^\infty \frac{\partial^2 u}{\partial x \partial y}(x,y)\sin\beta y dy +$$

$$+ \int_0^\infty X_2(x,y)\sin\beta y dy = 0;$$

$$\int_0^\infty \frac{\partial^2 v(x,y)}{\partial x^2}\sin\beta y dy = \frac{d^2}{dx^2}\int_0^\infty v(x,y)\sin\beta y dy = v_\beta''(x);$$

$$\int_0^\infty \frac{\partial^2 v(x,y)}{\partial y^2}\sin\beta y dy = \left[\frac{\partial v(x,y)}{\partial y}\sin\beta y\right]\Big|_{y=0}^{y=\infty} - \beta\int_0^\infty \frac{\partial v(x,y)}{\partial y}\cos\beta y dy =$$

$$= -\beta[v(x,y)\cos\beta y]\Big|_{y=0}^{y=\infty} - \beta^2\int_0^\infty v(x,y)\sin\beta y dy = \beta\chi(x) - \beta^2 v_\beta(x);$$

$$\int_0^\infty \frac{\partial^2 u(x,y)}{\partial x \partial y}\sin\beta y dy = \frac{d}{dx}\int_0^\infty \frac{\partial u(x,y)}{\partial x}\sin\beta y dy =$$

$$= \frac{d}{dx}\left([u(x,y)\sin\beta y]\Big|_{y=0}^{y=\infty} + \beta\int_0^\infty u(x,y)\cos\beta y dy \right) = -\beta\frac{d}{dx}u_\beta(x).$$

Then, the second equation in the transformation domain takes the form:

$$v_\beta''(x) - \frac{\beta^2(\kappa+1)}{\kappa-1}v_\beta(x) - \frac{2\beta}{\kappa-1}u_\beta'(x) = -\frac{\beta(\kappa+1)}{\kappa-1}\chi(x) - X_{2\beta}(x).$$

The following system of equations is derived:

$$\begin{cases} \dfrac{d^2u_\beta(x)}{dx^2} - \dfrac{\beta^2(\kappa-1)}{\kappa+1}u_\beta(x) + \dfrac{2\beta}{\kappa+1}\dfrac{dv_\beta(x)}{dx} = \dfrac{3-\kappa}{\kappa+1}\chi'(x) - \dfrac{\kappa-1}{\kappa+1}X_{1\beta}(x) \\ \dfrac{d^2v_\beta(x)}{dx^2} - \dfrac{\beta^2(\kappa+1)}{\kappa-1}v_\beta(x) - \dfrac{2\beta}{\kappa-1}\dfrac{du_\beta(x)}{dx} = -\beta\dfrac{\kappa+1}{\kappa-1}\chi(x) - X_{2\beta}(x) \end{cases}$$

In the case when the transverse crack in located inside the semi-strip, the integration is conducted by the generalized scheme:

$$\int_0^\infty \frac{\partial^2u(x,y)}{\partial y^2}\cos\beta y dy = \int_0^{B-0} \frac{\partial^2u(x,y)}{\partial y^2}\cos\beta y dy + \int_{B+0}^\infty \frac{\partial^2u(x,y)}{\partial y^2}\cos\beta y dy =$$

$$= \left[\frac{\partial u(x,y)}{\partial y}\cos\beta y\right]\Big|_{y=0}^{y=B-0} + \beta\int_0^{B-0}\frac{\partial u(x,y)}{\partial y}\sin\beta y dy + \left[\frac{\partial u(x,y)}{\partial y}\cos\beta y\right]\Big|_{y=B+0}^{y=\infty} +$$

$$+\beta\int_{B+0}^\infty \frac{\partial u(x,y)}{\partial y}\sin\beta y dy = \chi'(x) + \left\langle\frac{\partial u}{\partial y}(x,B)\right\rangle\cos\beta B +$$

$$+\beta[u(x,y)\sin\beta y]\Big|_{y=0}^{y=B-0} - \beta^2\int_0^{B-0} u(x,y)\cos\beta y dy +$$

$$+\beta[u(x,y)\sin\beta y]\Big|_{y=B+0}^{y=\infty} - \beta^2\int_{b+0}^\infty u(x,y)\cos\beta y dy = \chi'(x) - \psi'_2(x)\cos\beta B +$$

$$+\beta\langle u(x,B)\rangle\sin\beta B - \beta^2u_\beta(x) = \chi'(x) - \psi'_2(x)\cos\beta B + \beta\psi_1(x)\sin\beta B - \beta^2u_\beta(x)$$

$$\int_0^{\infty} \frac{\partial^2 v(x,y)}{\partial x \partial y} \cos \beta y dy = \frac{d}{dx} \left(\int_0^{B-0} \frac{\partial v(x,y)}{\partial y} \cos \beta y dy + \int_{B+0}^{\infty} \frac{\partial v(x,y)}{\partial y} \cos \beta y dy \right) =$$

$$= \frac{d}{dx} \left([v(x,y) \cos \beta y]|_{y=0}^{y=B-0} + \beta \int_0^{B-0} v(x,y) \sin \beta y dy + \right.$$

$$\left. + [v(x,y) \cos \beta y]|_{y=B+0}^{y=\infty} + \beta \int_{B+0}^{\infty} v(x,y) \sin \beta y dy \right) =$$

$$= \frac{d}{dx} \left(-\chi(x) + \langle v(x,B) \rangle \cos \beta B + \beta v_{\beta}(x) \right) = -\chi'(x) + \psi'_2(x) \cos \beta B + \beta v'_{\beta}(x);$$

$$\int_0^{\infty} \frac{\partial^2 v(x,y)}{\partial y^2} \sin \beta y dy = \int_0^{B-0} \frac{\partial^2 v(x,y)}{\partial y^2} \sin \beta y dy + \int_{B+0}^{\infty} \frac{\partial^2 v(x,y)}{\partial y^2} \sin \beta y dy =$$

$$= \left[\frac{\partial v(x,y)}{\partial y} \sin \beta y \right] \Big|_{y=0}^{y=B-0} - \beta \int_0^{B-0} \frac{\partial v(x,y)}{\partial y} \cos \beta y dy + \left[\frac{\partial v(x,y)}{\partial y} \sin \beta y \right] \Big|_{y=B+0}^{y=\infty} -$$

$$- \beta \int_{B+0}^{\infty} \frac{\partial v(x,y)}{\partial y} \cos \beta y dy = \left\langle \frac{\partial v}{\partial y}(x,B) \right\rangle \sin \beta B - \beta [v(x,y) \cos \beta y]|_{y=0}^{y=B-0} -$$

$$- \beta^2 \int_0^{B-0} v(x,y) \sin \beta y dy - \beta [v(x,y) \cos \beta y]|_{y=B+0}^{y=\infty} - \beta^2 \int_{B+0}^{\infty} v(x,y) \sin \beta y dy =$$

$$= \beta \chi(x) + \frac{\kappa - 3}{\kappa + 1} \psi'_1(x) \sin \beta B - \beta \langle v(x,B) \rangle \cos \beta B - \beta^2 v_{\beta}(x) =$$

$$= \beta \chi(x) + \frac{\kappa - 3}{\kappa + 1} \psi'_1(x) \sin \beta B - \beta \psi_2(x) \cos \beta B - \beta^2 v_{\beta}(x);$$

$$\int_0^{\infty} \frac{\partial^2 u(x,y)}{\partial x \partial y} \sin \beta y dy = \frac{d}{dx} \left(\int_0^{B-0} \frac{\partial u(x,y)}{\partial x} \sin \beta y dy + \int_{B+0}^{\infty} \frac{\partial u(x,y)}{\partial x} \sin \beta y dy \right) =$$

$$= \frac{d}{dx} \left([u(x,y) \sin \beta y]|_{y=0}^{y=B-0} - \beta \int_0^{B-0} u(x,y) \cos \beta y dy + \right.$$

$$\left. + [u(x,y) \sin \beta y]|_{y=B+0}^{y=\infty} - \beta \int_{B+0}^{\infty} u(x,y) \cos \beta y dy \right) =$$

$$= \frac{d}{dx} \left(\langle u(x,B) \rangle \sin \beta B - \beta u_{\beta}(x) \right) = \psi'_1(x) \sin \beta B - \beta u'_{\beta}(x).$$

As the result, the system of equilibrium equation in the transformation domain is written as

$$\begin{cases} u''_{\beta}(x) - \dfrac{\beta^2(\kappa\text{-}1)}{\kappa+1}u_{\beta}(x) + \dfrac{2\beta}{\kappa+1}v'_{\beta}(x) = \dfrac{3-\kappa}{\kappa+1}\chi'(x) - \\ \qquad -\beta\sin\beta B\dfrac{\kappa-1}{\kappa+1}\psi_1(x) + \cos\beta B\dfrac{\kappa-3}{\kappa+1}\psi'_2(x) \\ v''_{\beta}(x) - \dfrac{\beta^2(\kappa+1)}{\kappa-1}v_{\beta}(x) - \dfrac{2\beta}{\kappa-1}u'_{\beta}(x) = -\beta\dfrac{\kappa+1}{\kappa-1}\chi(x) - \\ \qquad -\sin\beta B\psi'_1(x) + \beta\dfrac{\kappa+1}{\kappa-1}\cos\beta B\psi_2(x) \end{cases}$$

In the case when the longitudinal crack in located inside the semi-strip the jumps in the transformation domain can be obtained by the following formulae:

$$\int_0^{\infty}\left\langle\frac{\partial u}{\partial x}(C,y)\right\rangle\cos\beta y dy = \frac{\kappa-3}{\kappa+1}\int_{b_0}^{b_1}\phi'_2(y)\cos\beta y dy =$$

$$= \frac{\kappa-3}{\kappa+1}\left\{[\phi_2(y)\cos\beta y]|_{y=b_0}^{y=b_1} + \beta\int_{b_0}^{b_1}\phi_2(y)\sin\beta y dy\right\} = \beta\frac{\kappa-3}{\kappa+1}\phi_{2\beta}$$

$$\int_0^{\infty}\left\langle\frac{\partial v}{\partial x}(C,y)\right\rangle\sin\beta y dy = -\int_{b_0}^{b_1}\phi'_1(y)\sin\beta y dy =$$

$$= -\left\{[\phi_1(y)\sin\beta y]|_{y=b_0}^{y=b_1} - \beta\int_{b_0}^{b_1}\phi_1(y)\cos\beta y dy\right\} = \beta\phi_{1\beta}$$

Appendix B: The Calculation of the Elements of Green's Matrix

The absolute convergence of all components of Green's matrix can be proved by Abel's test, since their asymptotic representation has the following form:

$$\sum_{n=0}^{\infty} \frac{[\sin\ \alpha_n z \cos \alpha_n z]}{n^4} \text{ and } \sum_{n=0}^{\infty} \frac{[\sin\ \alpha_n z \cos \alpha_n z]}{n^2}.$$

The formulae from [52] are used

$$\sum_{k=1}^{\infty} \frac{\cos kx}{k^2 + a^2} = \frac{\pi}{2a} \frac{cha(\pi - x)}{sha\pi} - \frac{1}{2a^2}, [0 < x < 2\pi]\ ([52], 1.445(2)),$$

$$\sum_{k=1}^{\infty} \frac{k \sin kx}{k^2 + a^2} = \frac{\pi}{2} \frac{sha(\pi - x)}{sha\pi}, [0 < x < 2\pi]\ ([52], 1.445(1)),$$

$$\sum_{k=1}^{\infty} \frac{\cos kx}{\left(k^2 + a^2\right)^2} = -\frac{1}{2a}\frac{d}{da}\left(\sum_{\lambda=1}^{\infty} \frac{\cos kx}{k^2 + a^2}\right) = -\frac{1}{2a}\frac{d}{da}\left(\frac{\pi}{2a}\frac{cha(\pi - x)}{sha\pi} - \frac{1}{2a^2}\right) =$$

$$= -\frac{1}{2a^4} + \frac{\pi}{4a^3 sh(\pi a)}[ch(a(\pi - x)) - a(\pi - x)sh(a(\pi - x))] +$$

$$+ \frac{\pi^4}{4a^2} \frac{ch(\pi a)ch(a(\pi - x))}{(sh(\pi a))^2}, [0 < x < 2\pi];$$

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$$\sum_{k=1}^{\infty}\frac{k\sin kx}{\left(k^2+a^2\right)^2}=-\frac{1}{2a}\frac{d}{da}\left(\sum_{\jmath=1}^{\infty}\frac{k\sin kx}{k^2+a^2}\right)=-\frac{1}{2a}\frac{d}{da}\left(\frac{\pi}{2}\frac{sha(\pi-x)}{sha\pi}\right)=$$
$$-\frac{\pi^2}{4a}\frac{sh(ax)}{\left(sh(a\pi)\right)^2}+\frac{\pi x}{8a}\frac{sh(ax)+sh(a(2\pi-x))}{\left(sh(a\pi)\right)^2},[0<x<2\pi].$$

Taking into consideration that $\alpha_n=\frac{\pi n}{a}$, and also the fact that cosine is even and sine is odd, the following formulae can be deduced:

$$\sum_{n=1}^{\infty}{}'\frac{\cos\alpha_n z}{{\alpha_n}^2+\beta^2}=\sum_{n=1}^{\infty}{}'\frac{\cos\alpha_n|z|}{{\alpha_n}^2+\beta^2}=\frac{ach(\beta(a-|z|))}{2\beta sh(a\beta)},$$

$$\sum_{n=1}^{\infty}{}'\frac{\sin\alpha_n z}{\left({\alpha_n}^2+\beta^2\right)^2}=\mathrm{sgn}(z)\sum_{n=1}^{\infty}{}'\frac{\sin\alpha_n|z|}{\left({\alpha_n}^2+\beta^2\right)^2}=$$
$$\frac{\mathrm{sgn}(z)}{8\beta(sh(2a\beta))}\left[azsh(\beta(2a-|z|))-2a^2sh(\beta|z|)+azsh(\beta|z|)\right],$$

$$\sum_{n=1}^{\infty}{}'\frac{\cos\alpha_n z}{\left({\alpha_n}^2+\beta^2\right)^2}=\sum_{n=1}^{\infty}{}'\frac{\cos\alpha_n|z|}{\left({\alpha_n}^2+\beta^2\right)^2}=\frac{a}{4\beta^3(ch(2a\beta)-1)}[sh(\beta(2a-|z|))+$$
$$+sh(\beta|z|)+\beta zch(\beta(2a-|z|))+2a\beta ch(\beta|z|)-\beta zch(\beta|z|)].$$

With the help of the following trigonometric formulae:

$$\sin x\cdot\sin y=-0.5(\cos(x+y)-\cos(x-y)),$$
$$\sin x\cdot\cos y=0.5(\sin(x+y)+\sin(x-y)),$$
$$\cos x\cdot\cos y=0.5(\cos(x+y)+\cos(x-y)),$$

the series in Green's matrix can be reduced to the previous three series.

As the result the following component-wise representation for Green's matrix-function is derived:

$$G_{11}(x,\xi)=\frac{\kappa ch(\beta(\xi-a+x))-\kappa ch(\beta(a-|\xi-x|))}{2\beta sh(a\beta)(\kappa-1)}-$$
$$-\frac{1}{2(sh(a\beta))^2(\kappa-1)}[sh(a\beta)(\xi sh(\beta(\xi-a+x))+xsh(\beta(\xi-a+x))-$$
$$-ash(\beta(\xi-a+x))-(a-|\xi-x|)ash(\beta(a-|\xi-x|)))-$$
$$-ach(a\beta)(ch(\beta(\xi-a+x))-ch(\beta(a-|\xi-x|)))],$$

$$G_{12}(x,\xi)=\frac{1}{2sh(a\beta)(\kappa+1)}\left[-\frac{ach(a\beta)}{\left((ch(a\beta))^2-1\right)(\kappa+1)}(sh(\beta(\xi-a+x))-\right.$$
$$-\ \mathrm{sgn}(x-\xi)sh(\beta(a-|\xi-x|)))+$$
$$+(ch(\beta(\xi-a+x))(\xi-a+x)-\ \mathrm{sgn}(x-\xi)ch(\beta(a-|\xi-x|))(a-|\xi-x|))],$$

$$G_{21}(x,\xi)=\frac{1}{2sh(a\beta)(\kappa-1)}\left[-\frac{ach(a\beta)}{\left((ch(a\beta))^2-1\right)(\kappa-1)}(sh(\beta(\xi-a+x))+\right.$$
$$+\ \mathrm{sgn}(x-\xi)sh(\beta(a-|\xi-x|)))+$$
$$+(ch(\beta(\xi-a+x))(\xi-a+x)+\ \mathrm{sgn}(x-\xi)ch(\beta(a-|\xi-x|))(a-|\xi-x|))],$$

$$G_{22}(x,\xi)=-\frac{ch(\beta(\xi-a+x))+ch(\beta(a-|\xi-x|))}{2\beta sh(a\beta)}+$$
$$+\frac{1}{2\beta(sh(a\beta))^2(\kappa+1)}[sh(a\beta)(ch(\beta(\xi-a+x))+ch(\beta(a-|\xi-x|))+$$
$$+\beta((a-x-\xi)sh(\beta(\xi-a+x))-(a-|\xi-x|)sh(\beta(a-|\xi-x|)))+$$
$$+a\beta ch(a\beta)(ch(\beta(\xi-a+x))+ch(\beta(a-|\xi-x|)))],$$

where $G_{i,\,j}(x,\xi)$ is the element of Green's matrix placed in ith row, jth column.

Appendix C: The Finding of the Coefficients $c_i, i = \overline{1,4}$

After rewriting of the system (4.6) in a scalar form the following expression is obtained:

$$\begin{cases} \dfrac{\kappa(c_1 - c_3)}{2\beta(\kappa - 1)} = T_1 \\ \dfrac{\kappa(c_2 - c_4)}{2\beta(\kappa + 1)} = T_2 \\ \dfrac{c_1 e^{a\beta}(\kappa - a\beta)}{2\beta(\kappa - 1)} - \dfrac{ac_2 e^{a\beta}}{2(\kappa + 1)} - \dfrac{c_3 e^{-a\beta}(a\beta + \kappa)}{2\beta(\kappa - 1)} + \dfrac{ac_4 e^{-a\beta}}{2(\kappa + 1)} = T_3 \\ \dfrac{ac_1 e^{a\beta}}{2(\kappa - 1)} + \dfrac{c_2 e^{a\beta}(a\beta + \kappa)}{2\beta(\kappa + 1)} - \dfrac{ac_3 e^{-a\beta}}{2(\kappa - 1)} + \dfrac{c_4 e^{-a\beta}(\kappa - a\beta)}{2\beta(\kappa + 1)} = T_4 \end{cases} \tag{C.1}$$

where $T_1 = -u_\beta^1(0), T_2 = -v_\beta^1(0), T_3 = -u_\beta^1(a), T_4 = -v_\beta^1(a)$.

Since the determinant of the system (C.1) is nonzero, then the solutions $Y_1(x)$ are $Y_2(x)$ linearly independent.

The solution of the system (C.1) is uniquely determined and it has the following form:

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$$c_1=\frac{\beta(\kappa-1)}{\kappa\left[\kappa^2(ch(2a\beta)-1)-2a^2\beta^2\right]}\left\{T_1\left[\kappa^2\left(e^{-2a\beta}-1\right)-2a\beta(a\beta+\kappa)\right]+2a^2\beta^2T_2+\right.$$
$$\left.+2\kappa T_3[\kappa sh(a\beta)+a\beta ch(a\beta)]+2a\beta\kappa T_4 sh(a\beta)\right\},$$
$$c_2=-\frac{\beta(\kappa+1)}{\kappa\left[\kappa^2(ch(2a\beta)-1)-2a^2\beta^2\right]}\left\{-2a^2\beta^2T_1+T_2\left[\kappa^2\left(1-e^{-2a\beta}\right)+2a\beta(a\beta-\kappa)\right]+\right.$$
$$\left.+2a\beta\kappa T_3 sh(a\beta)+2\kappa T_4[-\kappa sh(a\beta)+a\beta ch(a\beta)]\right\},$$
$$c_3=\frac{\beta(\kappa-1)}{\kappa\left[\kappa^2(ch(2a\beta)-1)-2a^2\beta^2\right]}\left\{T_1\left[\kappa^2\left(1-e^{-2a\beta}\right)+2a\beta(a\beta-\kappa)\right]+2a^2\beta^2T_2+\right.$$
$$\left.+2\kappa T_3[\kappa sh(a\beta)+a\beta ch(a\beta)]+2a\beta\kappa T_4 sh(a\beta)\right\},$$
$$c_4=\frac{\beta(\kappa+1)}{\kappa\left[\kappa^2(ch(2a\beta)-1)-2a^2\beta^2\right]}\left\{2a^2\beta^2T_1+T_2\left[\kappa^2\left(1-e^{-2a\beta}\right)+2a\beta(a\beta+\kappa)\right]-\right.$$
$$\left.\{-2a\beta\kappa T_3 sh(a\beta)-2\kappa T_4[-\kappa sh(a\beta)+a\beta ch(a\beta)]\right\}.$$

(C.2)

Appendix D: The Summing Up of the Weakly-Convergent Parts of the Integrals

Consider the integral $\int\limits_0^\infty a(x)dx$, which is weakly-convergent. The following method is used for the separation of its weakly-convergent part: the integral $\int\limits_0^\infty a(x)dx$ is expanded into two summands $\int\limits_0^\infty a(x)dx = \int\limits_0^A a(x)dx + \int\limits_A^\infty a(x)dx$. The integrand function in the second derived integral is replaced by its asymptotic representation when $x \to \infty$, and then the summand $\int\limits_0^A \widetilde{a}(x)dx$ is added and subtracted. Here, $\widetilde{a}(x)$ is the asymptotic representation of the integrand function $a(x)$. So,

$$\int\limits_0^\infty a(x)dx = \int\limits_0^\infty \widetilde{a}(x)dx + \left(\int\limits_0^A a(x)dx - \int\limits_0^A \widetilde{a}(x)dx \right), \quad A \to \infty \tag{D.1}$$

Improper integral in this representation can be summed up with the help of the following formulae [52]:

$$\int\limits_0^\infty x^n e^{-\beta x} \sin bx dx = (-1)^n \frac{\partial^n}{\partial \beta^n} \left(\frac{b}{b^2 + \beta^2} \right) [\mathrm{Re}\,\beta > 0, b > 0], ([52], 3.944(11))$$

$$\int\limits_0^\infty x^n e^{-\beta x} \cos bx dx = (-1)^n \frac{\partial^n}{\partial \beta^n} \left(\frac{\beta}{b^2 + \beta^2} \right) [\mathrm{Re}\,\beta > 0, b > 0], ([52], 3.944(12))$$

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$$\int_0^\infty e^{-px}\sin qx\frac{dx}{x}=arctg\frac{q}{p}\,[p>0],([52],3.941(1))$$

$$\int_0^\infty e^{-\beta x}\cos bxdx=\frac{\beta}{b^2+\beta^2},$$

$$\int_0^\infty xe^{-\beta x}\cos bxdx=\frac{\beta^2-b^2}{\left(b^2+\beta^2\right)^2},$$

$$\int_0^\infty x^2e^{-\beta x}\cos bxdx=\frac{2\beta\left(\beta^2-3b^2\right)}{\left(b^2+\beta^2\right)^3},$$

$$\int_0^\infty x^3e^{-\beta x}\cos bxdx=\frac{6\left(\beta^4-6b^2\beta^2+b^4\right)}{\left(b^2+\beta^2\right)^4};$$

$$\int_0^\infty e^{-\beta x}\sin bxdx=\frac{b}{b^2+\beta^2},$$

$$\int_0^\infty xe^{-\beta x}\sin bxdx=\frac{2b\beta}{\left(b^2+\beta^2\right)^2},$$

$$\int_0^\infty x^2e^{-\beta x}\sin bxdx=\frac{2b\left(3\beta^2-b^2\right)}{\left(b^2+\beta^2\right)^3},$$

$$\int_0^\infty x^3e^{-\beta x}\sin bxdx=\frac{24b\beta\left(\beta^2-b^2\right)}{\left(b^2+\beta^2\right)^4};$$

$$\int_0^\infty\left(e^{-\beta x}-e^{-\gamma x}\cos ax\right)\frac{dx}{x}$$

$$=\frac{1}{2}\ln\left(\frac{a^2+\gamma^2}{\beta^2}\right),[\mathrm{Re}\,\beta>0,\ \mathrm{Re}\,\gamma>0],([52],3.951(8))$$

$$\int_0^\infty\frac{e^{-\mu x}-e^{-\nu x}}{x}dx=\ln\frac{\nu}{\mu},[\mathrm{Re}\,\mu>0,\ \mathrm{Re}\,\nu>0],([52],3.434(2))$$

$$\int_0^\infty\frac{\sin(ax)}{x}dx=\frac{\pi}{2}\,\mathrm{sgn}\,a.([52],3.721(1))$$

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